

Enterprise Campus: Hierarchical Design Models

The hierarchical network design model breaks the complex flat network into multiple smaller and more manageable networks. Each level or tier in the hierarchy is focused on a specific set of roles. This design approach offers network designers a high degree of flexibility to optimize and select the right network hardware, software, and features to perform specific roles for the different network layers.

A typical hierarchical enterprise campus network design includes the following three layers:

- **Core layer:** Provides optimal transport between sites and high-performance routing. Due to the criticality of the core layer, the design principles of the core should provide an appropriate level of resilience that offers the ability to recover quickly and smoothly after any network failure event with the core block.

- **Distribution layer:** Provides policy-based connectivity and boundary control between the access and core layers.

- **Access layer:** Provides workgroup/user access to the network.

The two primary and common hierarchical design architectures of enterprise campus networks are the three-tier and two-tier layers models.

Three-Tier Model

This design model, illustrated in [Figure 3-1](#), is typically used in large enterprise campus networks, which are constructed of multiple functional distribution layer blocks.

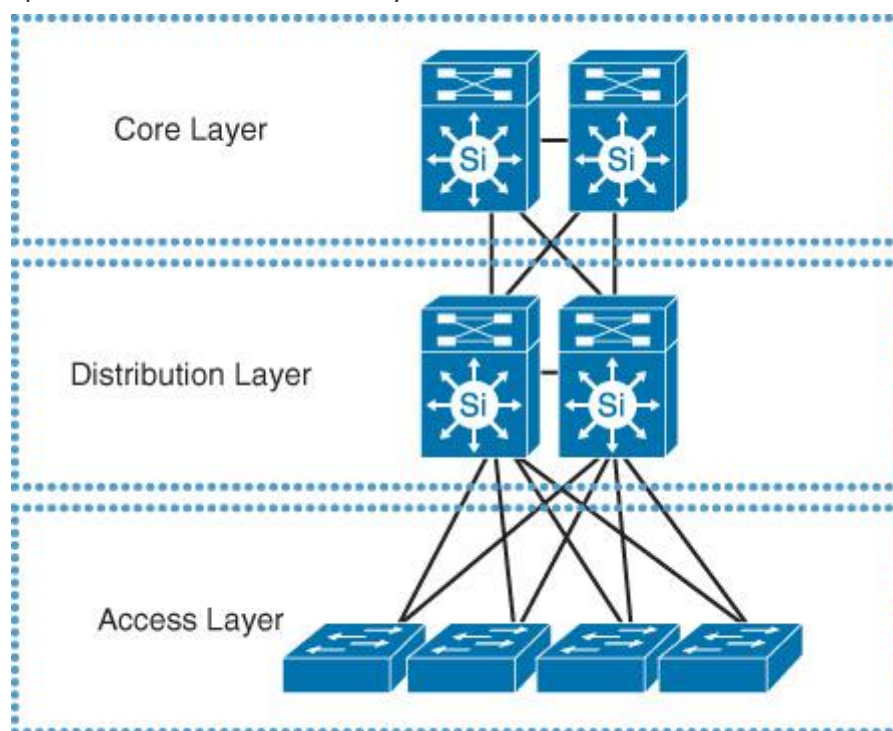


Figure 3-1 Three-Tier Network Design Model

Two-Tier Model

This design model, illustrated in [Figure 3-2](#), is more suitable for small to medium-size campus networks (ideally not more than three functional disruption blocks to be interconnected), where the core and distribution functions can be combined into one layer, also known as *collapsed core-distribution architecture*.

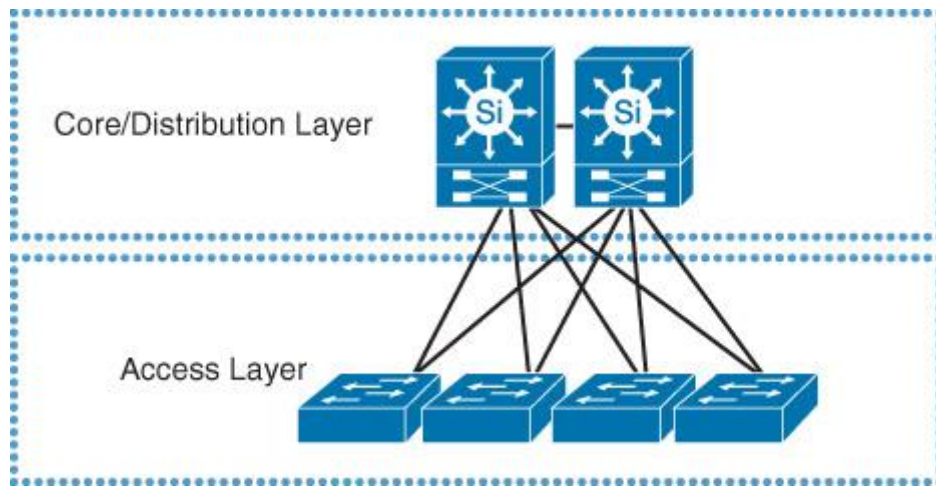


Figure 3-2 Two-Tier Network Design Model

Note

The term *functional distribution block* refers to any block in the campus network that has its own distribution layer such as user access block, WAN block, or data center block.

Enterprise Campus: Modularity

By applying the hierarchical design model across the multiple functional blocks of the enterprise campus network, a more scalable and modular campus architecture (commonly referred to as *building blocks*) can be achieved. This modular enterprise campus architecture offers a high level of design flexibility that makes it more responsive to evolving business needs. As highlighted earlier in this book, modular design makes the network more scalable and manageable by promoting fault domain isolation and more deterministic traffic patterns. As a result, network changes and upgrades can be performed in a controlled and staged manner, allowing greater stability and flexibility in the maintenance and operation of the campus network. [Figure 3-3](#) depicts a typical campus network along with the different functional modules as part of the modular enterprise architecture design.

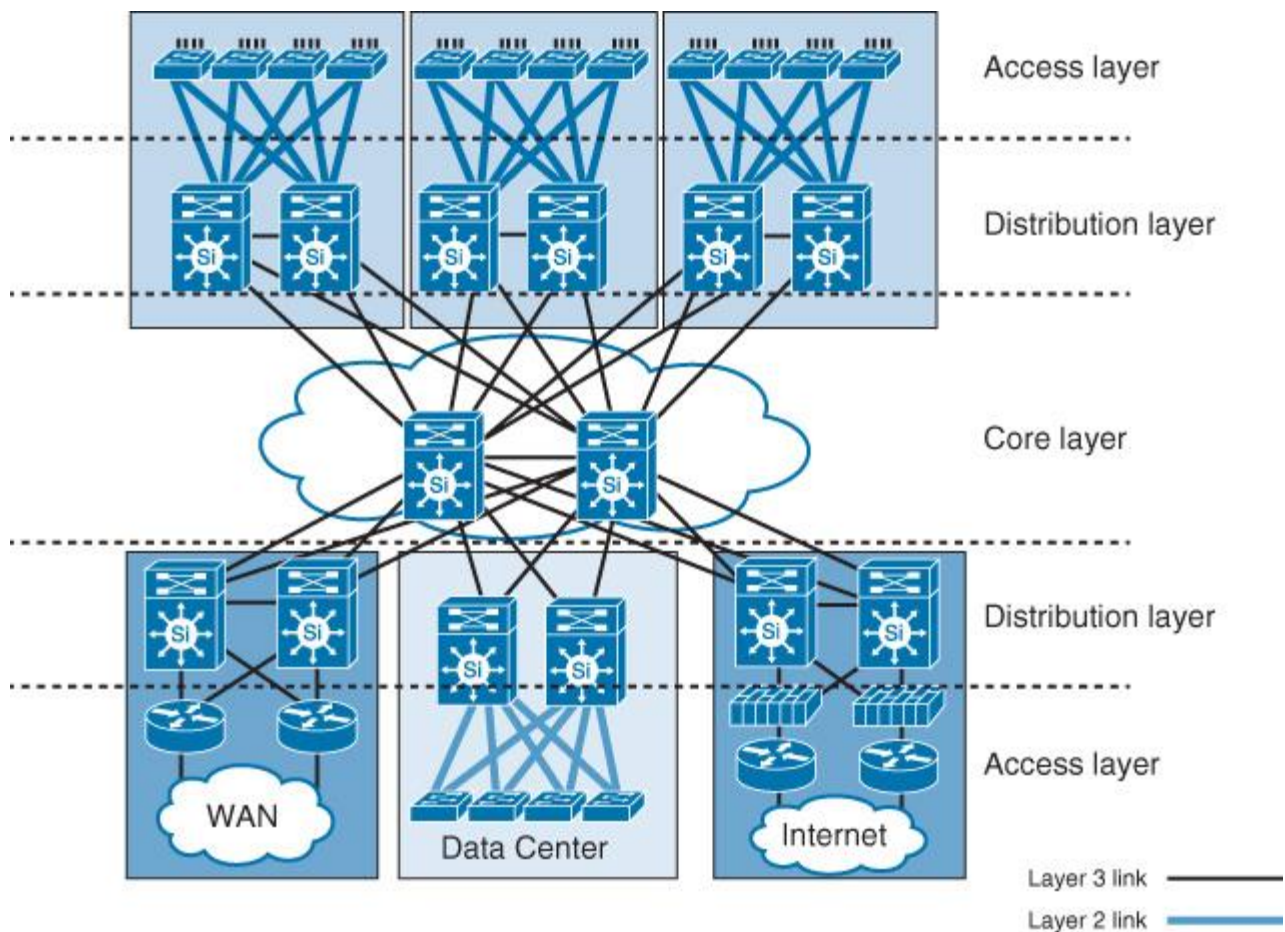


Figure 3-3 Typical Modular Enterprise Campus Architecture

Note

Within each functional block of the modular enterprise architecture, to achieve the optimal structured design, you should apply the same hierarchal network design principle.

When Is the Core Block Required?

A separate core provides the capability to scale the size of the enterprise campus network in a structured fashion that minimizes overall complexity when the size of the network grows (multiple campus distribution blocks) and the number of interconnections tying the multiple enterprise campus functional blocks increases significantly (typically leads to physical and control plane complexities), as exemplified in [Figure 3-4](#). In other words, not every design requires a separate core.

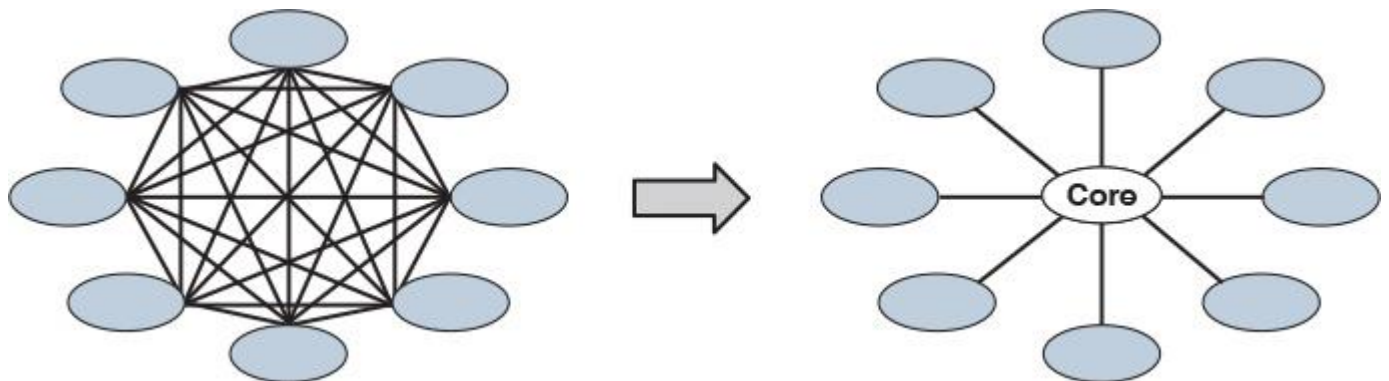


Figure 3-4 Network Connectivity Without Core Versus With Core

Besides the previously mentioned technical considerations, as a network designer you should always aim to provide a business-driven network design with a future vision based on the principle “build today with tomorrow in mind.” Taking this principle into account, one of the primary influencing factors with regard to

selecting two-tier versus three-tier network architecture is the type of site or network (remote branch, regional HQ, secondary or main campus), which will help you, to a certain extent, identify the nature of the site and its potential future scale (from a network design point of view). For instance, it is rare that a typical (small to medium-size) remote site requires a three-tier architecture even when future growth is considered. In contrast, a regional HQ site or a secondary campus network of an enterprise can have a high potential to grow significantly in size (number of users and number of distribution blocks). Therefore, a core layer or three-tier architecture can be a feasible option here. This is from a hypothetical design point of view; the actual answer must always align with the business goals and plans (for example if the enterprise is planning to merge or acquire any new business); it can also derive from the projected percentage of the yearly organic business growth. Again, as a network designer, you can decide based on the current size and the projected growth, taking into account the type of the targeted site, business nature, priorities, and design constraints such as cost. For example, if the business priority is to expand without spending extra on buying additional network hardware platforms (reduce capital expenditure [capex]), in this case the cost savings is going to be a design constraint and a business priority, and the network designer in this type of scenario must find an alternative design solution such as the collapsed architecture (two-tier model) even though technically it might not be the optimal solution.

That being said, sometimes (when possible) you need to gain the support from the business first, to drive the design in the right direction. By highlighting and explaining to the IT leaders of the organization the extra cost and challenges of operating a network that was either not designed optimally with regard to their projected business expansion plans, or the network was designed for yesterday's requirements and it will not be capable enough to handle today's requirements. Consequently, this may help to influence the business decision as the additional cost needed to consider three-tier architecture will be justified to the business in this case (long-term operating expenditure [opex] versus short-term capex). In other words, sometimes businesses focus only on the solution capex without considering that opex can probably cost them more on the long run if the solution was not architected and designed properly to meet their current and future requirements

Access-Distribution Design Model

Chapter 2, "Enterprise Layer 2 and Layer 3 Design," discussed different Layer 2 design models that are applicable to the campus LAN design, in particular to the access-distribution layer. Technically, each design model has different design attributes. Therefore, network designers must understand the characteristics of each design model to be able to choose and apply the most feasible model based on the design requirements.

The list that follows describes the three primary and common design models for the access layer to distribution layer connectivity. The main difference between these design models is where the Layer 2 and Layer 3 boundary is placed and how and where Layer 3 gateway services are handled:

■ **Classical multitier STP based:** This model is the classical or traditional way of connecting access to the distribution layer in the campus network. In this model, the access layer switches usually operate in Layer 2 mode only, and the distribution layer switches operate in Layer 2 and Layer 3 modes. As discussed earlier in this book, the primary limitation of this design model is the reliance on Spanning Tree Protocol (STP) and First Hop Redundancy Protocol (FHRP). For more information, see Chapter 2.

■ **Routed access:** In this design model, access layer switches act as Layer 3 routing nodes, providing both Layer 2 and Layer 3 forwarding. In other words, the demarcation point between Layer 2 and Layer 3 is moved from the distribution layer to the access layer. Based on that, the Layer 2 trunk links from access to distribution are replaced with Layer 3 point-to-point routed links, as illustrated in Figure 3-5.

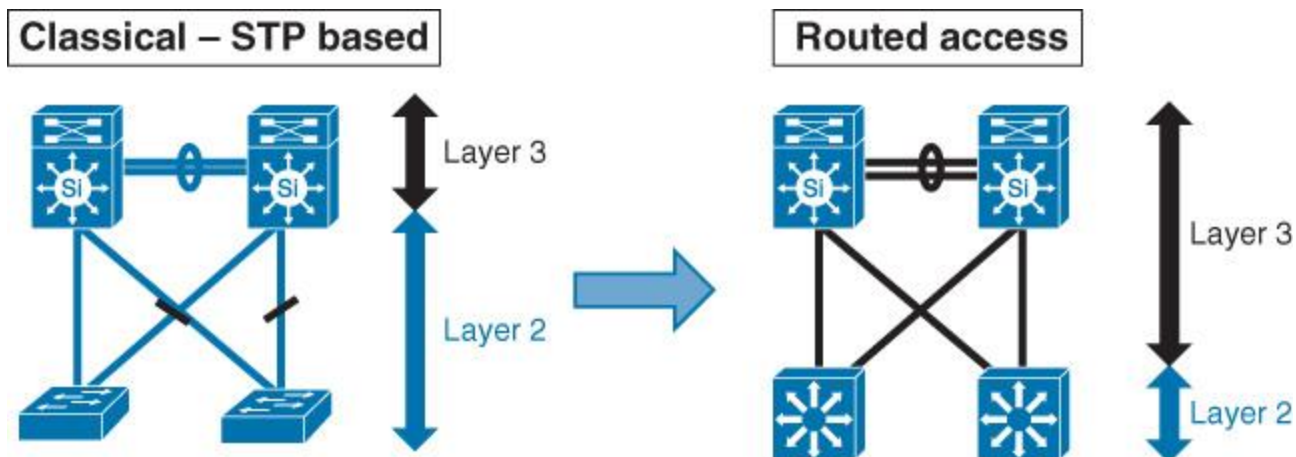


Figure 3-5 Routed Access Layer

The routed access design model has several advantages compared to the multitier classical STP-based access-distribution design model, including the following:

- Simpler and easier to troubleshoot, you can use a standard routing troubleshooting techniques, and you will have fewer protocols to manage and troubleshoot across the network
- Eliminate the reliance on STP and FHRP and rely on the equal-cost multipath (EMCP) of the used routing protocol to utilize all the available uplinks, which can increase the overall network performance
- Minimize convergence time during a link or node failure

Note

The routed access design model does not support spanning Layer 2 VLANs across multiple access switches, and this might not be a good choice for some networks. Although expanding Layer 2 over routed infrastructure is achievable using other different overlay technologies, this might add complexity to the design, or the required features may not be supported with the existing platforms for the access or distribution layer switches.

■ **Switch clustering:** As discussed in [Chapter 2](#), this design model provides the simplest and most flexible design compared to the other models discussed already. As illustrated in [Figure 3-6](#), by introducing the switch clustering concept across the different functional modules of the enterprise campus architecture, network designers can simplify and enhance the design to a large degree. This offers a higher level of node and path resiliency, along with significantly optimized network convergence time.

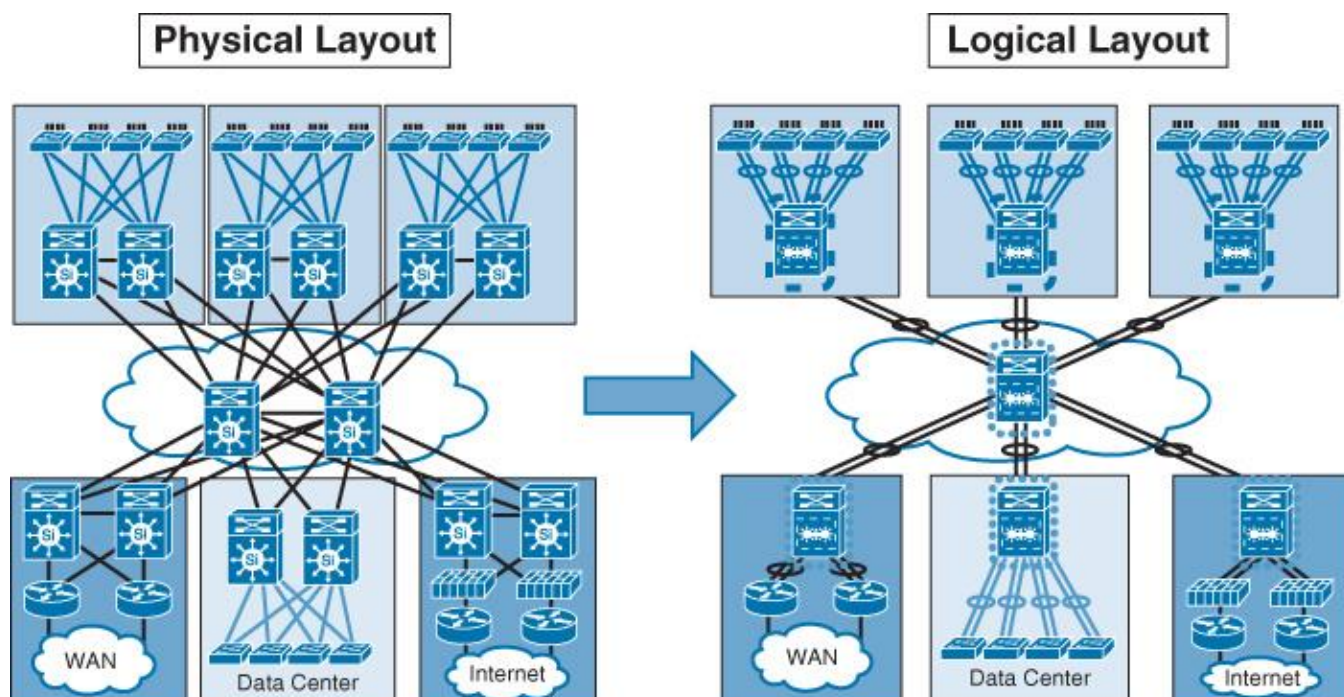


Figure 3-6 Switch Clustering Concept

The left side of [Figure 3-6](#) represents the physical connectivity, and the right side shows the logical view of this architecture, which is based on the switch clustering design model across the entire modular campus network.

[Table 3-1](#) compares the different access-distribution connectivity design models from different design angles.

	Multitier STP Based	Routed Access	Switch Clustering
Design flexibility	Limited (topology dependent)	Limited (For example, Flexible spanning Layer 2 over different access switches requires an overlay technology)	
Scalability	Supports scale up and limited scale out (topology dependent)	Supports both scale up and scale out	Scale up and limited scale out (typically limited to 2 distribution switches per cluster)
Layer 3 gateway services	Distribution layer (FHRP based)	Access layer (Layer 3 routing based)	Distribution layer (may or may not require FHRP*)
Multichassis link aggregation (mLAG)	Not supported	Not supported (instead relies on Layer 3 ECMP)	Supported
Access-to-distribution convergence time	Dependent on STP and FHRP timers (relatively slow)	Interior Gateway Protocol (IGP) dependent, commonly fast	Fast
Operational complexity	Complex (multiple control protocols to deal with [for example, STP, FHRP])	Moderate (Advanced routing design expertise may be required)	Simple

* Some switch clustering technologies, such as Cisco Nexus vPC, use FHRP (Hot Standby Router Protocol [HSRP]). However, from a forwarding plane point of view, both upstream switches (vPC peers) do forward traffic, unlike the classical behavior, which is based on active-standby.

Table 3-1 Comparing Access-Distribution Connectivity Models

Note

All the design models discussed in this section are valid design options. However, the design choice must be driven by the requirements and design constraints, such as cost, which can influence which option you can select. For example,

an access switch with Layer 3 capabilities is more expensive than a switch with Layer 2 capabilities only. This factor will be a valid tiebreaker if cost is a concern from the perspective of business requirements.

Enterprise Campus: Layer 3 Routing Design Considerations

The hierarchal enterprise campus architecture can facilitate achieving more structured hierarchal Layer 3 routing design, which is the key to achieving routing scalability in large networks. This reduces, to a large extent, the number of Layer 3 nodes and adjacencies in any given routing domain within each tier of the hierarchal enterprise campus network [27].

In a typical hierarchal enterprise campus network, the distribution block (layer) is considered the demarcation point between Layer 2 and Layer 3 domains. This is where Layer 3 uplinks participate in the campus core routing, using either an interior routing protocol (IGP) or Border Gateway Protocol (BGP), which can help to interconnect multiple campus distribution blocks together for end-to-end IP connectivity.

By contrast, with the routed access design model, Layer 3 routing is extended to the access layer switches. Consequently, the selection of the routing protocol is important for a redundant and reliable IP/routing reachability within the campus, considering scalability and the ability of the network to grow with minimal changes and impact to the network and routing design. All the Layer 3 routing design considerations discussed in previous chapters must be considered when applying any routing protocol to a campus LAN. [Figure 3-7](#) illustrates a typical ideal routing design that aligns the IGP design (Open Shortest Path First [OSPF]) with the enterprise campus hierarchal architecture, along with the different functional modules.

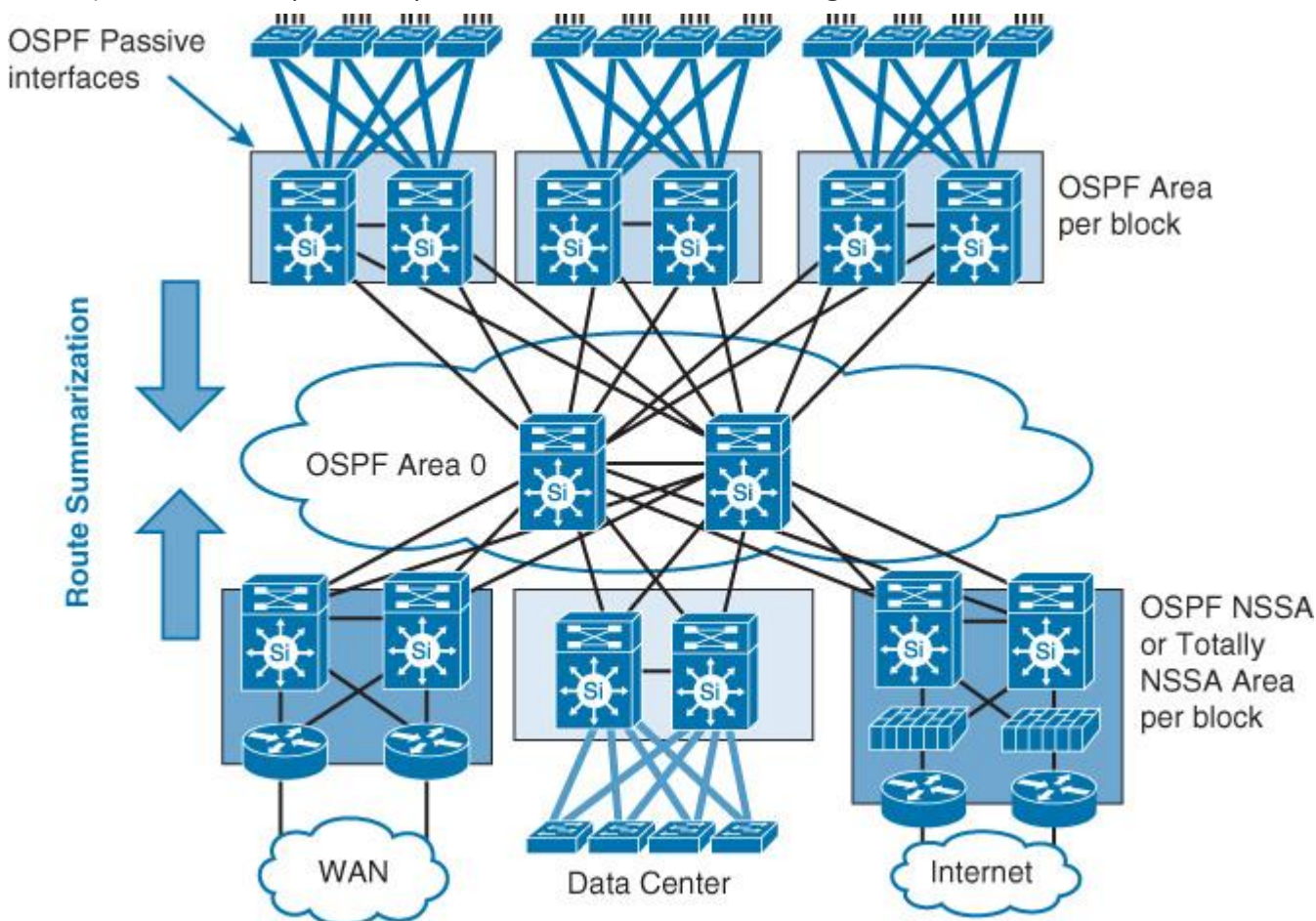


Figure 3-7 Campus Network: Layer 3 Routing

Note

In the preceding example, the data center and other modules coexist within the enterprise campus to illustrate a generic IGP design over multiple modules interconnected over one core infrastructure (typical large campus design). However, in some designs, the data center is interconnected over a WAN or dedicated fiber links with the campus network. Also, the other blocks, such as Internet and WAN blocks, might be coresident at the data center's physical

location as well. In this case, you can use the same IGP design concept, and you can treat the WAN interconnect as an external link, as illustrated in [Figure 3-8](#). Also, the enterprise core routing (that is, OSPF backbone area) can be extended over the WAN. In other words, all the concepts of IGP and BGP design discussed earlier in this book have to be considered to make the right design decision. (There is no single standard design that you can use in different scenarios.)

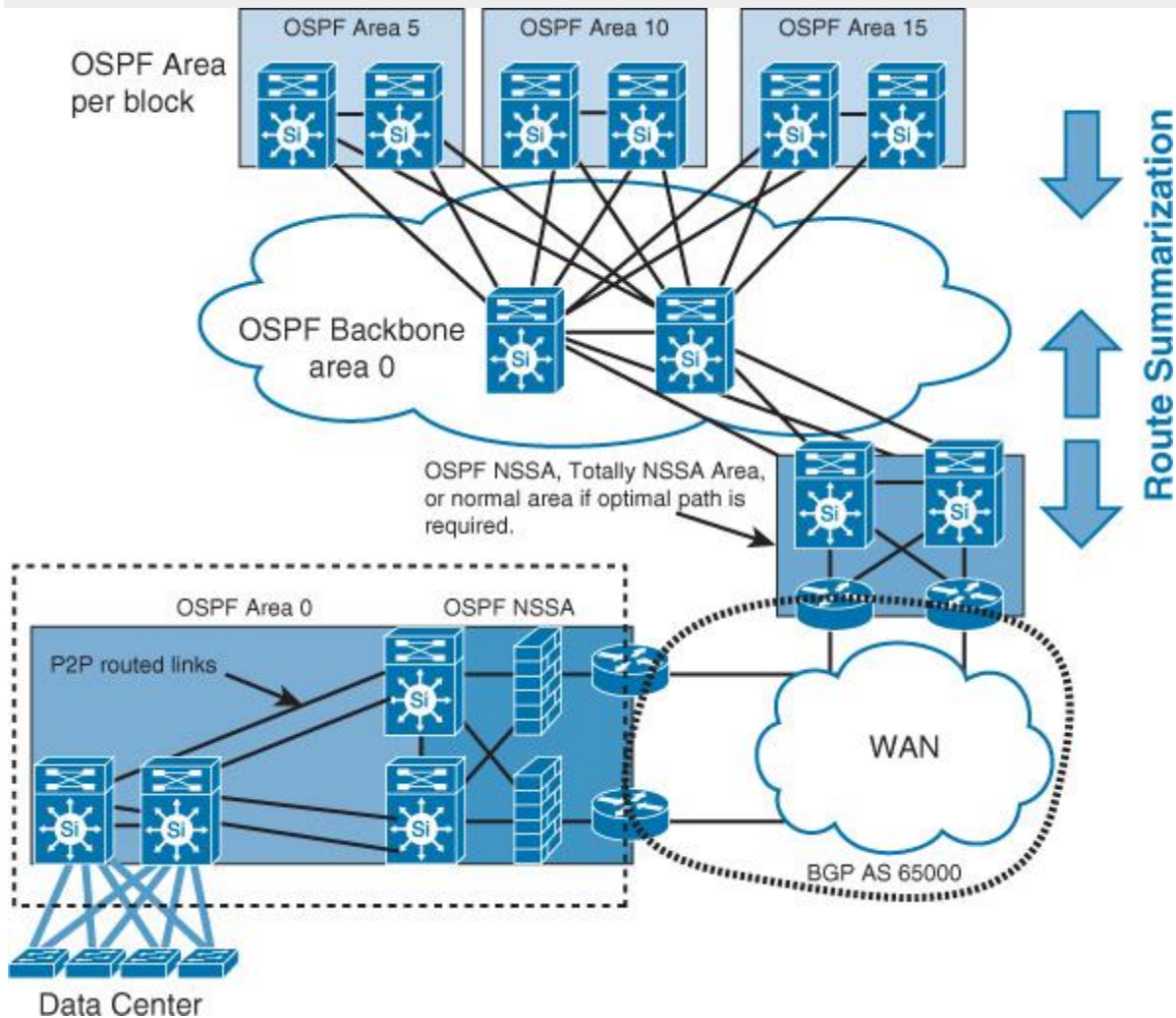


Figure 3-8 *Campus Network: Layer 3 Design with WAN Core*

EIGRP Versus Link State as a Campus IGP

As discussed in [Chapter 2](#), each protocol has its own characteristics, especially when applied to different network topologies. For example, Enhanced Interior Gateway Routing Protocol (EIGRP) offers a more flexible, scalable, and easier-to-control design over “hub-and-spoke” topology compared to link state. In addition, although EIGRP is considered more flexible on multitiered network topologies such as three-tier campus architecture, link-state routing protocols have still proven to be powerful, scalable, and reliable protocols in this type of network, especially OSPF, which is one of the most commonly implemented protocols used in campus networks. Furthermore, in large-scale campus networks, if EIGRP is not designed properly with regard to information hiding and EIGRP query scope containment (discussed in [Chapter 2](#)), any topology change may lead to a large floods of EIGRP queries. In addition, the network will be more prone to EIGRP stuck-in-active (SIA) impacts, such as a longer time to converge following a failure event and as a SIA timer puts an upper boundary on convergence times.

Consequently, each design has its own requirements, priorities, and constraints; and network designers must evaluate the design scenario and balance between the technical (protocol characteristics) and nontechnical (business priorities, future plans, staff knowledge, and so on) aspects when making design decisions.

[Table 3-2](#) provides a summarized comparison between the two common and primary IGPs (algorithms) used in large-scale hierarchal enterprise campus networks.

Design Consideration	EIGRP (DUAL)	Link State (Dijkstra)
Architecture flexibility	High (natively supports multitier architectures with routes summarization)	High, with limitations (The more tiers the network has, the less flexible the design can be.)
Scalability	High	High
Convergence time (protocol level)*	Fast (ideally with route summarization)	Fast (ideally with topology hiding, route summarization, and timers tuning)
MPLS-TE support	No	Yes

* This design aspect is covered in more detail in Chapter 9, “Network High-Availability Design.”

Table 3-2 *Link State Versus EIGRP in the Campus*

Enterprise Campus Network Virtualization

Virtualization in IT generally refers to the concept of having two or more instances of a system component or function such as operating system, network services, control plane, or applications. Typically, these instances are represented in a logical virtualized manner instead of being physical.

Virtualization can generally be classified into two primary models:

■ **Many to one:** In this model, multiple physical resources appear as a single logical unit. The classical example of many-to-one virtualization is the switch clustering concept discussed earlier. Also, firewall clustering, and FHRP with a single virtual IP (VIP) that front ends a pair of physical upstream network nodes (switches or routers) can be considered as other examples of the many-to-one virtualization model.

■ **One to many:** In this model, a single physical resource can appear as many logical units, such as virtualizing an x86 server, where the software (hypervisor) hosts multiple virtual machines (VMs) to run on the same physical server. The concept of network function virtualization (NFV) can also be considered as a one-to-many system virtualization model.

Drivers to Consider Network Virtualization

To meet the current expectations of business and IT leaders, a more responsive IT infrastructure is required. Therefore, network infrastructures need to move from the classical architecture (that is, based on providing basic interconnectivity between different siloed departments within the enterprise network) into a more flexible, resilient, and adaptive architecture that can support and accelerate business initiatives and remove inefficiencies. The IT and the network infrastructure will become like a service delivery business unit that can quickly adopt and deliver services. In other words, it will become a “business enabler.” This is why network virtualization is considered one of the primary principles that enables IT infrastructures to become more dynamic and responsive to the new and the rapidly changing requirements of today’s enterprises.

The following are the primary drivers of modern enterprise networks, which can motivate enterprise businesses to adopt the concept of network virtualization:

■ **Cost efficiency and design flexibility:** Network virtualization provides a level of abstraction from the physical network infrastructure that can offer cost-effective network designs along with a higher degree of design flexibility, where multiple logical networks can be provisioned over one common physical infrastructure. This ultimately will lead to lower capex because of the reduction in device complexity and number of devices. Similarly, it will open lower because the operations team will have fewer devices to manage.

■ **Support a simplified and flexible integrated security:** Network virtualization also promotes flexible security designs by allowing the use of separate security policies per logical or virtualized entity, where users' groups and services can be logically separated.

■ **Design and operational simplicity:** Network virtualization simplifies the design and provision of path and traffic isolation per application, group, service, and various other logical instances that require end-to-end path isolation.

Note

It is important that network designers understand the drivers toward adopting network virtualization from a business point of view, along with the strengths and weaknesses of each design model. This ensures that when a network virtualization concept is considered on a given area within the network or across the entire network, it will deliver the promised value (and not to be used only because it is easy to implement or it is an innovative approach). As discussed earlier in this book, the design that does not address the business's functional requirements is considered a poor design; consider the design principle "no gold plating" discussed in [Chapter 1, "Network Design Requirements: Analysis and Design Principles."](#)

Note

One of the main concerns about network virtualization is the concept of *fate sharing*, because any failure in the physical network can lead to a failure of multiple virtual networks running over the same physical infrastructure. Therefore, when the network virtualization concept is used, ideally a reliable and highly available network design should be considered as well. Besides the concerns about virtual network availability, there is always a concern about network virtualization (multitenant environment) where multiple virtual networks (VNs) operate over a single physical network infrastructure and each VN probably has different traffic requirements (different applications and utilization patterns). Therefore, there is a higher potential of having traffic congestion and degraded application quality and user experience if there is no efficient planning with regard to the available bandwidth, number of VNs, traffic volume per VN, applications in use, and the characteristics of the applications. In other words, if there is no adequate bandwidth available and quality of service (QoS) policies to optimize and control traffic behaviors, one VN may overutilize the available bandwidth of the underlying physical network infrastructure. This will usually lead to traffic congestion because other VNs are using the same underlying physical network infrastructure, resulting in what is known as *fate sharing*.

This section covers the primary network virtualization technologies and techniques that you can use to serve different requirements by highlighting the pros and cons of each technology and design approach. This can help network designers (CCDE candidates) to select the best suitable design after identifying and evaluating the different design requirements (business and functional requirements). This section primarily focuses on network virtualization over the enterprise campus network. [Chapter 4, "Enterprise Edge Architecture Design,"](#) expands on this topic to cover network virtualization design options and considerations over the WAN.

Network Virtualization Design Elements

As illustrated in [Figure 3-9](#), the main elements in an end-to-end network virtualization design are as follows:

■ **Edge control:** This element represents the network access point. Typically, it is a host or end-user access (wired, wireless, or virtual private network [VPN]) to the network where the identification (authentication) for physical to logical network mapping can occur. For example, a contracting employee might be assigned to VLAN X, whereas internal staff is assigned to VLAN Y.

■ **Transport virtualization:** This element represents the transport path that will carry different virtualized networks over one common physical infrastructure, such as an overlay technology like a generic routing encapsulation (GRE) tunnel. The terms *path isolation* and *path separation* are commonly used to refer to transport virtualization. Therefore, these terms are used interchangeably throughout this book.

■ **Services virtualization:** This element represents the extension of the network virtualization concept to the services edge, which can be shared services among different logically isolated groups, such as an Internet link or a file server located in the data center that must be accessed by only one logical group (business unit).

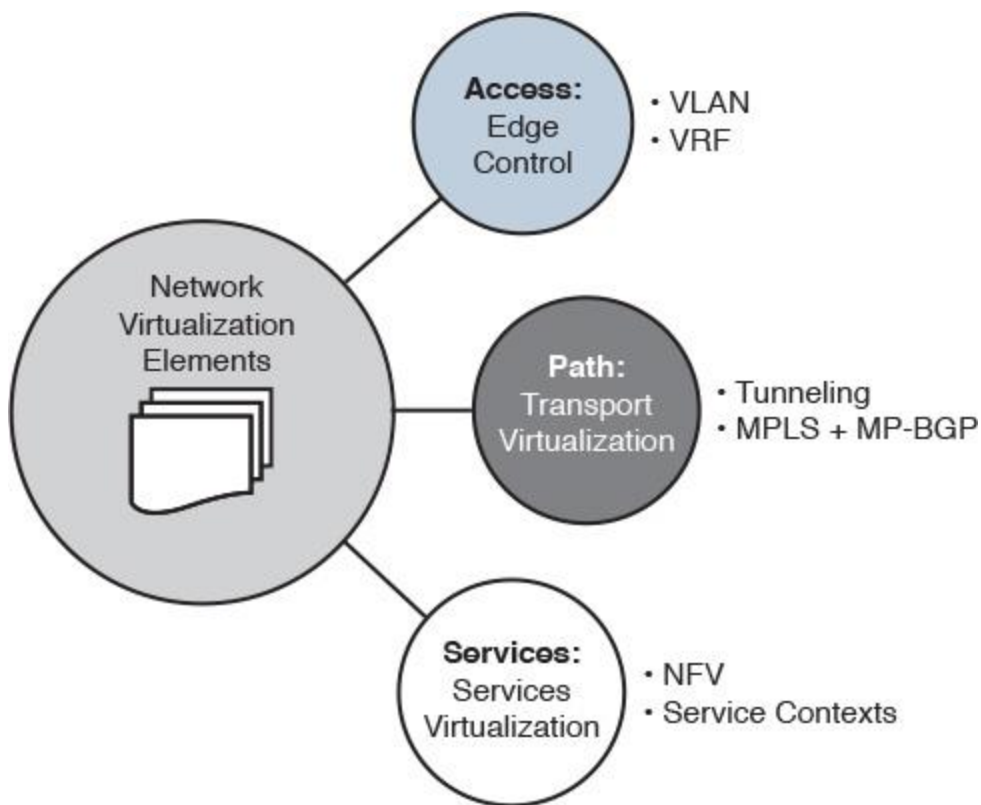


Figure 3-9 Network Virtualization Elements

Enterprise Network Virtualization Deployment Models

Now that you know the different elements that, individually or collectively, can be considered as the foundational elements to create network virtualization within the enterprise network architecture, this section covers how you can use these elements with different design techniques and approaches to deploy network virtualization across the enterprise campus. This section also compares these different design techniques and approaches.

Network virtualization can be categorized into the following three primary models, each of which has different techniques that can serve different requirements:

- Device virtualization
- Path isolation
- Services virtualization

Moreover, you can use the techniques of the different models individually to serve certain requirements or combined together to achieve one cohesive end-to-end network virtualization solution. Therefore, network designers must have a good understanding of the different techniques and approaches, along with their attributes, to select the most suitable virtualization technologies and design approach for delivering value to the business.

Device Virtualization

Also known as *device partitioning*, device virtualization represents the ability to virtualize the data plane, control plane, or both, in a certain network node, such as a switch or a router. Using device level virtualization by itself will help to achieve separation at Layer 2, Layer 3, or both, on a local device level. The following are the primary techniques used to achieve device level network virtualization:

- **Virtual LAN (VLAN):** VLAN is the most common Layer 2 network virtualization technique. It is used in every network where one single switch can be divided into multiple logical Layer 2 broadcast domains that are virtually separated from other VLANs. You can use VLANs at the network edge to place an endpoint into a certain virtual network. Each VLAN has its own MAC forwarding table and spanning-tree instance (Per-VLAN Spanning Tree [PVST]).

■ **Virtual routing and forwarding (VRF):** VRFs are conceptually similar to VLANs, but from a control plane and forwarding perspective on a Layer 3 device. VRFs can be combined with VLANs to provide a virtualized Layer 3 gateway service per VLAN. As illustrated in [Figure 3-10](#), each VLAN over a 802.1Q trunk can be mapped to a different subinterface that is assigned to a unique VRF, where each VRF maintains its own forwarding and routing instance and potentially leverages different VRF-aware routing protocols (for example, OSPF or EIGRP instance per VRF).

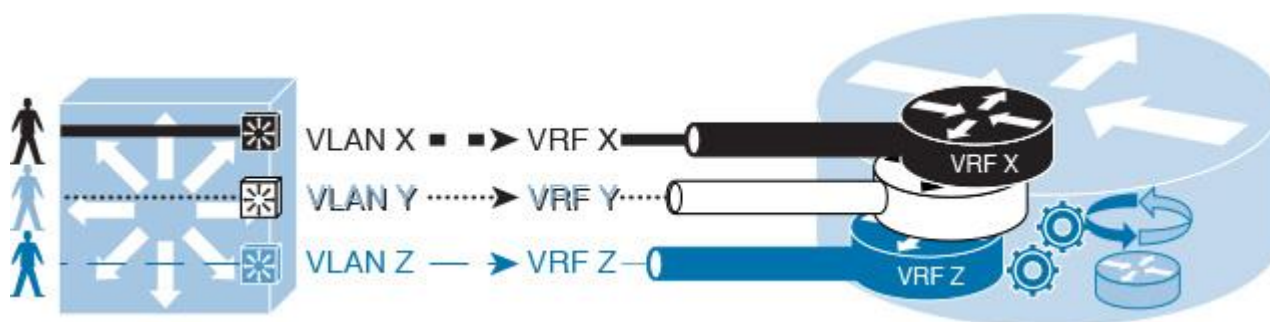


Figure 3-10 Virtual Routing and Forwarding

Path Isolation

Path isolation refers to the concept of maintaining end-to-end logical path transport separation across the network. The end-to-end path separation can be achieved using the following main design approaches:

■ **Hop by hop:** This design approach, as illustrated in [Figure 3-11](#), is based on deploying end-to-end (VLANs + 802.1Q trunk links + VRFs) per device in the traffic path. This design approach offers a simple and reliable path separation solution. However, for large-scale dynamic networks (large number of virtualized networks), it will be a complicated solution to manage. This complexity is associated with design scalability limitation.

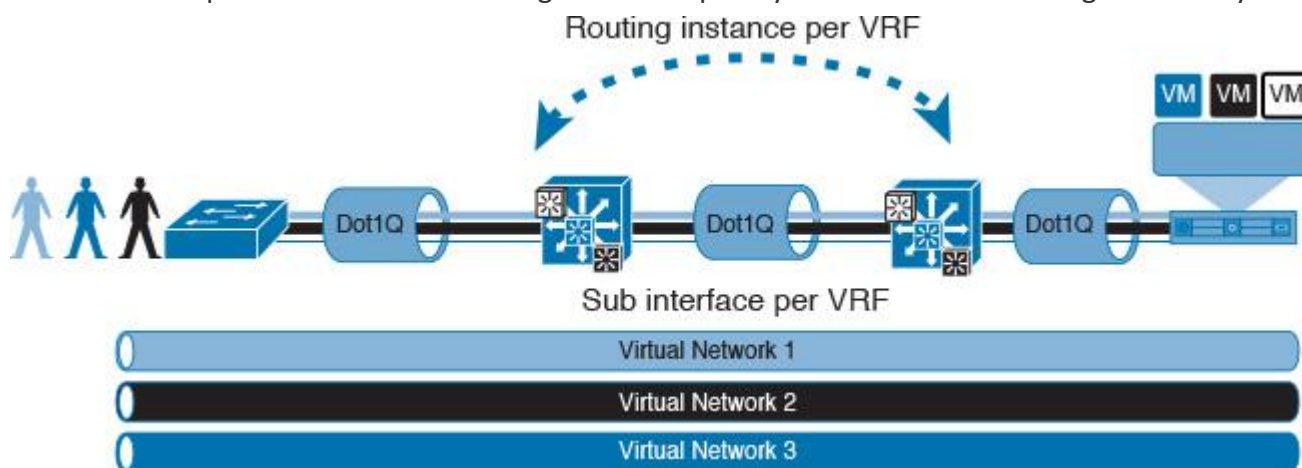


Figure 3-11 Hop-by-Hop Path Virtualization

■ **Multihop:** This approach is based on using tunneling and other overlay technologies to provide end-to-end path isolation and carry the virtualized traffic across the network. The most common proven methods include the following:

■ **Tunneling:** Tunneling, such as GRE or multipoint GRE (mGRE) (dynamic multipoint VPN [DMVPN]), will eliminate the reliance on deploying end-to-end VRFs and 802.1Q trunks across the enterprise network, because the vitalized traffic will be carried over the tunnel. This method offers a higher level of scalability as compared to the previous option and with simpler operation to some extent. This design is ideally suitable for scenarios where only a part of the network needs to have path isolation across the network.

However, for large-scale networks with multiple logical groups or business units to be separated across the enterprise, the tunneling approach can add complexity to the design and operations. For example, if the design requires path isolation for a group of users across two “distribution blocks,” tunneling can be a good fit, combined with VRFs. However, mGRE can provide the same transport and path isolation goal for larger

networks with lower design and operational complexities. (See the section “[WAN Virtualization](#),” in [Chapter 4](#) for a detailed comparison between the different path separation approaches over different types of tunneling mechanisms.)

■ **MPLS VPN:** By converting the enterprise to be like a service provider type of network, where the core is Multiprotocol Label Switching (MPLS) enabled and the distribution layer switches to act as provider edge (PE) devices. As in service provider networks, each PE (distribution block) will exchange VPN routing over MP-BGP sessions, as shown in [Figure 3-12](#). (The route reflector [RR] concept can be introduced, as well, to reduce the complexity of full-mesh MP-BGP peering sessions.)

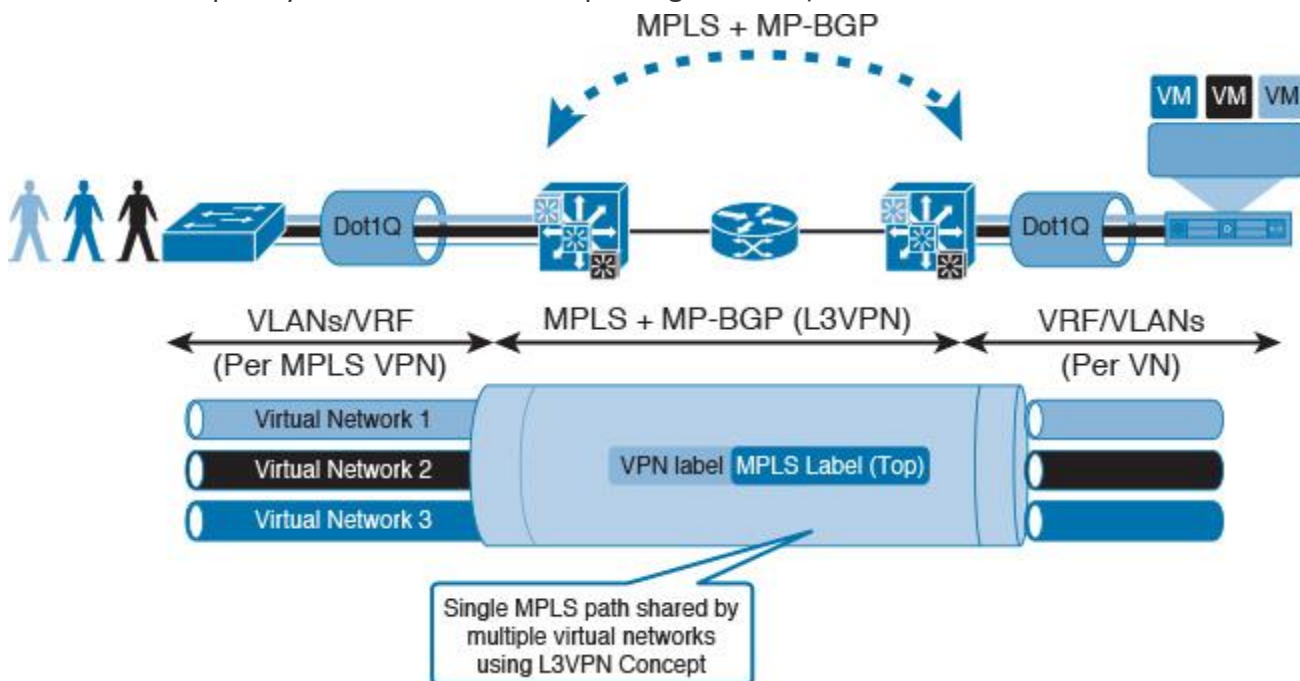


Figure 3-12 *MPLS VPN-Based Path Virtualization*

Furthermore, L2VPN capabilities can be introduced in this architecture, such as Ethernet over MPLS (EoMPLS), to provide extended Layer 2 communications across different distribution blocks if required. With this design approach, the end-to-end virtualization and traffic separation can be simplified to a very large extent with a high degree of scalability. (All the MPLS design considerations and concepts covered in the Service Provider part—[Chapter 5](#), “[Service Provider Network Architecture Design](#),” and [Chapter 6](#), “[Service Provider MPLS VPN Services Design](#),”—in this book are applicable if this design model is adopted by the enterprise.)

[Figure 3-13](#) illustrates a summary of the different enterprise campus network’s virtualization design techniques.

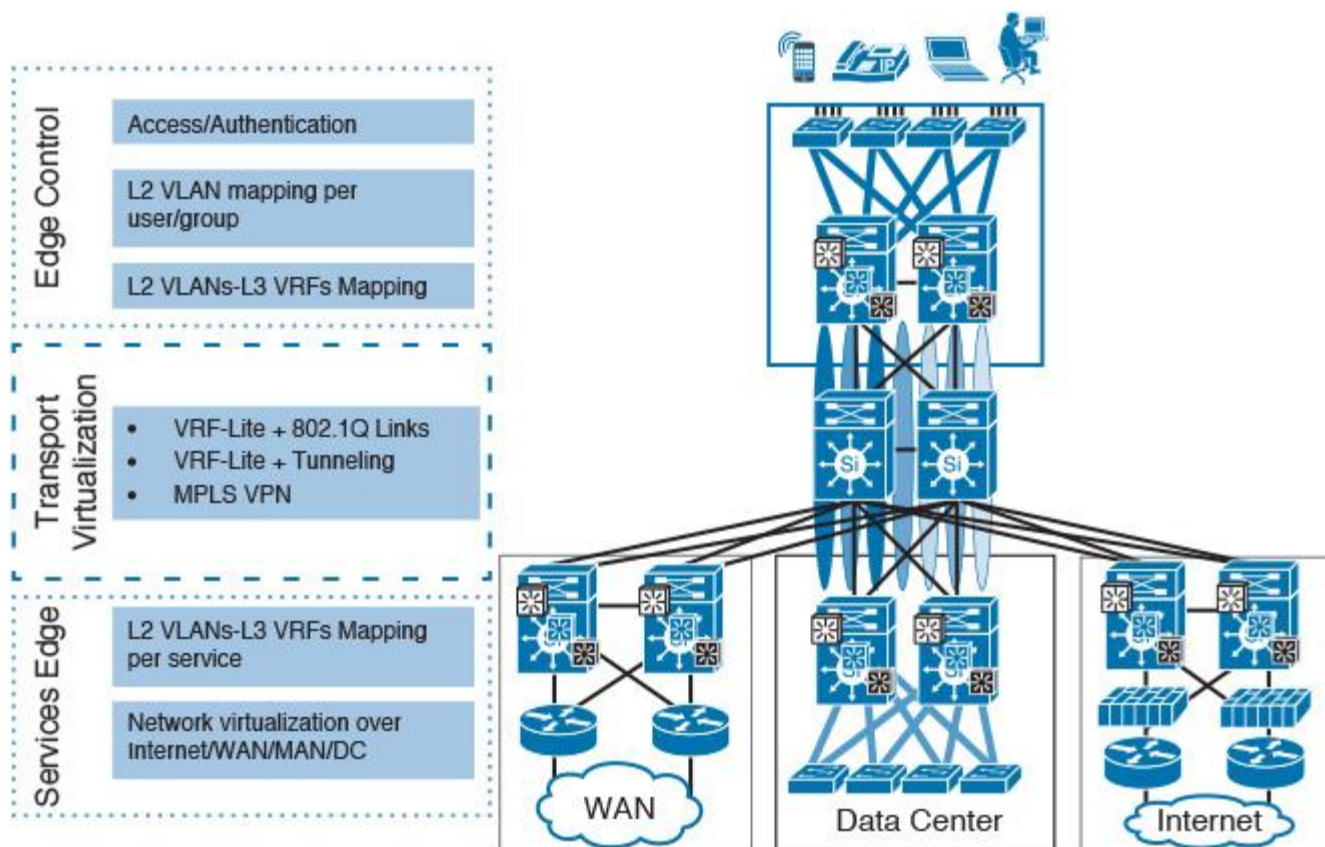


Figure 3-13 Enterprise Campus Network Virtualization Techniques

As mentioned earlier in this section, it is important for network designers to understand the differences between the various network virtualization techniques. [Table 3-3](#) compares these different techniques in a summarized way from different design angles.

	End to End (VLAN + 802.1Q + VRF)	VLANs + VRFs + GRE Tunnels	VLANs + VRFs + mGRE Tunnels	MPLS Core with MP-BGP
Scalability	Low	Low	Moderate	High
Operational complexity	High	Moderate	Moderate	Moderate to high
Design flexibility	Low	Moderate	Moderate	High
Architecture	Per hop end-to-end virtualization	P2P (multihop end-to-end virtualization)	P2MP (multihop end-to-end virtualization)	MPLS-L3VPN-based virtualization
Operation staff routing expertise	Basic	Medium	Medium	Advanced
Ideal for	Limited NV scope in terms of size and complexity	Interconnecting specific blocks with NV or as an interim solution	Medium to large overlaid NV design	Large to very large (global scale) end-to-end NV design

Table 3-3 Network Virtualization Techniques Comparison

Service Virtualization

One of the main goals of virtualization is to separate services access into different logical groups, such as user groups or departments. However, in some scenarios, there may be a mix of these services in term of service access, in which some of these services must only be accessed by a certain group and others are to be shared among different groups, such as a file server in the data center or Internet access, as shown in [Figure 3-14](#).

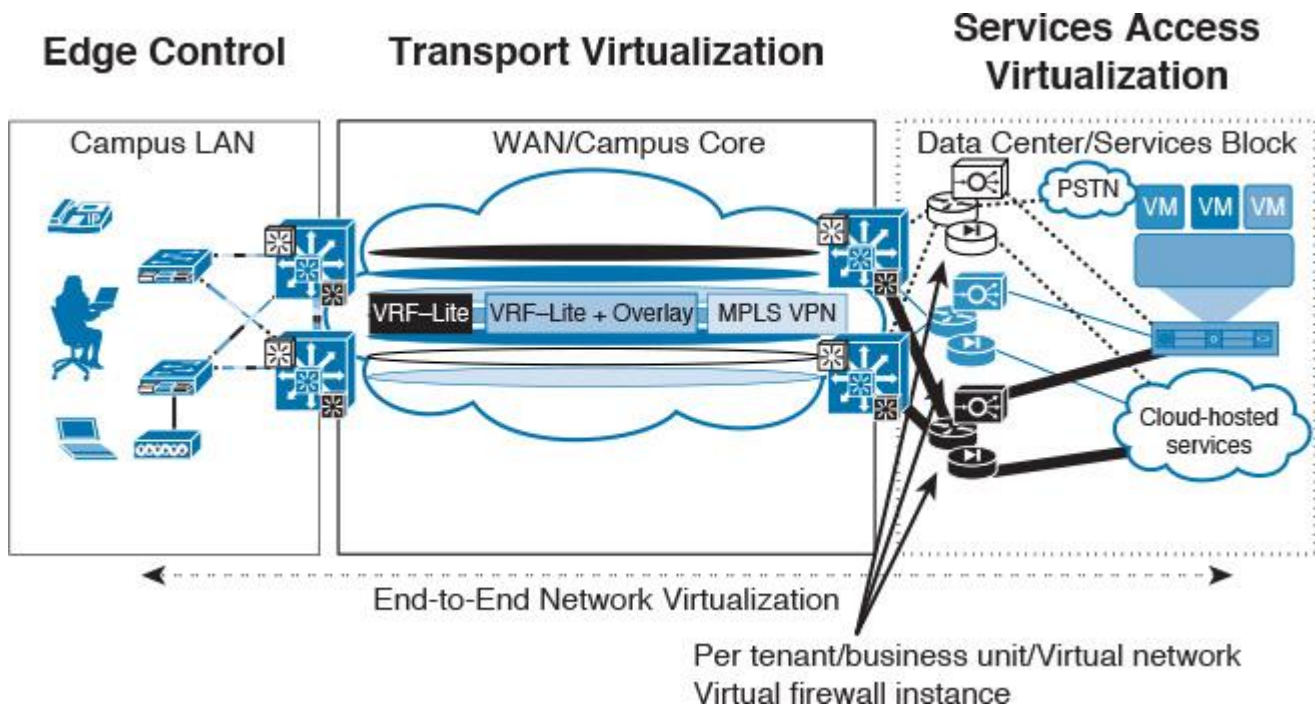


Figure 3-14 End-to-end Path and Services Virtualization

Therefore, in scenarios like this where service access has to be separated per virtual network or group, the concept of network virtualization must be extended to the services access edge, such as a server with multiple VMs or an Internet edge router with single or multiple Internet links.

Note

The virtualization of a network can be extended to other network service appliances, such as firewalls. For instance, you can have a separate virtual firewall per virtual network, to facilitate access control between the virtual user

network and the virtualized services and workload, as shown in [Figure 3-15](#). The virtualization of network services appliance can be considered as a “one-to-many” network device level virtualization.

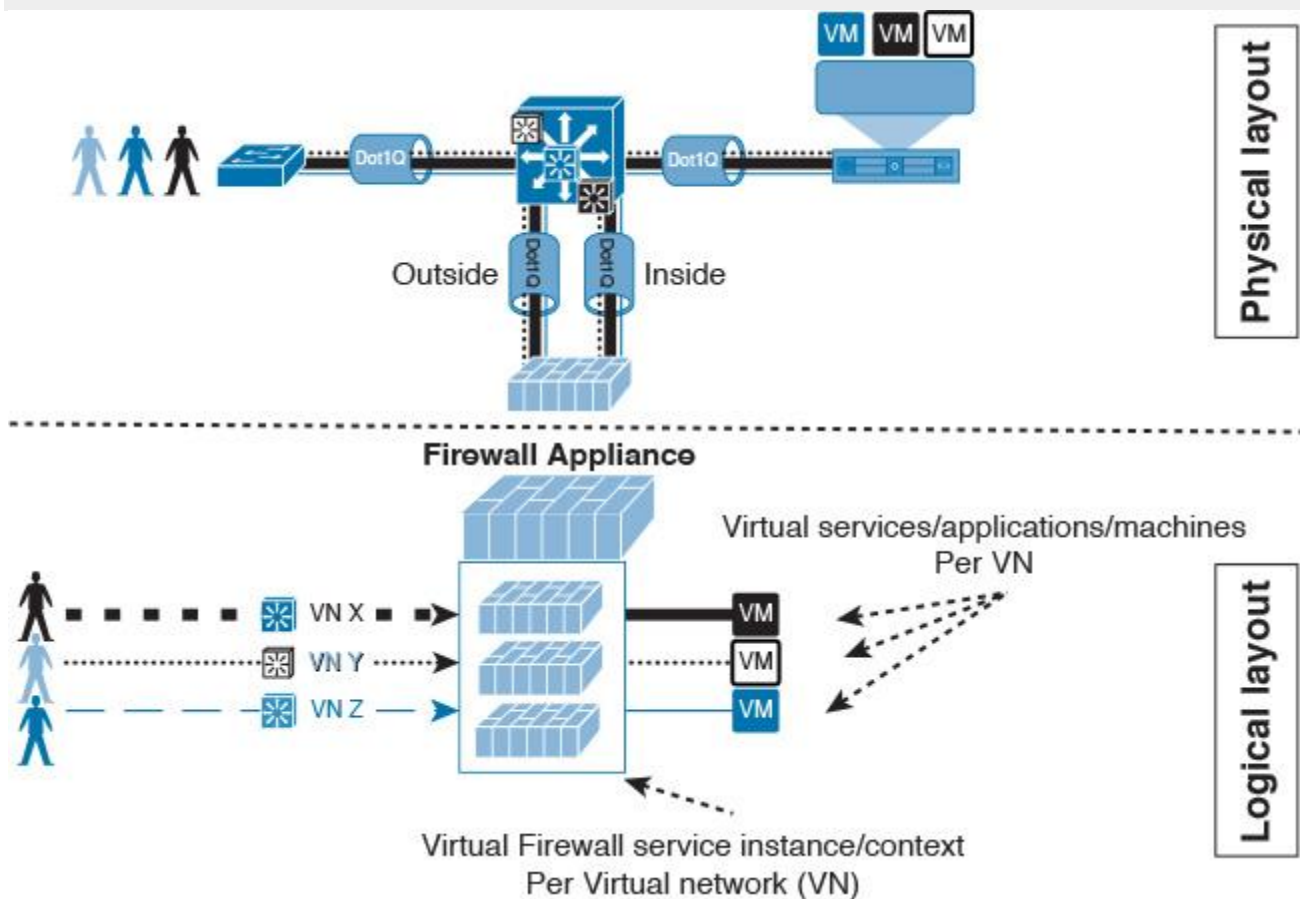


Figure 3-15 Firewall Virtual Instances

Furthermore, in multitenant network environments, multiple security contexts offer a flexible and cost-effective solution for enterprises (and for service providers). This approach enables network operators to partition a single pair of redundant firewalls or a single firewall cluster into multiple virtual firewall instances per business unit or tenant. Each tenant can then deploy and manage its own security policies and service access, which are virtually separated. This approach also allows controlled intertenant communication. For example, in a typical multitenant enterprise campus network environment with MPLS VPN (L3VPN) enabled at the core, traffic between different tenants (VPNs) is normally routed via a firewalling service for security and control (who can access what), as illustrated in [Figure 3-16](#).

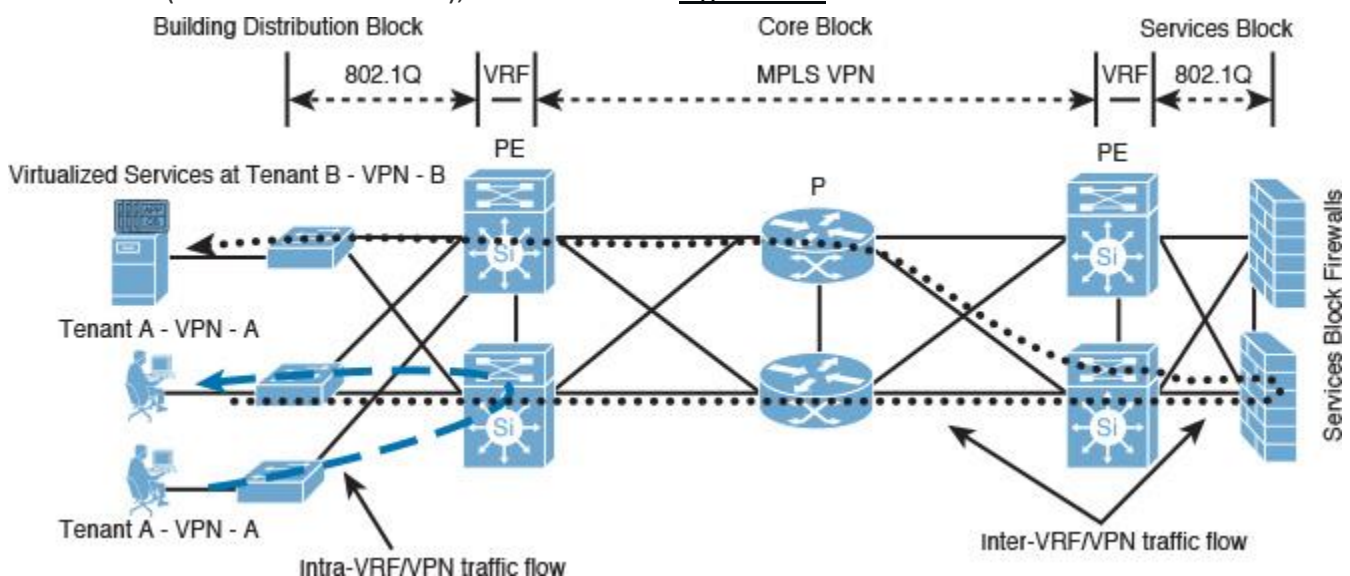


Figure 3-16 Intertenant Services Access Traffic Flow

Figure 3-17 zooms in on the firewall services contexts to show a more detailed view (logical/virtualized view) of the traffic flow between the different tenants/VPNs (A and B), where each tenant has its own virtual firewall service instance located at the services block (or at the data center) of the enterprise campus network.

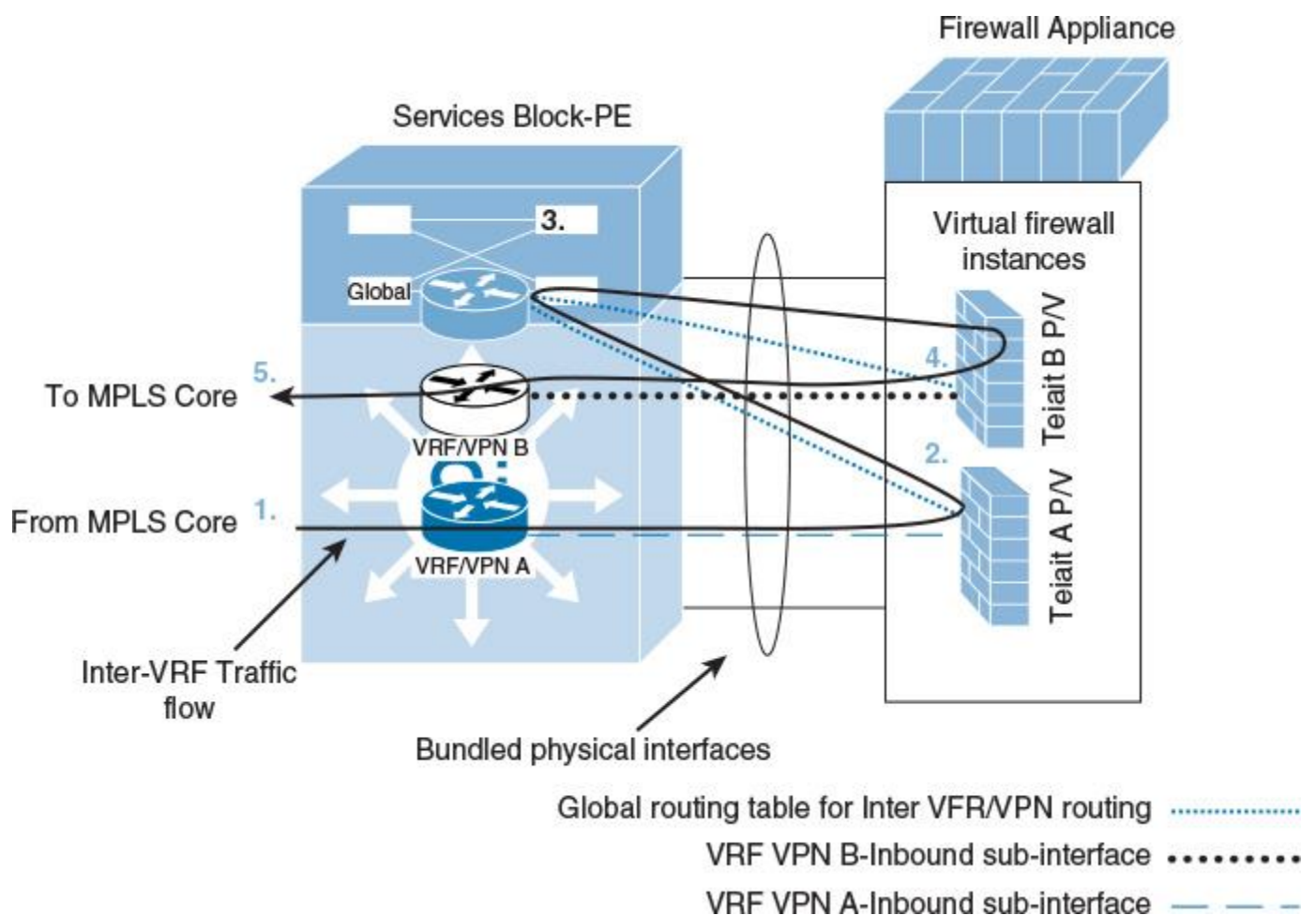


Figure 3-17 Intertenant Services Access Traffic Flow with Virtual Firewall Instances

In addition, the following are the common techniques that facilitate accessing shared applications and network services in multitenant environments:

■ **VRF-Aware Network Address Translation (NAT):** One of the common requirements in today's multitenant environments with network and service' virtualization enabled, is to provide each virtual (tenant) network the ability to access certain services (shared services) either hosted on premise (such as at the emperies data center or services block) or hosted externally (in a public cloud). Also, providing Internet access to the different tenants (virtual) networks, is a common example of today's multitenant network requirements. To maintain traffic separation between the different tenants (virtual networks) where private IP address overlapping is a common attribute in this type of environment, NAT is considered one of the common and cost-effective solutions to provide NAT per tenant without compromising path separation requirements between the different tenants' networks (virtual networks). When NAT is combined with different virtual network instances (VRFs), it is commonly referred to as *VRF-Aware NAT*, as shown in Figure 3-18.

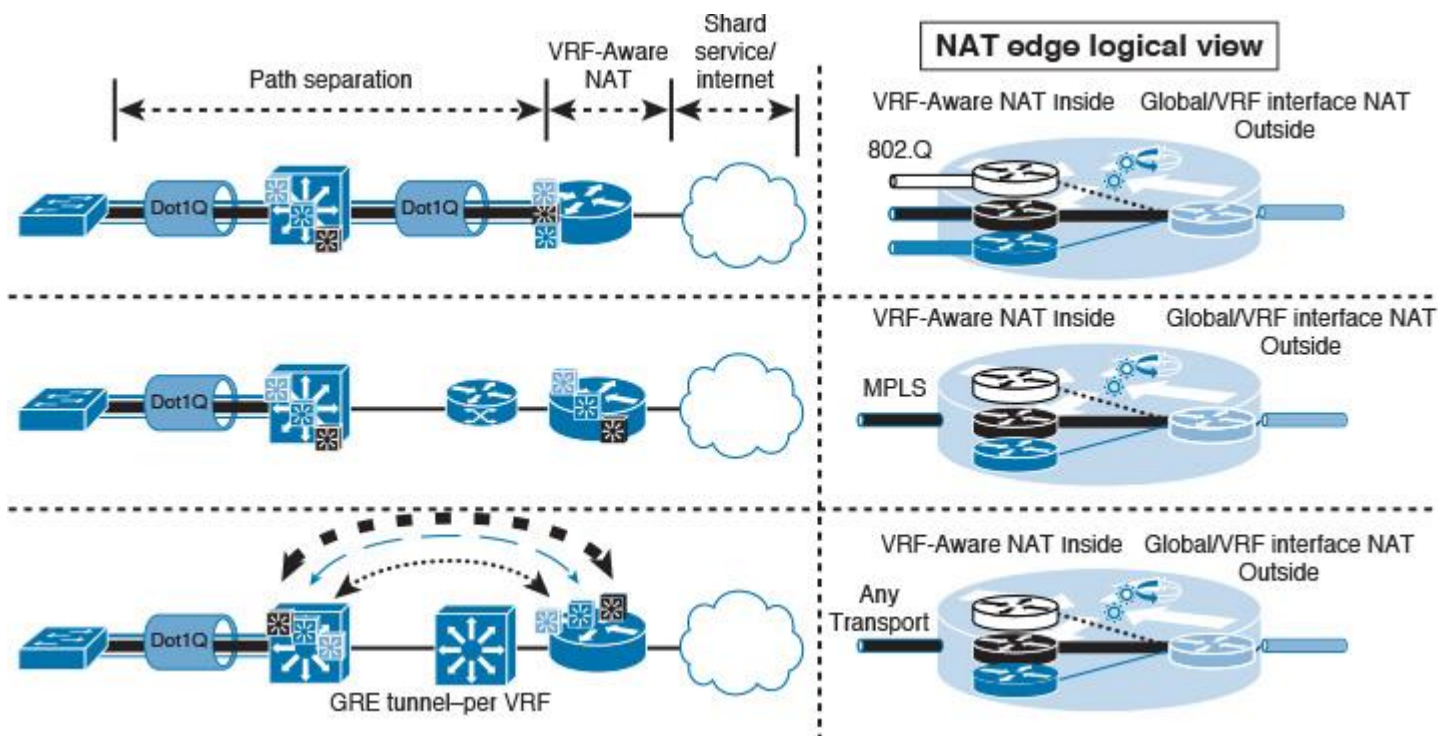


Figure 3-18 VRF-Aware NAT

Note

VRF-aware service infrastructure (VASI) refers to the ability of an infrastructure or a network node, such as a router, to facilitate the application of features and management services (such as encryption and NAT) between VRFs internally within the same node, using virtual interfaces. For two VRFs to communicate internally within a network node (router), a VASI virtual interface pair can be configured. Each interface in this pair must be associated with a different VRF so that those two virtual interfaces can be logically wired,¹ as illustrated in Figure 3-19. This capability is available in some high-end Cisco routers.

1. "Configuring VRF-Aware Service Infrastructure," http://www.cisco.com/c/en/us/td/docs/ios_xr_sw/iosxr_r3-8/vfw/configuration/guide/vfc38/vfc38vas.pdf

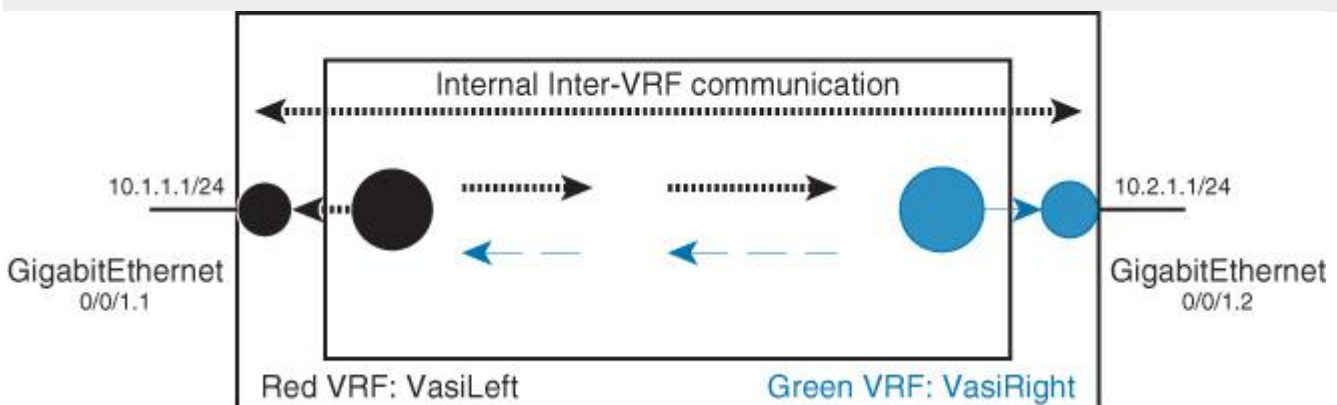


Figure 3-19 VRF-aware Services Infrastructure

■ **Network function virtualization (NFV):** The concept of NFV is based on virtualizing network functions that typically require a dedicated physical node, appliances, or interfaces. In other words, NFV can potentially take any network function typically residing in purpose-built hardware and abstract it from that hardware. As depicted in Figure 3-20, this concept offers businesses several benefits, including the following:

- Reduce the total cost of ownership (TCO) by reducing the required number and diversity of specialized appliances
- Reduce operational cost (for example, less power and space)
- Offer a cost-effective capital investment

- Reduce the level of complexity of integration and network operations
- Reduce time to market for the business by offering the ability to enable specialized network services (Especially in multitenant where a separate network function/service per tenant can be provisioned faster)

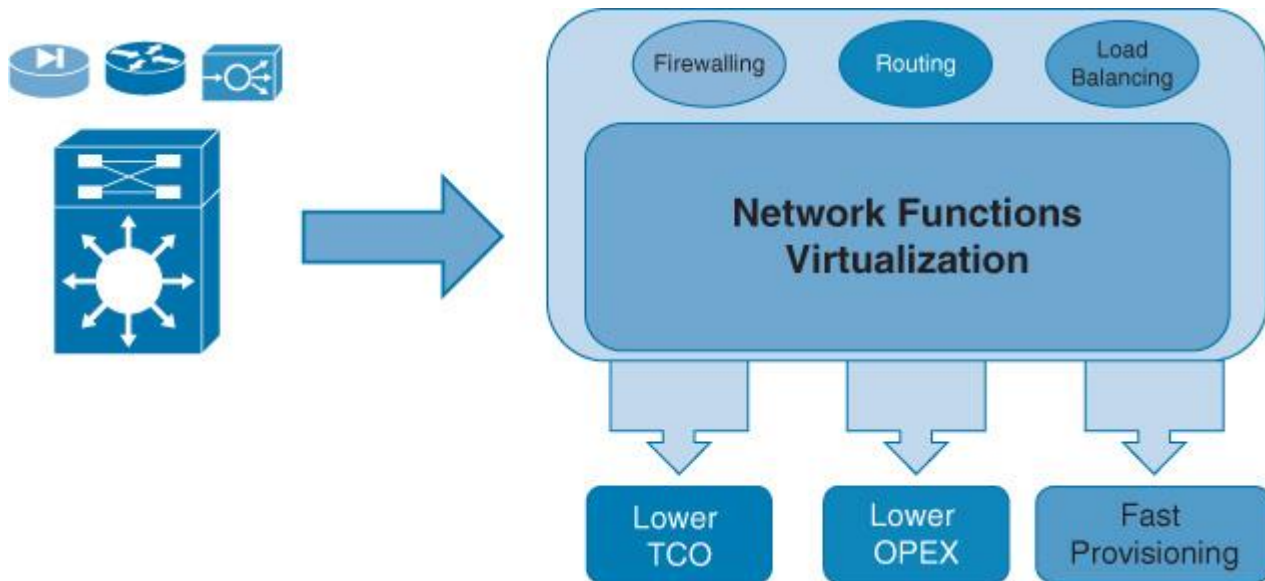


Figure 3-20 NFV Benefits

This concept helps businesses to adopt and deploy new services quickly (faster time to market), and is consequently considered a business innovation enabler. This is simply because purpose-built hardware functionalities have now been virtualized, and it is a matter of service enablement rather than relying on new hardware (along with infrastructure integration complexities).

Note

The concept of NFV is commonly adopted by service provider networks nowadays. Nonetheless, this concept is applicable and usable in enterprise networks and enterprise data center networks that want to gain its benefits and flexibility.

Note

In large-scale networks with a very high volume of traffic (typically carrier grade), hardware resource utilization and limits must be considered.

Summary

The enterprise campus is one of the vital parts of the modular enterprise network. It is the medium that connects the end users and the different types of endpoints such as printers, video endpoints, and wireless access points to the enterprise network. Therefore, having the right structure and design layout that meets current and future requirements is critical, including the physical infrastructure layout, Layer 2, and Layer 3 designs. To achieve a scalable and flexible campus design, you should ideally base it on hierarchical and modular design principles that optimize the overall design architecture in terms of fault isolation, simplicity, and network convergence time. It should also offer a desirable level of flexibility to integrate other networks and new services and to grow in size.

However, the concept of network virtualization helps enterprises to utilize the same underlying physical infrastructure while maintaining access, and path and services access isolation, to meet certain business goals or functional security requirements. As a result, enterprises can lower capex and opex and reduce the time and effort required to provision a new service or a new logical network. However, the network designer must consider the different network virtualization design options, along with the strengths and weaknesses of each, to deploy the suitable network virtualization technique that meets current and future needs. These needs must take into account the different variables and constraints, such as staff knowledge and the hardware platform supported features and capabilities.

Further Reading

“Borderless Campus 1.0 Design Guide,” <http://www.cisco.com>

“Enterprise Campus 3.0 Architecture: Overview and Framework,” <http://www.cisco.com>

“Cisco Wired LAN, Cisco Validated Design,” <http://www.cisco.com>

“Network Virtualization Solutions, Design Guides,” <http://www.cisco.com>

“Network Services Virtualization,” <http://www.cisco.com>

Chapter 4. Enterprise Edge Architecture Design

Enterprise edge refers to the various enterprise network modules and components that facilitate efficient and secure communication between the different enterprise locations, including campuses, data centers, third-party external connections, remote sites, business partners, mobile users, and the Internet. Typically, enterprise edge can be divided into two primary modules as part of the modular enterprise architecture:

- WAN module

- Internet module

This chapter covers different design options and considerations for the WAN and Internet modules.

Note

The WAN and Internet enterprise modules may vary from design to design in terms of naming, size, and functionality. For instance, some designers combine the functions of these two modules into one module. In spite of this, the overall design principles and goals are similar to a large extent.

Enterprise WAN Module

The WAN module is the gateway of the enterprise network to the other remote sites (typical remote branches) and regional sites. As part of the modular enterprise network architecture, this module aggregates and houses all the WAN or MAN edge devices that extend the enterprise network to the remote sites using different transport media types and technologies. Enterprises require a WAN design that offers a common resource access experience to the remote sites, along with sufficient performance and reliability.

As organizations move into multinational or global business markets, they require a flexible network design that reduces the time needed to add new remote sites. They also require the deployment of new technologies that support emerging business applications and communications. In addition, it is becoming a common business requirement for users to have a consistent experience when connecting to the enterprise’s online resources (such as applications and files), whether they are at the company headquarters or at a remote site; therefore, the WAN design should be flexible enough to accommodate different business application requirements. It should also offer the ability to scale bandwidth or to add new sites or resilient links without any major change to the overall design architecture.

From an enterprise architectural point of view, the primary WAN module can be either coresident physically within the data center block of the enterprise or at the primary enterprise campus. In both cases, there is “no difference” for the WAN module architecture itself, because the aggregation layer of the WAN module will be connected to the core layer of either the enterprise campus or data center.

This chapter covers and compares the different design options that you can use to provide the most suitable WAN design to meet various business, functional, and technical requirements.

WAN Transports: Overview

One of the primary concerns of enterprise IT leaders is to manage costs and maintain reliable WAN infrastructures to meet their business goals. Furthermore, businesses are realizing that avoiding the complexities around the WAN transport between the different enterprise locations is becoming key to success in today’s high-tech competitive market where technology solutions became the primary business

enabler and facilitator. Most commonly, enterprise IT leaders are always concerned about adopting a WAN solution that is capable and agile enough to address the following considerations:

- Interconnect the different enterprise locations and remote sites that are geographically dispersed
- Meet enterprise security policy requirements by protecting enterprise traffic over the WAN (secure transport), to offer the desired end-to-end level of protection and privacy across the enterprise network
- Cost-effective and reliable WAN by providing flexible and reliable transport that meets the primary business objectives and critical enterprise application requirements and that supports the convergence of voice, data, and video to satisfy the minimum requirements of today's converged enterprise networks
- Support business evolution, change, and growth by offering the desired level of agility and scalability to meet the current and projected growth of remote sites with flexible bandwidth rates

Note

The preceding factors are not necessarily the standard or minimum requirements for every enterprise network. Even so, these factors are the generic and common concerns that most IT enterprises have. These concerns can vary from business to business. For instance, many businesses have no concern about having unsecured IP communications over their private WAN.

Furthermore, today's Internet service providers (ISPs) can offer a dramatically enhanced Internet bandwidth and price performance with significantly improved service reliability. From a business perspective, this can be seen as high-bandwidth capacity at a cheaper cost; however, the level of end-to-end service efficiency and the level of reliability of the Internet might not be suitable for many businesses. The main point is that nowadays the Internet with secure virtual private network (VPN) overlay is by many businesses as either their primary WAN transport or as a redundant WAN transport. The subsequent sections in this chapter cover it in more detail.

Consequently, there are multiple WAN topologies and transport models an enterprise can choose from, such as point to point, hub and spoke, and any to any. In addition, traffic over each of these topologies can be carried over Layer 2 or Layer 3 WAN transports either over a private WAN network such as a Multiprotocol Label Switching (MPLS) provider or overlaid over the Internet. Typically, each model has its strengths and weaknesses in some areas. Therefore, enterprise network designers must understand all the different aspects of each WAN transport and the supported topologies, to select the right solution that is the best fit for the enterprise business, application, and functional requirements.

Modern WAN Transports (Layer 2 Versus Layer 3)

The decision to select L2 or L3 for the enterprise WAN transport is completely a design decision, and ideally it has to be a business-driven decision. Therefore, it is hard to make a general recommendation that an L3 WAN is better than an L2 WAN for an enterprise. However, some factors can drive the decision in the right direction based on business, functional, and application requirements, along with the enterprise WAN layout and design constraints. This section highlights the advantages and disadvantages of both modern WAN transports (L2 and L3 based), and then compares them from different design angles. Once the differences between these two WAN transport options are identified, the job of selecting the right WAN technology or transport will be easier. It will simply be a matter of identifying the requirements and mapping them to the most suitable WAN transport model.

Layer 3 MPLS VPN as a WAN transport enables true any-to-any connectivity between any number of sites without the need for a full mesh of circuits or routing adjacencies. This can offer significant scalability improvements for enterprises with a large number of remote locations. With L3VPN, the provider typically exchanges routing information with enterprise WAN edge routers and forwards packets based on Layer 3 information (IP). In addition, L3VPN service providers (SPs) offer a complete control plane and forwarding plane separation per enterprise (customer), with each enterprise IP service assigned to its own virtual IP network (VPN) within the SP MPLS network. With regard to Layer 2 VPN services, however, the providers typically have no participation in the enterprise Layer 3 WAN control plane. This is because the provider forwards the traffic of any given enterprise based on its Layer 2 information, such as Ethernet MAC addresses.

Note

For the purpose of the CCDE exam, the scenario and the requirements always determine the right choice. There might be situations where both WAN options seem to be equally valid, but typically one of them should be more suitable than the other because of a constraint or a requirement given to you on the exam. However, in some cases, there might be more than one right answer or optimal and suboptimal answers. Therefore, you need to have the right justification to support your design decision (using the logic of *why*).

Layer 2 MPLS-Based WAN

Metro Ethernet (ME) services offered by ME SPs (Carrier Ethernet) are one of the most common Layer 2 WANs used by today's large enterprises. Layer 2 WAN (ME-based) offers two primary connectivity models for enterprises:

- E-Line, also known as *Ethernet Virtual Private Line* (EVPL), which provides a point-to-point service. With EVPL, the typical physical link is Ethernet (Fast Ethernet or Gigabit Ethernet), and the multiple circuits under one physical link are wired virtually over the ME provider cloud using VLANs as a service identifier.
- E-LAN provides multipoint or any-to-any connectivity, also known as *Virtual Private LAN Services* (VPLS), and offers any-to-any connectivity with high flexibility for the enterprise WAN.

From an enterprise perspective, the Layer 2 (ME) services provided by today's SPs over their MPLS core infrastructure appear either like a LAN switch for multipoint L2VPN services or like a simple passthrough link for the point-to-point L2VPN sites, as depicted in [Figure 4-1](#).

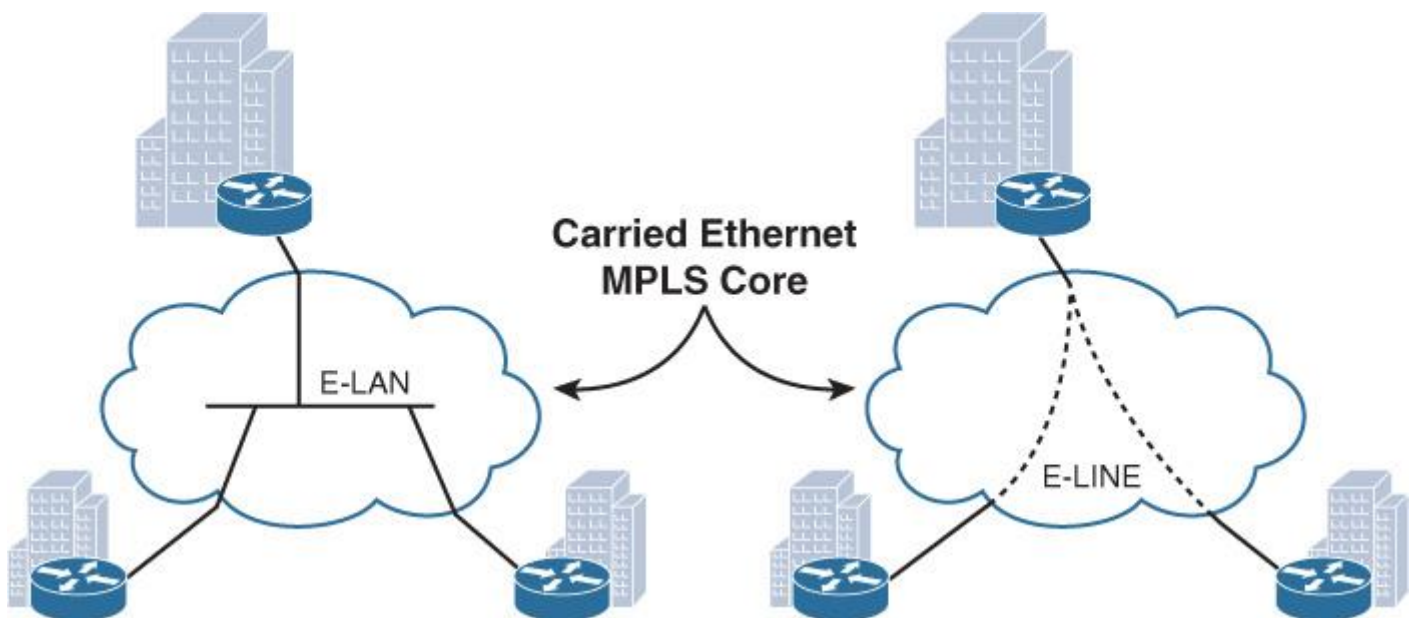


Figure 4-1 Layer 2 WAN MPLS Based

Note

Although E-Tree is another type of ME connectivity model, it is a variation from E-LAN to provide hub-and-spoke connectivity model.

Note

This section discusses these technologies from the enterprise point of view, as an L2 or L3 WAN solution. [Chapter 6](#), "[Service Provider MPLS VPN Services Design](#)," discusses it from an SP point of view.

Note

L2VPN SPs can preserve the access media using the legacy access media type (such as ATM and Frame Relay) if this is required by the business or if there is a lack of ME coverage by the SP in certain remote areas. In addition, this type of connectivity (mixed) is a common scenario during the migration phase from legacy to modern L2 WAN services.

Layer 2 MPLS-Based WAN Advantages and Limitations

[Table 4-1](#) highlights the primary advantages and limitations of a Layer 2 MPLS-based WAN.

Advantages

Bandwidth scalability: At the time of this writing, L2 WAN services can scale from 1 Mbps to 10 Gbp, which makes this transport highly scalable in terms of bandwidth requirements.

Performance and quality of service (QoS): By increasing bandwidth on a WAN circuit with L2WAN technologies, enterprises can have a low-latency and low-jitter WAN transport, which can be a good fit for converged networks (video, voice, and data). Furthermore, it can support end-to-end QoS classes as part of the service level agreement (SLA) offered by the SP (class of service [CoS] or differentiated services code point [DSCP] based).

Routing control: With an L2 WAN, the enterprise will have full control over the design, implementation, and operations of routing across the WAN. This can also be considered added flexibility. For instance, with this model, enterprises have the freedom to choose the desired WAN routing protocol and can deploy enterprise controlled WAN network virtualization.

Limitations

There might be limited service access coverage of the L2VPN (ME) in some locations.

For large-scale networks with many remote sites connected over a common E-LAN service, there will be design limitations and issues on higher levels (control plane), such as a large number of routing adjacencies.

Routing protocol limitations on certain topologies can introduce limitations here, as well, because not every enterprise can afford to change its WAN routing protocol (for example, a full-mesh VPLS-based WAN with Open Shortest Path First [OSPF], where special care has to be taken for OSPF).

Service offering availability: SPs globally are moving toward adapting networks to provide ME services; therefore, it is available in a large number of places.

Enterprise staff knowledge and expertise can be considered a limiting factor as well. For instance, the staff might not be able to design, deploy, or operate large-scale WAN routing (for example, using Enhanced Interior Gateway Routing Protocol [EIGRP] or Border Gateway Protocol [BGP]).

Topology flexibility: With an L2 WAN, enterprises can have different topology layouts as required, such as point to point, point to multipoint, and multipoint to multipoint.

Services flexibility: With an L2 WAN, the enterprise can run any advanced IP/non-IP services without any reliance on the SP's support, such as IPv6 or multicast.

Cost-effective: An L2 WAN, such as ME, replicates the cost model of Ethernet to the WAN.

Table 4-1 *Layer 2 MPLS-Based WAN Advantages and Limitations*

Note

There are other variations of ME services, such as Ethernet private line (EPL), that support Ethernet over xWDM (dense wavelength-division multiplexing [DWDM], coarse wavelength-division multiplexing [CWDM]), SONET, or dedicated Ethernet interconnects over fiber. However, this type of service is more expensive than the other ones that are offered by MPLS SPs, such as VPLS or EVPL (Ethernet virtual circuit [EVC]) as an L2VPN ME service.

Layer 3 MPLS-Based WAN

MPLS L3VPN enables enterprise customers to route traffic across the SP cloud as a transit L3 WAN network, with a simplified “one-hop” single routing session per link between the enterprise WAN edge router and provider edge routers. This means that the SP will typically offload all the enterprise WAN control plane complexities in terms of design and operations of the core WAN routing. As a result, enterprises will gain a significant savings in terms of operational expenses and speed up the time to add new remote sites, especially if there hundreds or thousands of remote sites that need to be interconnected. In addition, this model will help to simplify and optimize end-to-end WAN QoS design and bandwidth volume planning of the enterprise WAN connectivity. [Figure 4-2](#) illustrates a Layer 3 MPLS-based WAN. (The MPLS L3VPN cloud appears as single router [peer] from the remote site's point of view.)

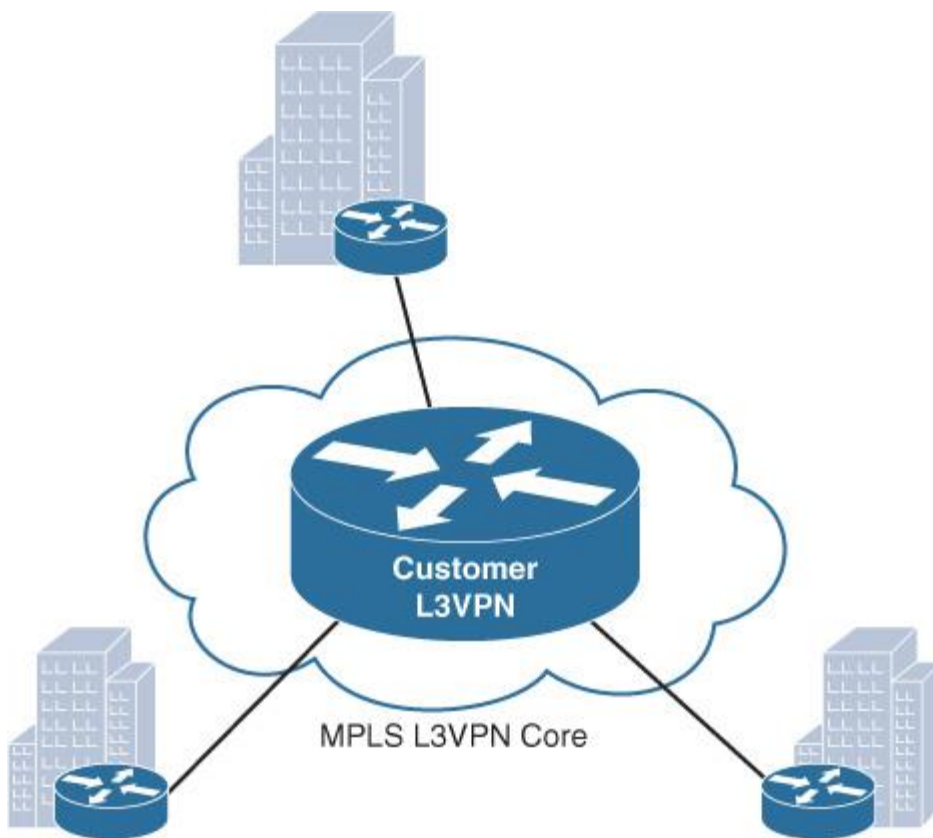


Figure 4-2 Layer 3 MPLS-Based WAN

Layer 3 MPLS-Based WAN Advantages and Limitations

Table 4-2 highlights the primary advantages and limitations of a Layer 3 MPLS-based WAN.

Advantages	Limitations
Bandwidth scalability: An MPLS L3VPN WAN can provide flexible bandwidth capacity and smooth upgrades as compared to other WAN transports.	Cost of bandwidth: Although L3WAN (L3VPN) offers a scalable and high speed of bandwidth, L2VPN (such as ME services) offers a higher bandwidth scale (for example, 10G wire rates) with a lower cost.
Performance and QoS: MPLS L3VPN can support consistent end-to-end QoS (DSCP driven) that enterprises can take advantage of, especially for real-time delay-sensitive traffic such as Voice over IP (VoIP).	IP addressing: When enterprises move from legacy L2 WAN technologies such as Frame Relay, re-addressing of the WAN interfaces is required.
Cost-effective: L3VPN can reduce the operational cost because there will be no need to operate and maintain a large and complex routed WAN core infrastructure.	SP dependencies: With L3VPN, there is always dependency on SP support to deploy new network services, such as using IPv6 or multicast.
Service offering availability: Most of the SPs globally offer MPLS L3VPN. Therefore, it is available in a large number of places.	Flexible topology: For nonstandard layouts, such as hub and spoke, there might be an additional cost involved for the additional VPNs to be provisioned by the SP.

Topology flexibility: L3 WAN supports different topology layouts as required, such as point to point, point to multipoint, and any to any.

LAN extension: L3 WAN is typically a Layer 3 transit network, which by nature does not allow Layer 2 extensions. Therefore, if a LAN extension is required, it must be achieved using additional technologies. This will typically be an overlay technology, such as self-deployed L2VPN over MPLS over generic routing encapsulation (GRE) over L3VPN WAN, which can add an additional layer of complexity to the network with regard to operations and troubleshooting. It also might not be a desirable solution for the requirements of some applications, due to the increased packets' header that may lead to fragmentation or serialization delay along the WAN path.

Access flexibility: With L3 WAN, the enterprise can be provisioned with any type of access media depending on the access availability of that location (for example, Ethernet, WiMAX, and VPN, over the Internet or 3G/4G).

Routing simplicity: With L3VPN, enterprises will typically offload the core routing to the SP. From the enterprise point of view, only one routing peer/session per link needs to be maintained.

Table 4-2 *Layer 3 MPLS-Based WAN Advantages and Limitations*

Internet as WAN Transport

Despite the fact that the nature of the Internet is a “best effort transport” that lacks end-to-end QoS support, the modern Internet can offer relatively high reliability and high-speed connectivity between various locations at a low cost. In addition, in today’s modern businesses, many enterprises are increasingly hosting their services in the cloud and embracing many cloud-based services and applications offered as *software as a service* (SaaS) such as Cisco WebEx and Microsoft Office365, which is changing the traffic pattern to be more toward the Internet.

Furthermore, the Internet can be a reasonable and cost-effective choice for remote sites as a primary transport when it is not feasible to connect over other WAN transport options or when there is a lack of WAN access coverage in certain remote areas. This design primarily relies on using VPN tunneling (overlay) techniques to connect remote sites to the hub site (enterprise WAN module located at the enterprise campus or data center) over the Internet. Ideally, this design is based on a hub-and-spoke connectivity using dynamic multipoint VPN (DMVPN) as the VPN overlay technology for the “Internet as WAN transport” design model, which offers the flexibility to provide any-to-any connectivity as well (direct spoke to spoke). However, point-to-point tunneling mechanisms such as the classical IPsec and GRE are still viable overlay

options to be considered (considering typical scalability limitations with peer-to-peer (P2P) tunnels [see [Table 4-4](#) for a detailed comparison of the different VPN mechanisms]). [Figure 4-3](#) depicts the different typical connectivity options available with the Internet as a WAN transport.

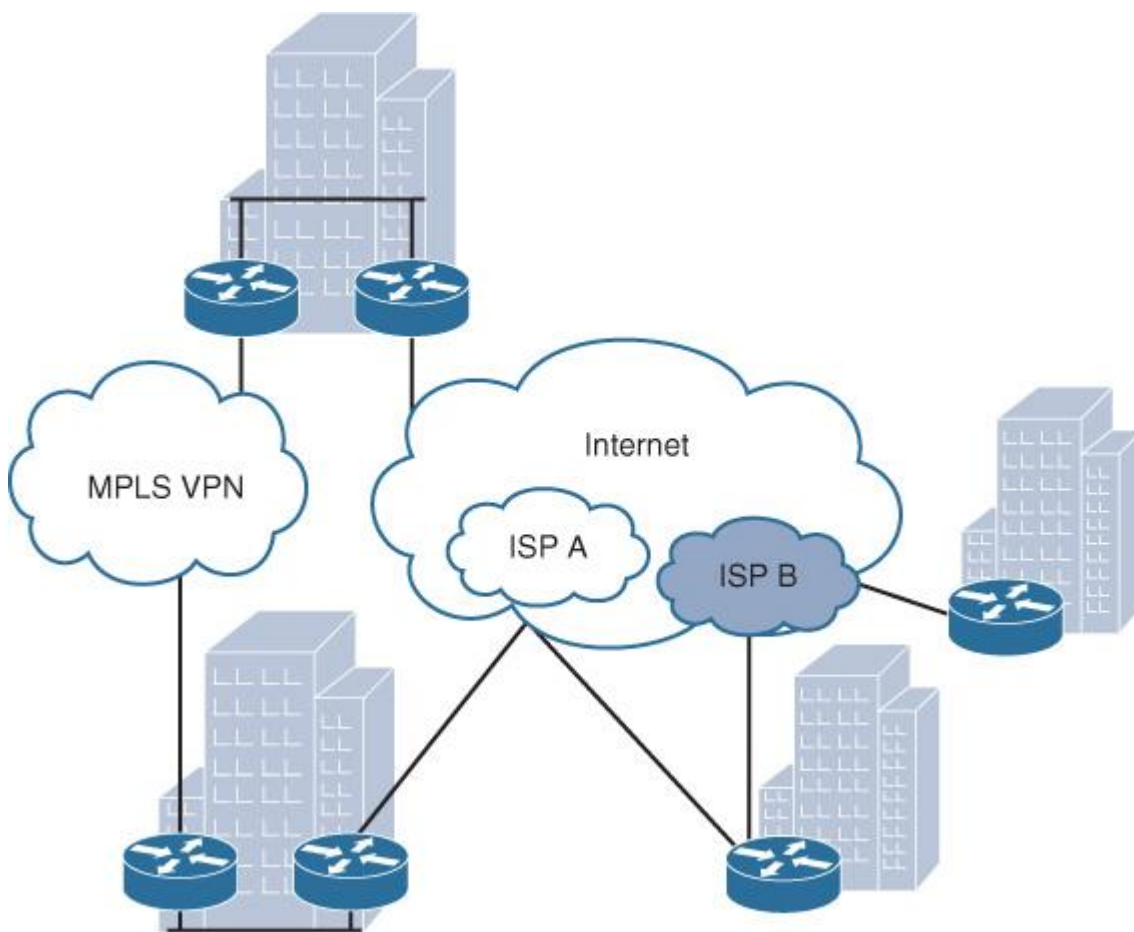


Figure 4-3 Internet as a WAN Transport

Note

The decision of when to use the Internet as a WAN transport and how to use it in terms of level of redundancy and whether to use it as a primary versus backup path depends on the different design requirements, design constraints, and business priorities (see [Table 4-4](#)).

Furthermore, Gartner Inc. report “Hybrid Will Be the New Normal for Next Generation Enterprise WAN” analyzes and demonstrates the importance of the integration of the Internet and MPLS WAN to deliver a cohesive hybrid WAN model to meet today’s modern businesses and applications’ trends and requirements such as Cloud-based services. “Network planners must establish a unified WAN with strong integration between these two networks to avoid application performance problems.”¹

1. Document ID:G00266397, www.gartner.com

Note

Cisco’s implementation of this (hybrid WAN) concept is called *intelligent WAN (IWAN)*, and combined with Cisco’s intelligent routing is also known as *performance routing (PfR)*, which offers more intelligent path control to protect critical applications and load balance traffic over multiple paths (for example MPLS + Internet) based on predefined criteria to optimize application performance and user experience.

Note

The connectivity to the Internet can be either directly via the enterprise WAN module or through the Internet model (edge). This decision is usually driven by the enterprise security policy, to determine where the actual tunnel termination must happen. For instance, there might be a dedicated demilitarized zone (DMZ) for VPN tunnel

termination at the enterprise Internet edge that has a backdoor link to the WAN distribution block to route the decapsulated DMVPN traffic.

Internet as WAN Transport Advantages and Limitations

Table 4-3 highlights the primary advantages and limitations of the Internet as a WAN transport model.

Advantages	Limitations
Offers low-cost WAN connectivity with relatively high bandwidth.	Reliability: Although today’s ISP can offer a relatively high level of Internet service reliability, it still cannot satisfy the strict level of service reliability required by some businesses.
Offload traffic from traversing any private WAN path that is destined to the Internet, such as accessing services hosted in a public cloud provider via the direct Internet access (split tunneling).	Consistent QoS: Because the Internet by nature is best effort IP transport, enterprises cannot maintain true end-to-end service differentiation and consistent QoS.
Flexible access because the Internet connectivity can be provisioned over various media types such as wireless, Long-Term Evolution (LTE), 3G, digital subscriber line (DSL), or Ethernet.	Operations complexity: Typically, using the Internet as a WAN transport requires an overlay mechanism such as GRE and DMVPN combined with protection (IPsec). These multiple technologies may add a layer of complexity when there an issue to troubleshoot or a new site to be added, especially if other advanced IP services are enabled such as IP multicast.
Provisioning an Internet service is usually relatively quicker than provisioning other WAN services. This enables large enterprises that have many small remote sites they want to add quickly with flexible media access types and wide geographic coverage to accelerate their time to market; this is a common scenario in retail businesses.	

Table 4-3 Internet as a WAN Transport Model Advantages and Limitations

WAN Transport Models Comparison

Table 4-4 summarizes the selection criteria of different WAN transports discussed earlier in this chapter (from an enterprise point of view), considering various design aspects.

	MPLS L2VPN WAN	MPLS L3VPN WAN	Internet as WAN
Topology	Flexible	Limited and very dependent on the SP	Limited (overlay technology dependent P2P or hub and spoke (DMVPN supports spoke to spoke))
Bandwidth	Very flexible (can vary between 1 Mbps to 10 Gbps)	Flexible (less than L2 MPLS-based WAN [ME])	Flexible with limitations, depending on the site location and connectivity provisioning type (DSL versus 3G versus 4G)
WAN core routing control	Enterprise managed and controlled	SP managed and controlled	Enterprise managed and controlled
Cost	Moderate	Usually more expensive than L2, especially when high bandwidth is required	Cheap
CoS	Depending on the SP, but can support CoS based on L2 marking and DSCP	End-to-end Layer 3 CoS (DSCP based)	End-to-end QoS guarantee not supported (only at the network edge)
Staff experience	Requires experienced staff to design and manage the core WAN routing	High level of routing expertise not required for the WAN	Requires experienced staff to design and manage the WAN routing and the overlay VPN setup
Remote site scalability	Introduce some routing/adjacency issues and limitations with a large number of sites	Can support very large scale of remote sites	Scalable to some extent (limited to the WAN router hardware capability [for example, supported number of VPN sessions])
Site physical connectivity	Limited options (for example, legacy Frame Relay, ME)	Very flexible, can be any type of access (legacy, Ethernet, VPN over Internet to SP MPLS)	Flexible (DSL, Ethernet, 3G/4G)

Table 4-4 Comparison of WAN Transport Models

By understanding the differences between each WAN transport and the capabilities and limitations of each transport, network designers should be able to make a more business-driven design decision by mapping the suitable WAN transport to the different design requirements (for example, business, functional, and application). In addition, network designers ideally must consider the answers to the following questions during the planning phase of the WAN transport selection:

- Who is responsible for the core WAN routing management?
- Who manages the customer edge (CE) WAN devices?
- How critical is the WAN connectivity to the business? What is the impact of an outage on the WAN connectivity to the business in terms of cost and functions?
- What is the number of remote sites and what is the percentage of the projected growth, if any?
- Are there any budget constraints?
- What are the required WAN capabilities to transport business applications over the WAN with the desired experience (such as QoS, IP multicast, or IPv6)?

WAN Module Design Options and Considerations

This section highlights the design considerations and the different WAN connectivity design options that pertain to the enterprise WAN module and remote site WAN connectivity as well [31].

Design Hierarchy of the Enterprise WAN Module

The main goal of the WAN module is to aggregate the connectivity and traffic of the enterprise WAN that connects the enterprise with various types of remote locations. Therefore, this module provides the traffic and connectivity aggregation of the extended remote sites. Applying the hierarchical design principle can maximize the flexibility and scalability of this enterprise module. In addition, this structured approach will simplify adding, removing, and integrating different network nodes and services such as WAN routers, firewalls, and WAN acceleration appliances. As illustrated in [Figure 4-4](#), applying hierarchy will enable each layer to perform certain functions using different features and tools in a more structured manner. Furthermore, the level of flexibility and adaptability offered by the hierarchal structure makes the WAN module design highly scalable and resilient.

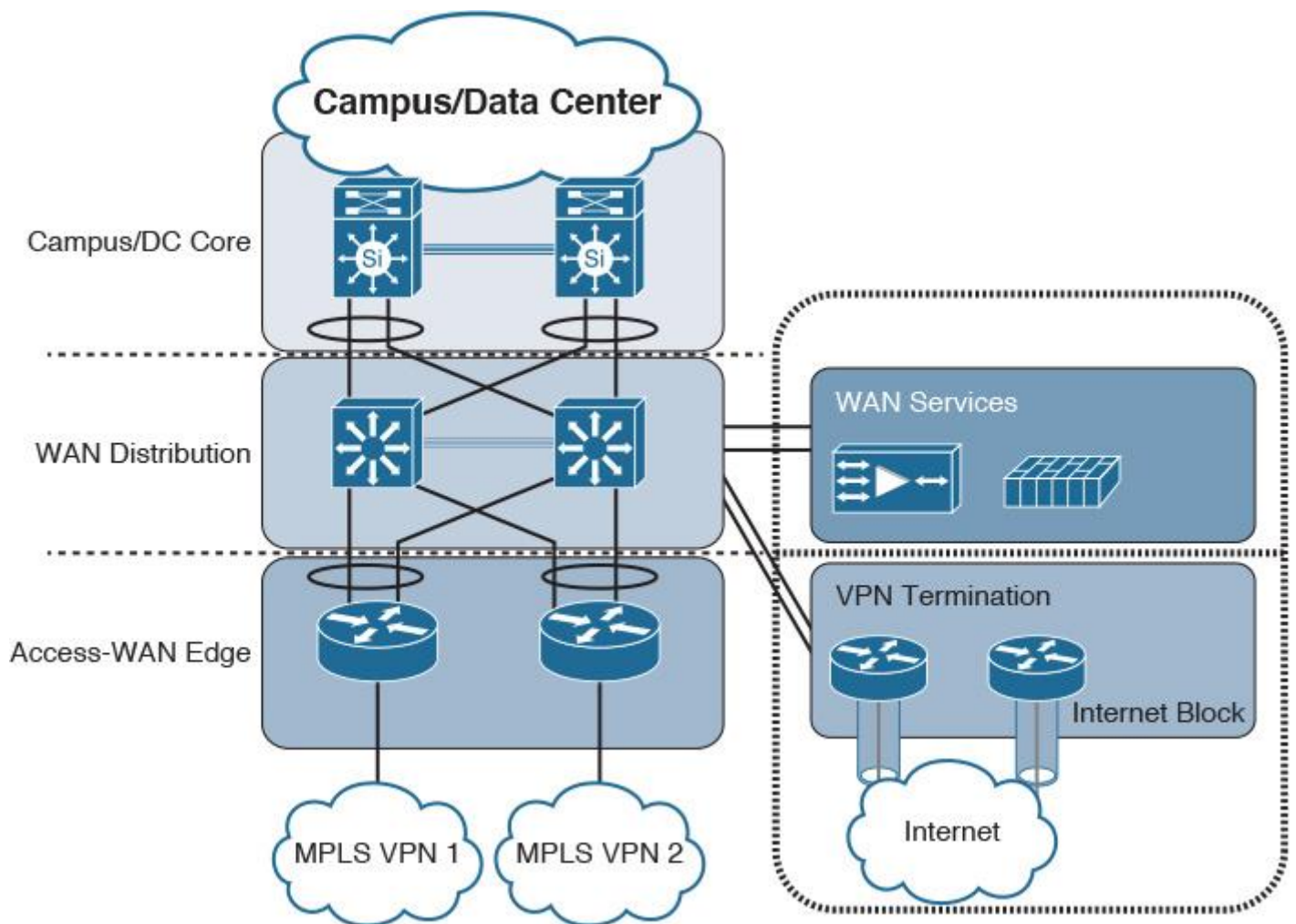


Figure 4-4 WAN Module Hierarchical Design

WAN Module Access to Aggregation Layer Design Options

The aggregation layer of the WAN module aggregates traffic and connectivity of access layer nodes, which is typically the WAN edge routers and other WAN services, such as firewalls and WAN acceleration appliances.

There are three common design options to interconnect WAN edge routers to the aggregation layer switches of the WAN module, illustrated in [Figure 4-5](#).

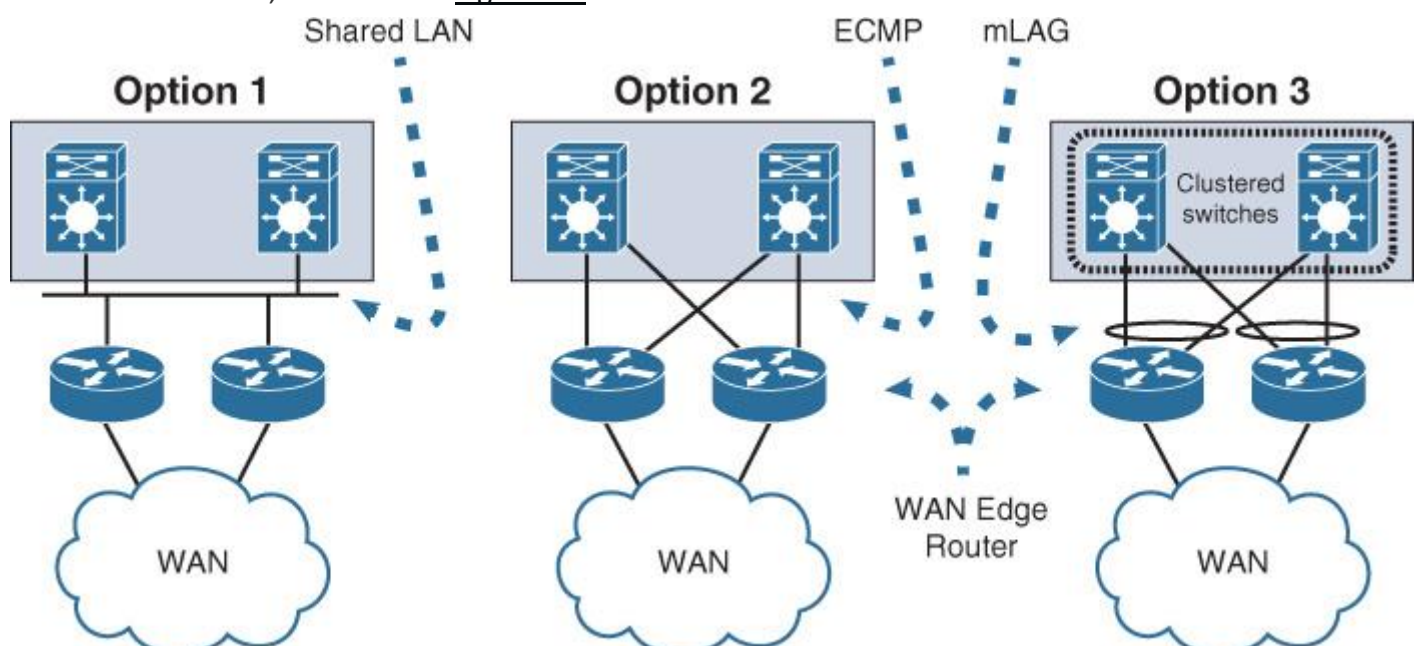


Figure 4-5 WAN Access-Distribution Connectivity Options

Both design options, 2 (equal-cost multipath [ECMP]) and 3 (multichassis link aggregation [mLAG]), offer a more flexible design compared to Option 1; however, there are some technical differences between these two design options. [Table 4-5](#) summarizes these differences. Although both design options (2 and 3) can achieve the same goal to a large extent, network designers should select the most suitable option based on the environment and required capabilities and features (optimal versus suboptimal).

Option 2	Option 3
ECMP based.	mLAG based.
Link redundancy based on redundant L3 links (more routing peers).	Link redundancy based on redundant L2 mLAG links (fewer routing peers).
Route reconvergence is required when one uplink fails.	No routing reconvergence is required if one mLAG member link fails.
Convergence time by default relies on routing protocol design and timers.	With mLAG, each flow typically utilizes one member link and will be limited to the capacity of that link (unless “flowlet” concept is used ²).
Support both scale-out and scale-up. You can use more than two aggregation layer nodes.	Support scale-up (limited scale-out as a maximum of 2 aggregation layer nodes per mLAG can be used).
The more links to be added the larger routing database.	The more member of links to be added, the larger the number of Address Resolution Protocol (ARP) entries.
ECMP flow-based load balancing.	Supports L3/L4 load-balancing hashing (load distribution) across the mLAG member links.

Table 4-5 ECMP Versus mLAG²

2. “Dynamic Load Balancing Without Packet Reordering,” IETF Draft, chen-nvo3-load-balancing, <http://www.ietf.org>

Option 1, however, has several design limitations. For instance, without a careful interior gateway protocol (IGP) tuning, this design option can lead to a slow network convergence at the WAN edge, which can result (from a business point of view) in undesirable outcomes after a failure event. In addition, this design option has a potential of instability and scalability when the network grows in terms of nodes connectivity and routing adjacencies (over a single shared LAN segment). In other words, this design option is the least resilient and scalable option among the other design options. Despite that, design option 1 can still meet some design requirements that do not need a tight convergence time or any scalability considerations, such as regional HQ WAN model with only a pair of WAN edge nodes and no future plan to increase the number of nodes or links. Therefore, the requirements always govern which design option is the best, factoring in whether any design constraint may influence the design choice.

WAN Edge Connectivity Design Options

Enterprises can consider several WAN edge connectivity design options, such as single-homed or dual-homed, and using single or dual edge routers. The most important consideration here is to identify the

business drivers and requirements that influence the selection of one option over others. Many variables will influence a business-driven design decision to select a specific WAN edge connectivity design option. The most common factors that drive this decision are as follows:

- Site type (for example, small branch versus data center versus regional office)
- Level of criticality (How much can the downtime cost? How critical is this site to the business if it goes offline for xamount of time?)
- Traffic load (For example, the load on the HQ data center is more than that of the regional data center.)
- Cost (Is cost-saving a top priority?)

Table 4-6 summarizes the various types of WAN edge connectivity design options depicted in Figure 4-6, along with the different considerations from network design perspective.

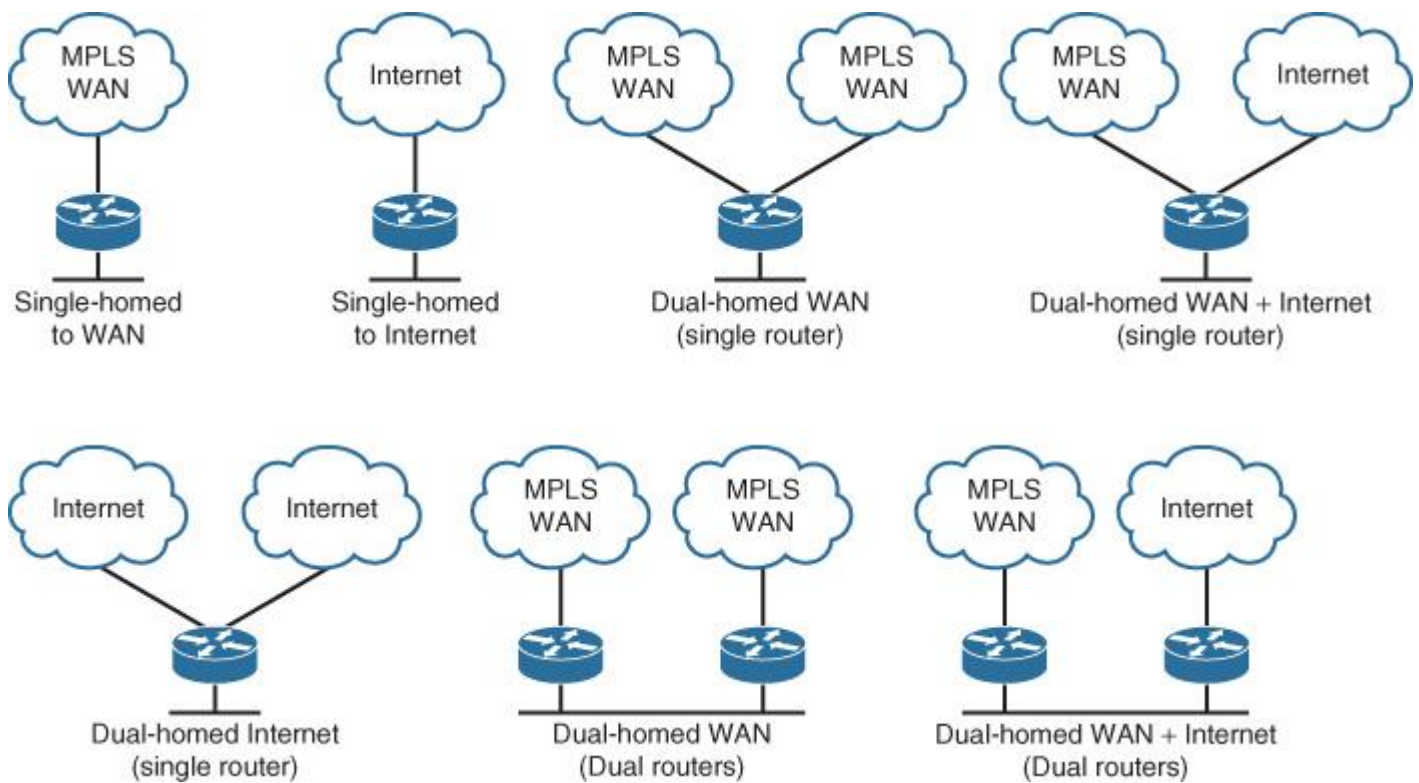


Figure 4-6 WAN Edge Connectivity Options

Connectivity Model	Redundancy	Reliability	Cost	Supported QoS model	Suitability
Single-homed to WAN	None	Moderate	Moderate	Consistent end to end	Small to medium-size branch sites with high traffic volume
Single-homed to Internet	None	Low	Low	Internet edge only	Small branch sites with low fault-tolerance requirements
Dual-homed WAN (single router)	Link redundancy only	Moderate	High	Consistent End to end	Medium-size to large, critical, or regional remote sites
Dual-homed WAN + Internet (single router)	Link redundancy only	Moderate (lower than MPLS)	Moderate	Consistent End to end over the MPLS path	Medium-size to large or regional remote sites
Dual-homed Internet (single router)	Link redundancy only	Moderate (lower than MPLS + Internet)	Moderate	Internet edge only	Small to medium-size remote site
Dual-homed WAN (dual routers)	Link and device redundancy	Very high (single provider versus dual providers)	Very high	Consistent End to end	Hub, HQ, DC, or large regional site
Dual-homed WAN + Internet (dual routers)	Link and device redundancy	High	Moderate to high	Consistent End to end over the MPLS path	Hub, large remote or regional sites

Table 4-6 WAN Edge Connectivity Design Options

Single WAN Provider Versus Dual Providers

The previous section discussed the various WAN edge design options and the characteristics of each option from a design point of view. This section focuses on the dual WAN edge connectivity, and takes it a step further to compare the impact of connecting a multihomed site to a single SP versus two different SPs, as summarized in [Table 4-7](#).

	Single Service Provider	Dual (Different) Service Providers
Design simplicity (consistency, features)	Simple, consistent (for example, SLA, QoS design)	Can be inconsistent and more complex (for example, different SLAs, different QoS models, different routing protocols)
Cost	Fixed	May lead to a better competitive pricing
Availability	SP outage can lead to a WAN blackout	Offers a higher degree of WAN reliability and availability
Operational complexity	Simpler (consistent)	More complex (for example, dealing with different SLAs and maybe routing protocols)

Table 4-7 Single Provider Versus Dual Provider

Note

Large enterprises with large geographic distribution can mix between the connectivity options (single versus dual WAN) by using single and dual providers, based on the criticality of the site and business needs. For instance, regional hub sites and data centers can be dual homed to two providers. In addition, this mixed connectivity design approach, where some remote sites are single-homed to a single provider while others are multihomed to dual providers (typically larger sites such as data centers or regional HQs), can offer a transit path during a link failure, as depicted in the scenario in [Figure 4-7](#). Ideally the transit site should be located within the same geographic area or country (in the case of global organizations) to mitigate any latency or cost-related issues, when applicable, by reducing the number

of international paths traffic has to traverse. In addition, the second provider in [Figure 4-7](#) can be an Internet-based transport such as DMVPN over Internet.

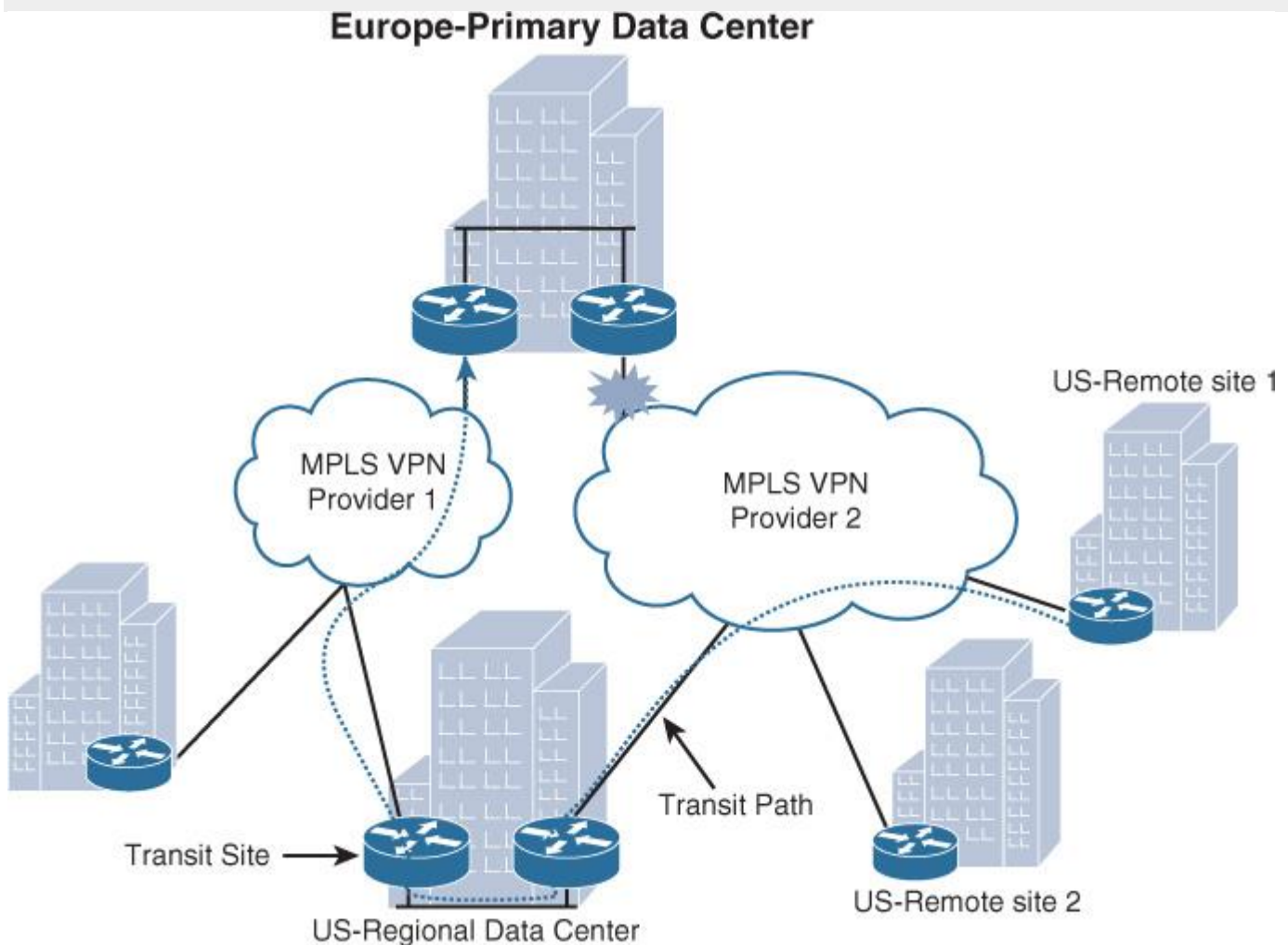


Figure 4-7 *Transit Path Scenario with Dual WAN Providers*

Remote Site (Branch) WAN Design Considerations

The WAN edge design options of a remote site can be based on any of the design options described in the previous section (see [Table 4-6](#)), where single or dual WAN edge routers can be used based on the requirements of each particular site. Most commonly, in large enterprises, remote sites are categorized based on different criteria such as size, criticality, location, and typically all the sites under the same categorization follow same design standards.

Note

The edge node is usually either a CE node (for MPLS L3 or Layer 2 WAN) or a VPN spoke node. In some cases, a single WAN edge router can perform the role of both a CE router and VPN spoke router.

However, the level of availability is something that can be determined based on different variables, as discussed in the previous section, such as the level of criticality of the remote site. The rule of thumb for remote site availability is that “the network ideally should tolerate single failure conditions,” either the failure of any single WAN link or any single network device at the hub/HQ WAN side (by considering control plane or overlay failover techniques). However, as discussed earlier in his book, the different business drivers, constraints, and the level of site criticality can drive the level of availability of any given remote site. In other words, remote site availability is not always a requirement or a component that must be considered in the design. In general, remote sites with single-router, dual links must be able to tolerate the loss of either of the WAN links. Remote sites with dual router, dual links can tolerate the loss of either a WAN edge router or a WAN link (multiple-failure scenarios).

In addition, from a design perspective, the selected WAN connectivity option has a significant influence on the LAN design of a remote site, as does the size of the site (in terms of the number of users and endpoints connected to the network). In general, the design models of a remote site fit into two primary models, as depicted in [Figure 4-8](#) and [Figure 4-9 \(a and b\)](#) and compared in [Table 4-8](#):

■ Single-tier design model

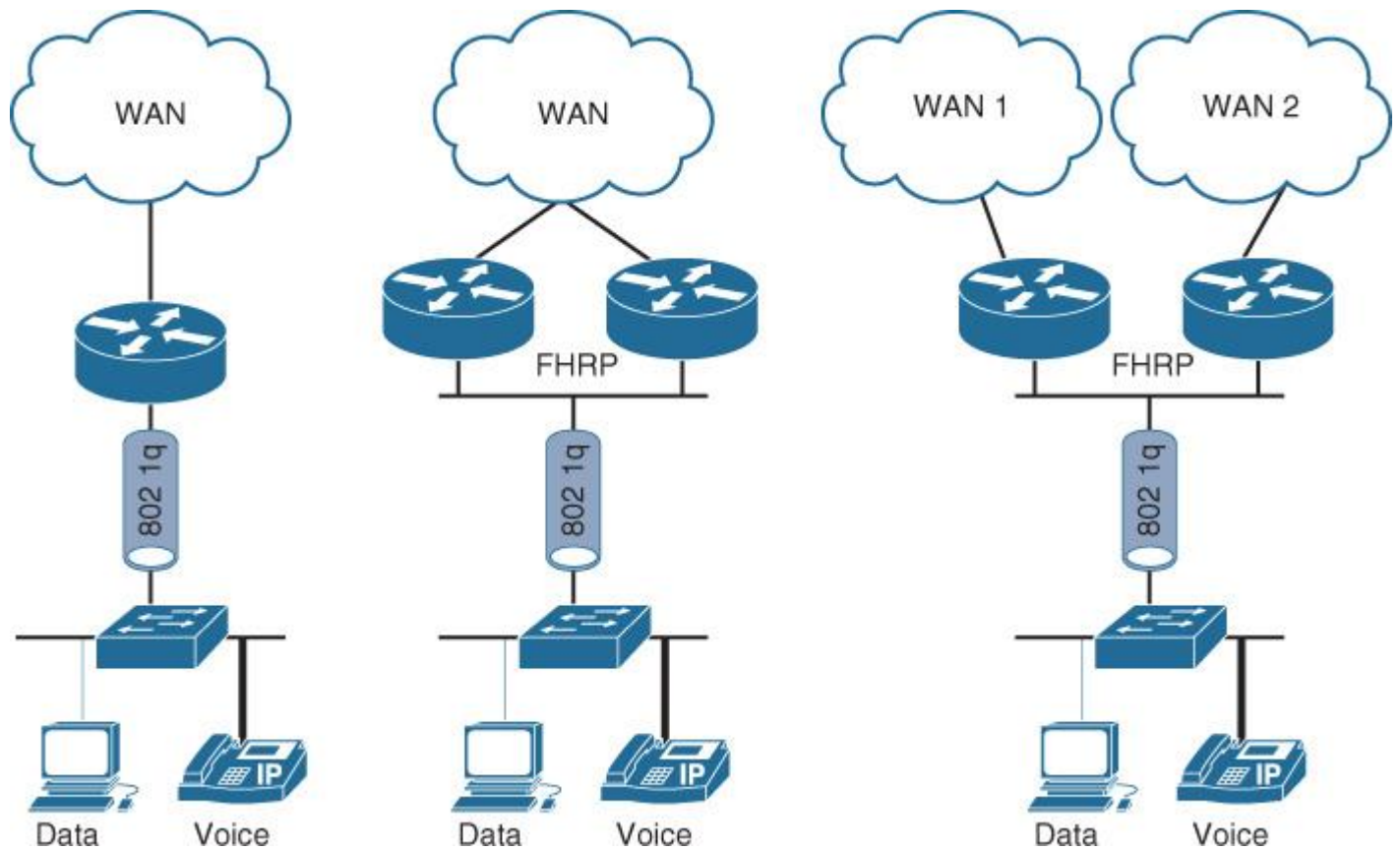


Figure 4-8 Single-Tier Remote Site Design Model

■ Multitier design model

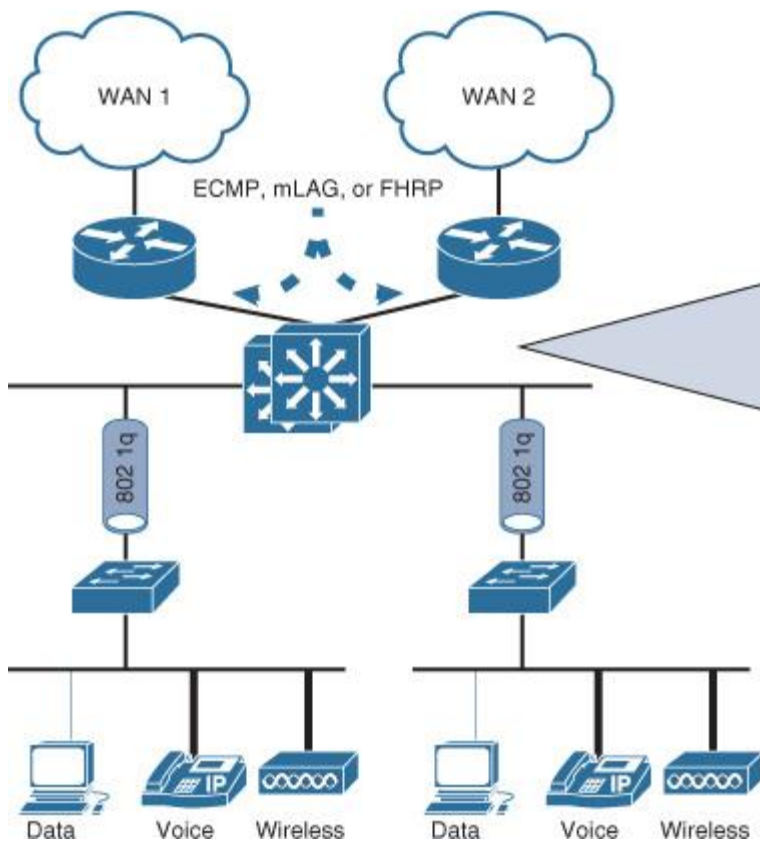


Figure 4-9a Multitier Remote Site: Dual WAN Router Design Model

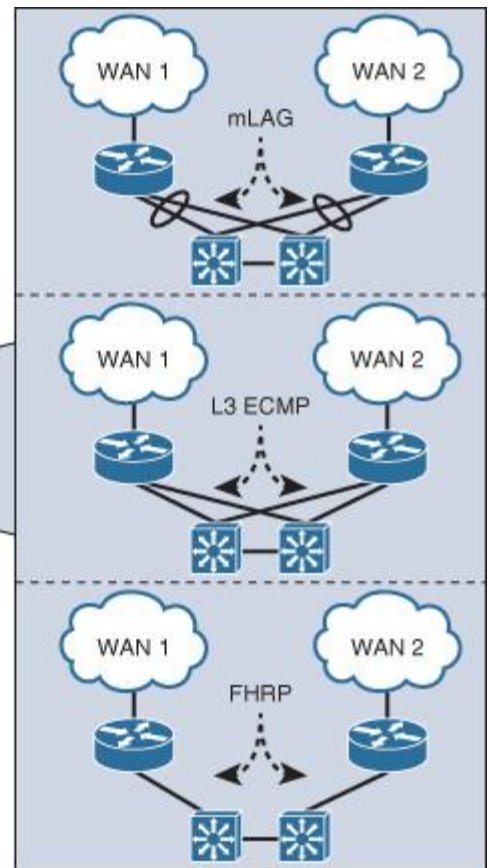


Figure 4-9b Multitier Remote Site: Single WAN Router Design Model

	Single-Tier Design Model	Multitier Design Model
LAN scalability	Very limited	More scalable
First-hop Layer 3 service	WAN edge router	Distribution layer switches
LAN flexibility	Very limited	More flexible
LAN to WAN edge Layer 3 connectivity options	FHRP (First Hop Redundancy Protocol), in dual edge routers scenario	IGP over ECMP FHRP mLAG with FHRP mLAG with IGP
Supported number of endpoints	Small	Small to medium-sized
Supported WAN connectivity options	Single WAN, single router Dual WAN, single router Dual WAN, dual routers	Single WAN, single router Dual WAN, single router Dual WAN, dual routers

Table 4-8 Remote Site Design Models

Note

The WAN connectivity options in [Table 4-8](#) apply for both private enterprise WAN or overlaid WAN over the Internet transport.

Internet as WAN Transport (DMVPN Based)

As discussed earlier in this section, using the Internet as a WAN transport in conjunction with DMVPN as the overlay transport offers several benefits to the business and enterprise WAN design, in particular for the remote site WAN connectivity, including the following:

- Offers a cost-effective and reliable (to a large extent) WAN connectivity over the Internet
- Reduces the time to add new remote sites over various media access types such as DMVPN over Internet over LTE, 3G, or DSL, combined with the support of zero-touch configuration of hub routers when introducing new spokes to the network
- Provides automatic full-mesh connectivity with simple configuration of hub and spoke
- Supports (any-to-any) spoke-to-spoke direct connectivity fashion
- Supports dynamically addressed spokes
- Supports provisioning behind devices performing Network Address Translation (NAT)
- Features such as automatic IPsec triggering for building an IPsec tunnel
- Supports multiple flexible hub-and-spoke design options that serve different design goals, scales, and requirements, as illustrated in [Figure 4-10](#)

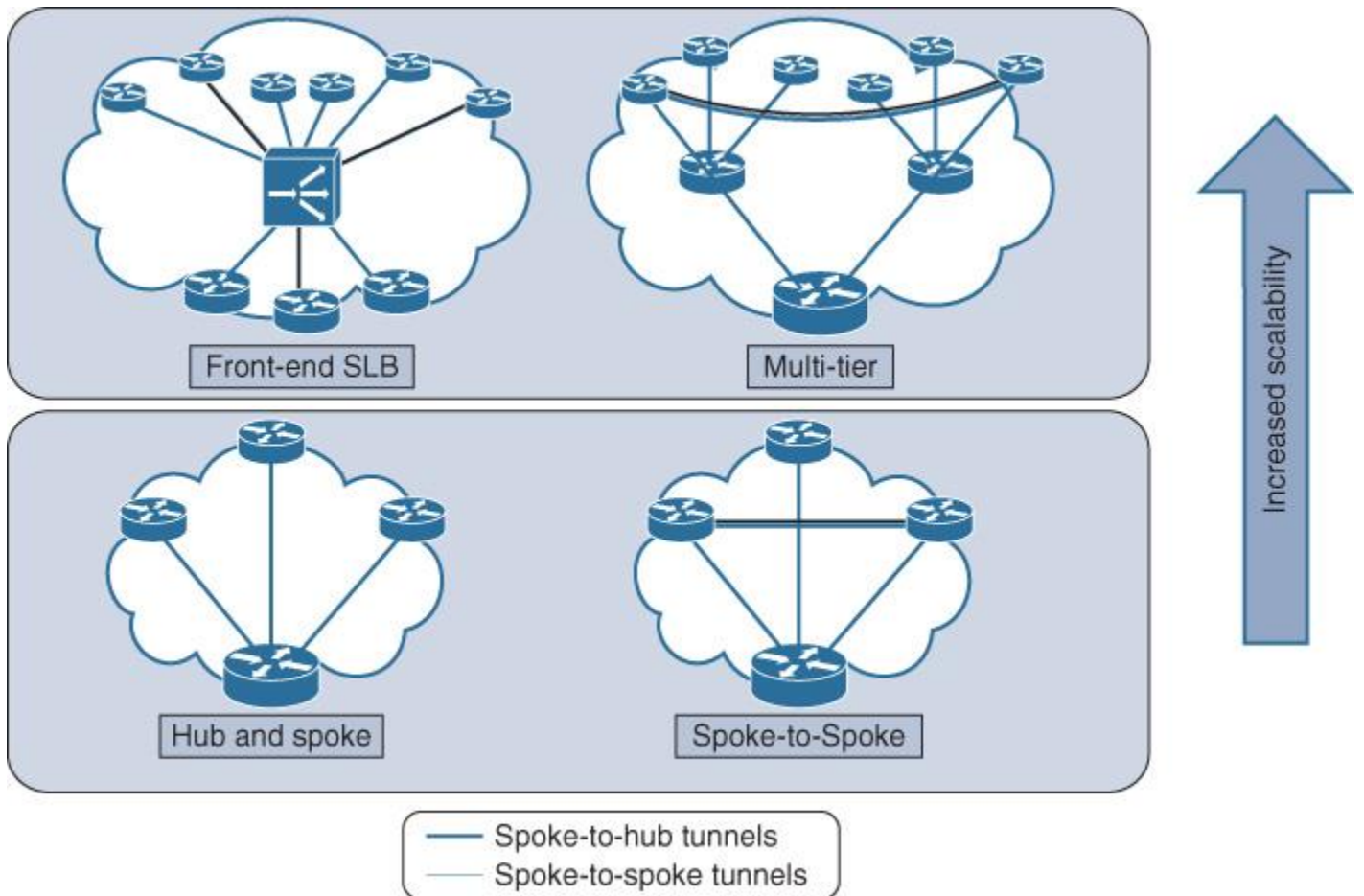


Figure 4-10 DMVPN Connectivity Models

Note

Routing over GRE tunnels with large routing tables may require adjustments (normally lowering) to the maximum transmission unit (MTU) value of the tunnel interface.

As illustrated in [Figure 4-11](#), Three primary connectivity options comprise this design model (remote sites DMVPN-based WAN).

- Single router, single link
- Single router, dual links
- Dual routers, dual links

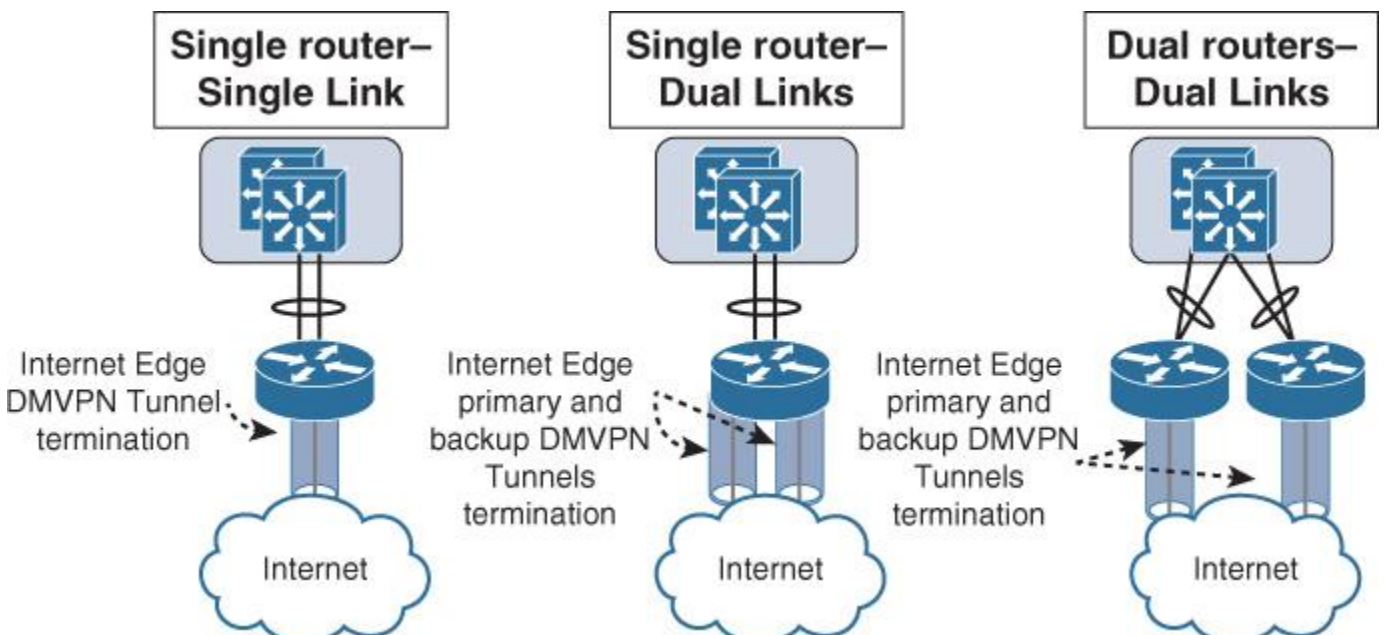


Figure 4-11 Remote Site DMVPN-Based WAN Connectivity Options

These connectivity options are primarily driven by the desired level of redundancy over the Internet. For instance, the *dual routers, dual links* connectivity option eliminates any single point of failure at the WAN/Internet edge, and with the *single router, dual links*, the edge router is posing single point of failure to the design even though there are two redundant Internet links. Therefore, based on the different design requirements and constraints, the best “relevant” option can be selected. For example, if a retailer has a small remote site with a very small number of users that can perform transactions and save them locally in case of any WAN outage, adding redundancy may be considered overengineering in this particular scenario because it will not add significant value to the business, and it may be seen as additional unnecessary cost from the business point of view.

Enterprise WAN Module Design Options

Previous sections in this chapter discussed the design elements and components of the enterprise WAN module. This section highlights the common proven design models you can use as a foundational reference for the enterprise WAN module design, based on the scale of the remote sites [32].

Note

As stated, you can use these design models as foundational reference architecture and scale them based on the requirements. For instance, the design Option 1 model can easily be migrated to design Option 2 when the number of remote sites increases and requires a higher level of redundancy. Similarly, the number of edge access nodes (WAN/Internet edge routers) can be scaled out depending on the design requirements. For instance, an enterprise may consider design Option 1 with an additional redundant edge route to a second MPLS WAN, while the Internet edge router is to be used only as a third level of redundancy with a tunneling mechanism.

Note

The number of remote sites in the following categorization is a rough estimation only (based on the current Cisco Validated Design [CVD] at the time of this writing). Typically, this number varies based on several variables discussed earlier in this book, such as hardware limitations and routing design in terms of number of routes.

Option 1: Small to Medium

This design option illustrated in [Figure 4-12](#) has the following characteristic:

- Dual redundant edge routers.
- Single WAN connectivity (primary path).
- Single Internet connectivity (backup path over VPN tunnel).
- Ideally, each of the WAN and Internet routers are dual-homed to the WAN module aggregation clustered switches using Layer 3 over mLAG (or Layer 3 ECMP in case of no switch clustering).
- This design model ideally can support a small to medium number of remote sites (ideally for a few hundred, as a max, taking into account hardware limitations as well).

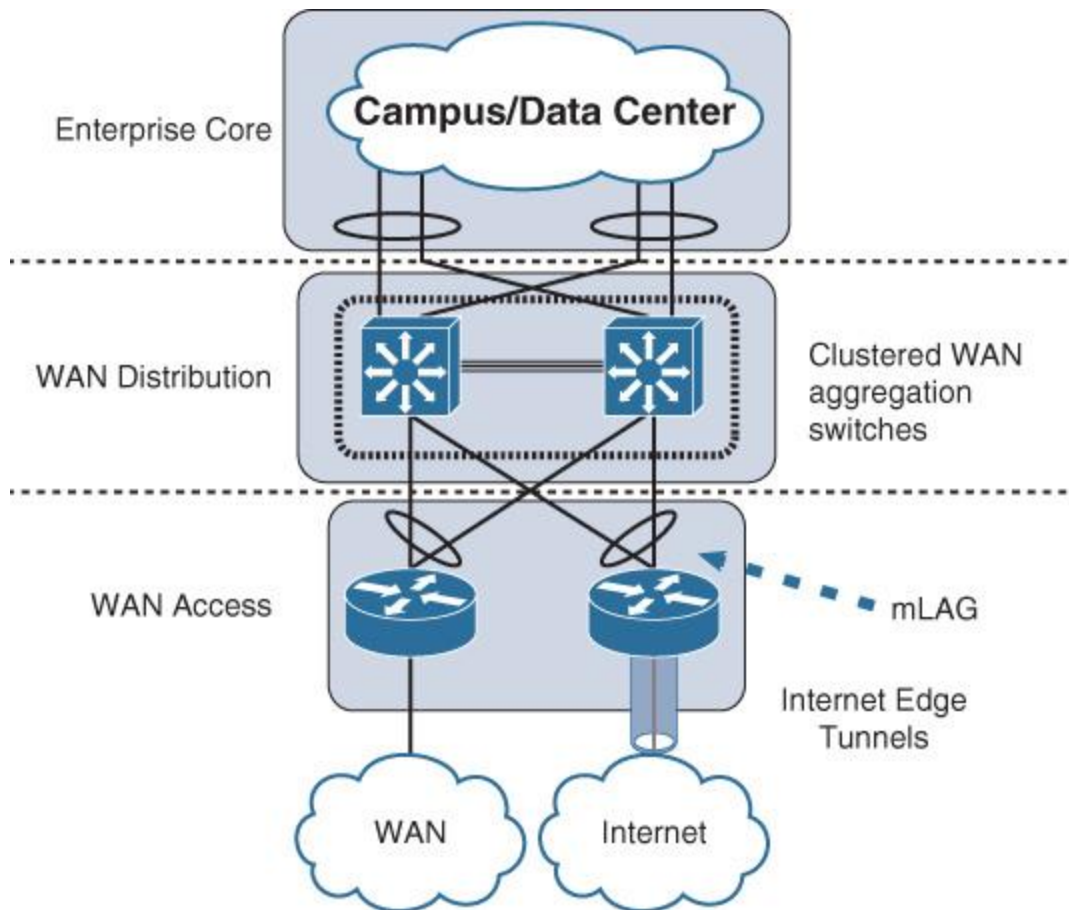


Figure 4-12 Enterprise WAN Module Design Option 1

Option 2: Medium to Large

This design option, illustrated in [Figure 4-13](#), has the following characteristics:

- Dual WAN connectivity and dual WAN routers.
- Dual Internet connectivity and dual Internet routers (typically backup path over VPN tunnel as well as primary for VPN-only remote sites).
- Each of the WAN and Internet routers is dual-homed to the WAN module aggregation clustered switches using Layer 3 over mLAG mLAG (or Layer 3 ECMP in case of no switch clustering).
- This design model supports a medium to large number of remote sites (can support up to few thousand remote sites depending on the hardware capabilities of WAN routers and VPN termination routers).

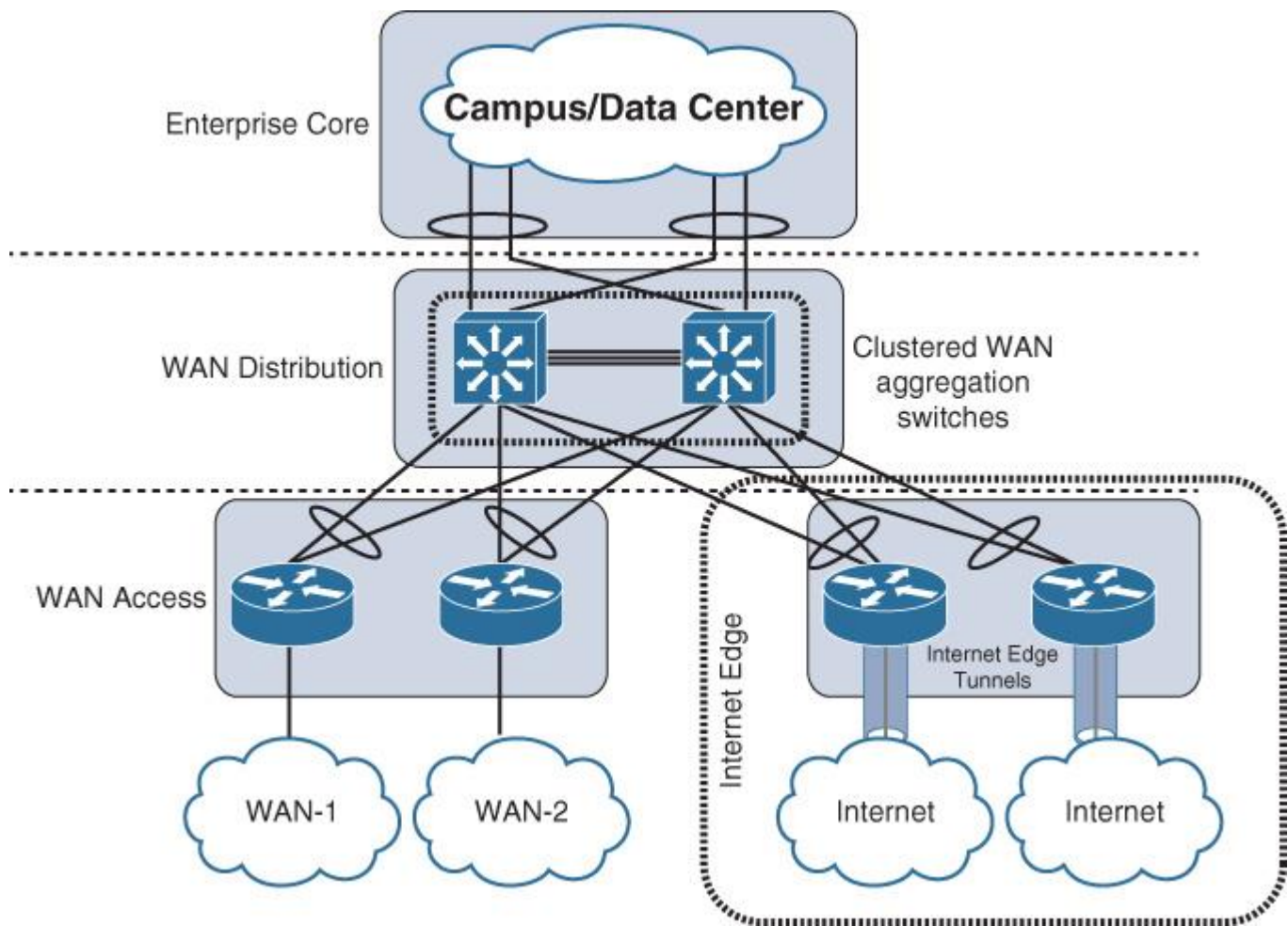


Figure 4-13 Enterprise WAN Module Design Option 2

Table 4-9 highlights the supported remote site WAN connectivity design options with regard to the two enterprise WAN design modules discussed earlier.

	WAN Module Design Option 1	WAN Module Design Option 2
Scalability with regard to the number of remote sites*	Small to medium	High
Support remote sites with single WAN link (MPLS only)	Yes	Yes
Support remote sites with single WAN link (over Internet)	Yes (single hub, single point of failure)	Yes
Support remote sites with dual WAN links (MPLS only)	No	Yes
Support remote sites with dual WAN link (MPLS + Internet)	Yes	Yes
Support remote sites with dual WAN links (over Internet only)	Yes (single hub, single point of failure, which eliminates the benefit of the redundant Internet links of the remote site)	Yes

* Available hardware resources such as memory and CPU, as well as routing design in terms of prefix aggregation and timers tuning, are primary influencing factors to the solution scalability.

Table 4-9 Supported Remote Site WAN Connectivity Design Options

Option 3: Large to Very Large

This architecture, illustrated in [Figure 4-14](#), targets very large-scale routed WAN deployments. This architecture encompasses branch, metro connectivity, and global core backbones.

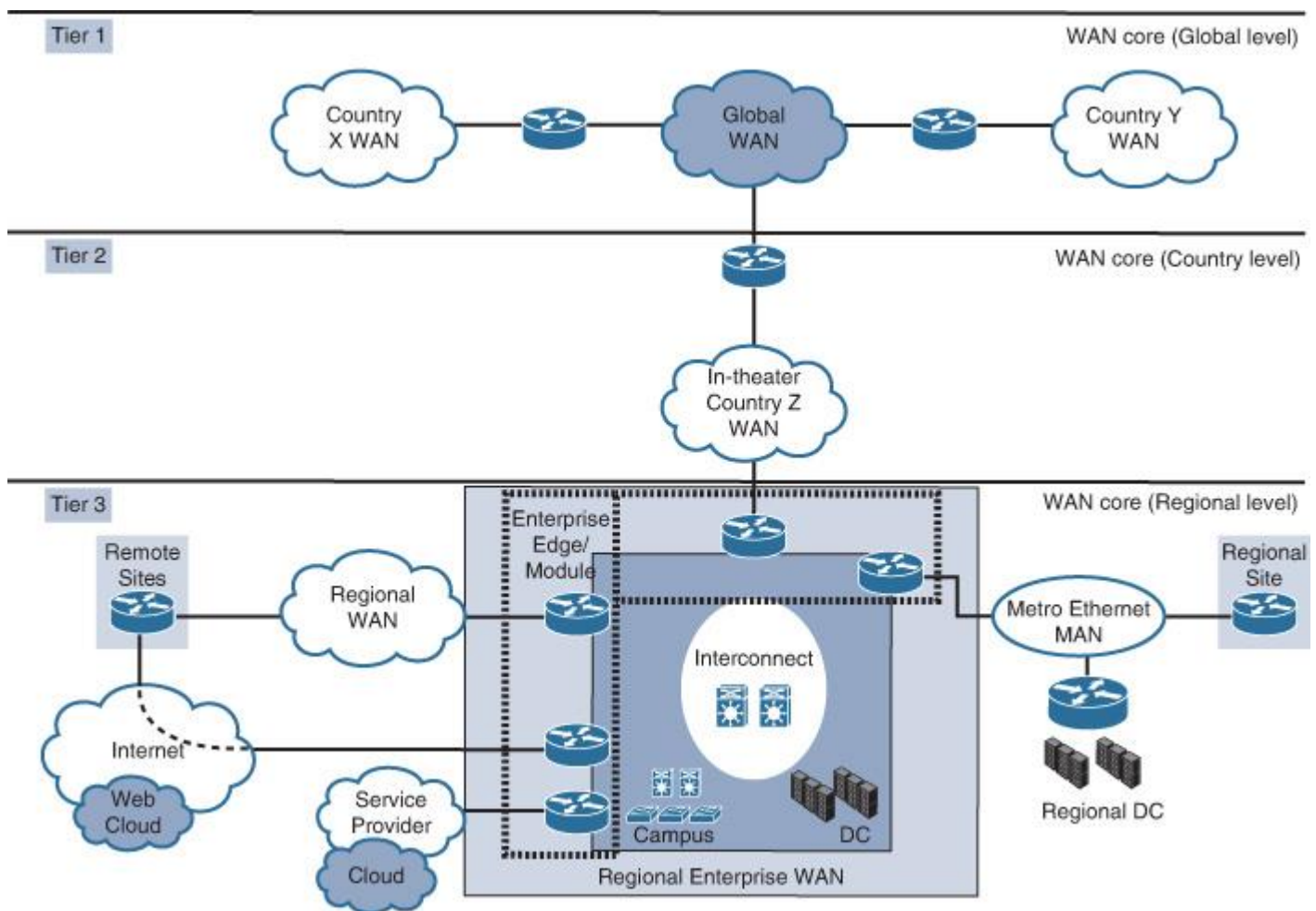


Figure 4-14 Enterprise WAN Module Design Option 3

This architecture consists of five primary modules:

- **Regional WAN:** Connects branch offices and aggregates remote locations
- **Regional MAN:** Connects remote offices and data centers across metro area transports
- **WAN core:** Interconnects regional networks and data centers within a country or theater or globally (provides connectivity between regional enterprises, interconnects within a theater and globally between theaters)
- **Enterprise edge:** Connects the enterprise network to other external networks and services (Internet service, mobile service)
- **Enterprise interconnect:** Used as an interconnection and aggregation point for all modules (provides connectivity between the regional WANs, MANs, data centers, enterprise edge, and campus networks)

This hierarchical structure offers flexibility for the design to be separated into different element tiers that are suitable to different environments. When a global footprint is required, all the elements of this architecture will likely apply. Whereas with a footprint that is solely within a single theater (region or country), it will not require the global core. However, it can be added when there is a requirement to expand into other regions [28]. The design of each element, such as the regional WAN and remote sites, should follow the design options discussed earlier in this chapter. For example, the regional “enterprise WAN module” can be based on any of the WAN design options discussed earlier in this chapter.

WAN Virtualization and Overlays Design Considerations and Techniques

Chapter 3, “Enterprise Campus Architecture Design,” covered drivers and advantages of considering network virtualization techniques within the enterprise, in particular the enterprise campus network. This section focuses on how the enterprise can extend network virtualization across the WAN to maintain end-to-end

path isolation for the different logical networks and groups. Network virtualization over the WAN (WAN virtualization) can be achieved using various approaches and techniques. This section highlights these various design options and the characteristics of each, and suggests uses that can fit different business requirements. One of the primary and common foundational technologies used by enterprises to facilitate achieving WAN virtualization is overlay technologies. In fact, overlay (also referred to as *self-deployed VPN*) technologies are adopted by businesses to serve different purposes, from enterprise an network design perspective, such as the following:

- Build a cost-effective WAN model also known as *Internet as WAN transport model* (virtual private WAN). You can use this model either as a primary or backup WAN transport.
- Provide a mechanism to maintain end-to-end path isolation for logical groups or entities across the enterprise WAN.
- Secure IP communications over the private enterprise WAN or Internet.
- Provide a controlled remote-access method for mobile users or third-party entities that might require access to certain internal resources.
- Provide overlaid transport for services and applications that are not supported by the underlay transport infrastructure, such as multicast-based applications and IPv6 applications over “unicast-only” IPv4 IP network.

Therefore, it is critical that network designers have a very good understanding of the different overlay (VPN) options in terms of supported design models, strengths, limitations, and suitable use cases.

Table 4-10 compares the different overlay (VPN) technologies that commonly used to extend enterprise connectivity and facilitate network virtualization over IP transport networks.

	Remote Access VPN (Client-based)			
	DMVPN	IPsec	GRE	
Targeted transport network	Public Internet transport	Private WAN and public Internet	Private WAN and public Internet	Private WAN and public Internet
Supported network topology	Hub-spoke (client to server)	Hub-spoke, spoke to spoke	Point to point	Point to point
Routing technique	Reverse-route injection	Static and dynamic routing	Reverse-route injection	Static and dynamic routing
Encryption style	Peer-to-peer protection	Peer-to-peer protection (with IPsec)	Peer-to-peer protection	Peer-to-peer protection (with IPsec)
IP multicast	Multicast replication at hub (if multicast is supported by the VPN client or may requires GRE with IPsec)	Multicast replication at hub	Point-to-point (with GRE)	Point to point
Scalability	Moderate	High	Low	Low
Design flexibility	Limited to client/server communication style	High	Low	Moderate
Operational complexity	Moderate (only for remote-access users)	Moderate	High	High
Network virtualization techniques	Limited (VRF-aware remote access)	Flexible (end to end)	Limited (VRF-aware IPsec)	Flexible (end to end)

Table 4-10 Network Overlay Technologies Comparison

Note

The primary scalability limiting factor of any VPN solution is the support number of sessions by the used hardware platform.

Note

GET VPN is an encryption mechanism that enables you to preserve IP header information that supports true “any-to-any” encrypted IP connectivity model. Therefore, it is commonly used over private transport networks such a private WAN instead of the other IP tunneling mechanisms. Having said that, GETVPN is not always the ideal or optimal overlay and encryption solution over the private WAN. For example, if the existing WAN platforms of an organization do not

support GETVPN (and the business has no intention or plan to upgrade any network hardware/software), then you need to deal with the design constraints and consider other options here, such as IPsec with GRE or mGRE.

These different VPN technologies highlighted in [Table 4-10](#) are the foundation of achieving WAN virtualization. However, modern large-scale enterprises can use other approaches to maintain end-to-end path separation, such as “self-deployed” MPLS L3VPN. The following section classifies, discusses, and compares all the different primary technologies and design options that can help you achieve WAN virtualization to suit different design requirements.

WAN Virtualization

Introducing virtualization and path isolation over the WAN transport is commonly driven by the adoption of the network virtualization concept by the enterprise within the campus LAN, branches, or data center network. Therefore, to maintain end-to-end path isolation, network virtualization must be extended over the WAN transport in a manner that does not compromise path-isolation requirements. From a WAN design point of view, two primary WAN connectivity models drive the overall WAN virtualization design choices:

■ **Customer- or enterprise-controlled WAN:** Also known as *self-deployed*, this model provides full control for the enterprise to use the desired core routing design and the type of virtualization techniques that meet their requirements, such as MPLS in the core or tunneling with multiple VRFs. Furthermore, all the techniques discussed in the campus enterprise virtualizations section are applicable. Typically, this model is based on the fact that the enterprise controls the WAN core infrastructure or transport, as depicted in [Figure 4-15](#).

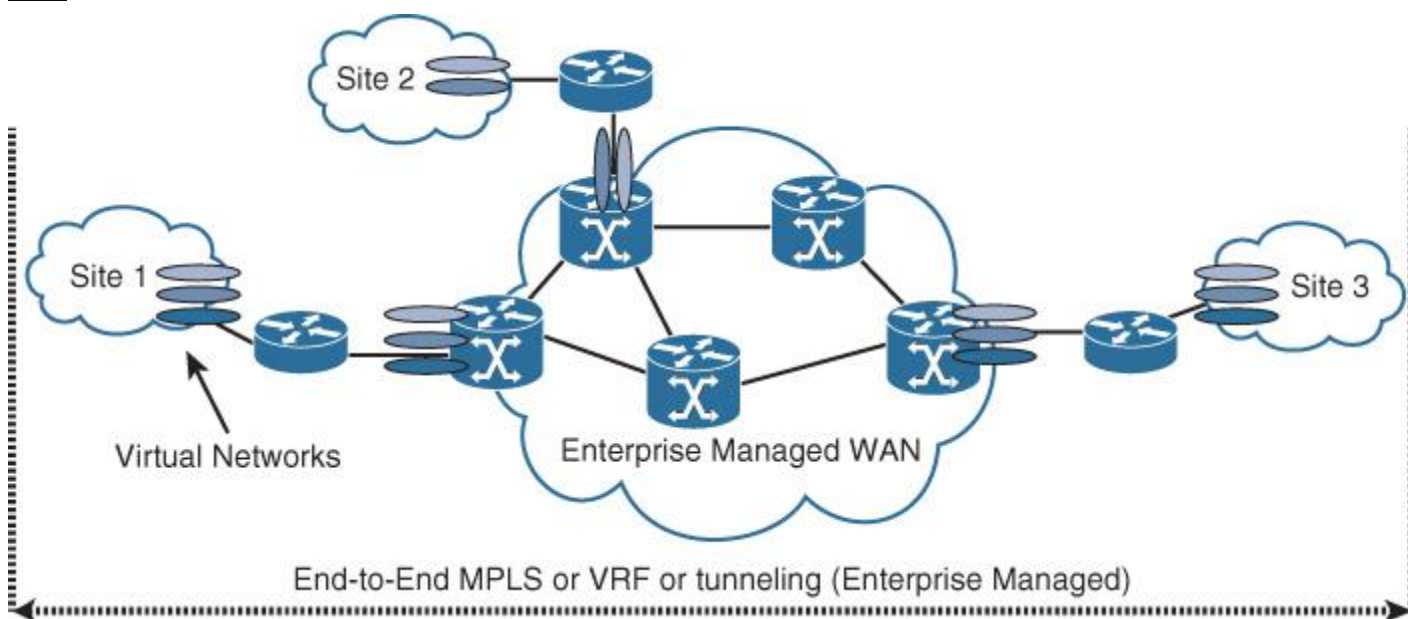


Figure 4-15 Enterprise-Managed WAN

Note

If the WAN SP in the middle provides L2 WAN transport, it can be categorized under the enterprise-controlled WAN model, because the enterprise will have the control and freedom, to a large extent, to deploy the desired end-to-end

WAN virtualization techniques based on the business and technical requirements, such as MPLS-enabled virtualization or subinterfaces with VRFs, as illustrated in [Figure 4-16](#).

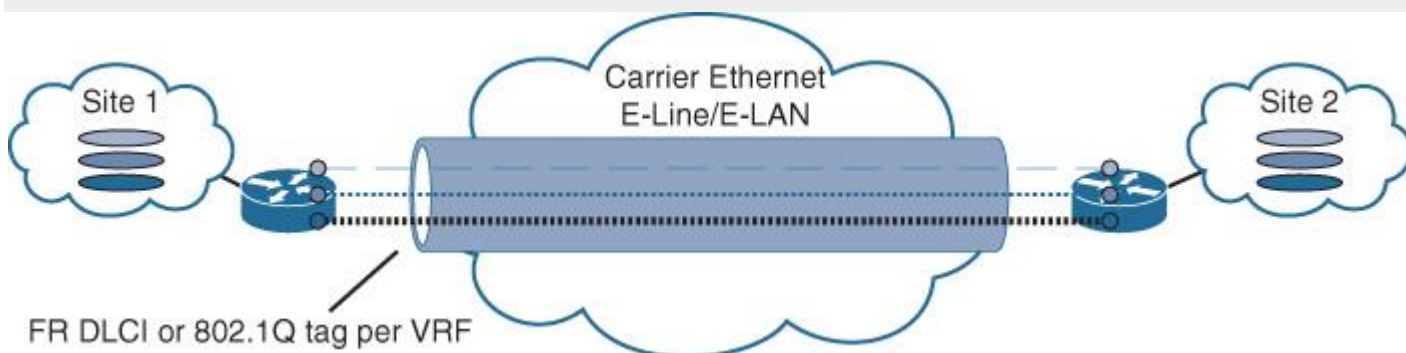


Figure 4-16 Enterprise-Managed WAN over L2VPN Cloud

■ **SP-controlled WAN:** This model, compared to the previous model, provides the least control for the enterprise when it comes to routing an end-to-end network virtualization over an SP-controlled WAN transport, such as MPLS L3VPN, as depicted in [Figure 4-17](#). Therefore, enterprises need to either extend the virtualization to the SP (to the PE node) or build an overlay over the SP managed network between their CE nodes to facilitate the formation of the required end-to-end network virtualization. This approach is commonly referred to as *over the top*.

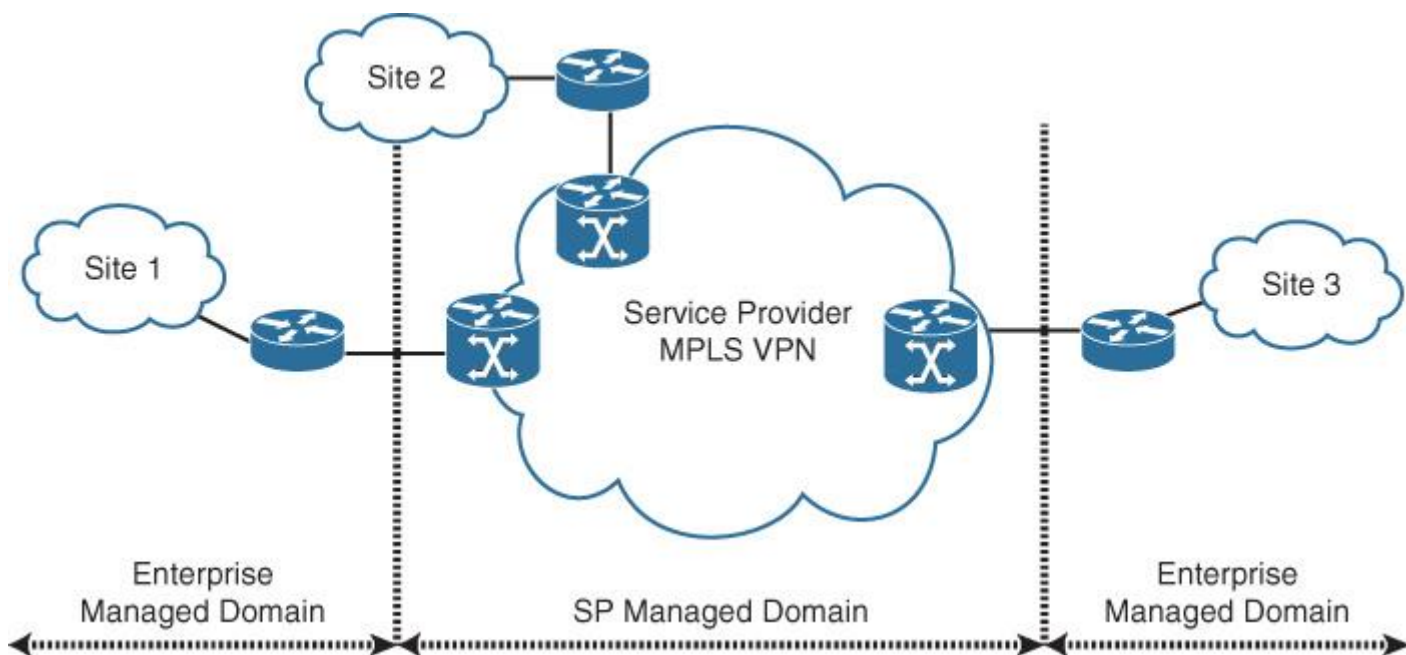


Figure 4-17 Unmanaged Enterprise WAN

Over-the-Top WAN Virtualization Design Options (Service Provider Coordinated/Dependent)

The following design options require coordination with the SP to support extending the enterprise network virtualization over the MPLS L3VPN provider network. The two common and proven approaches used to extend network virtualization of an enterprise over unmanaged L3VPN SP network are as follows:

■ **Back-to-back VRFs to provider PE:** This approach is based on using the concept of Multi-VRF CE. This approach provides L3 path virtualization extension without exchanging labels over IP tunnels or physical interfaces (subinterfaces) with the provider PE, as illustrated in [Figure 4-18](#). Typically, a routing instance (process) per VRF is required at each CE and PE node to exchange routing information per virtual network.

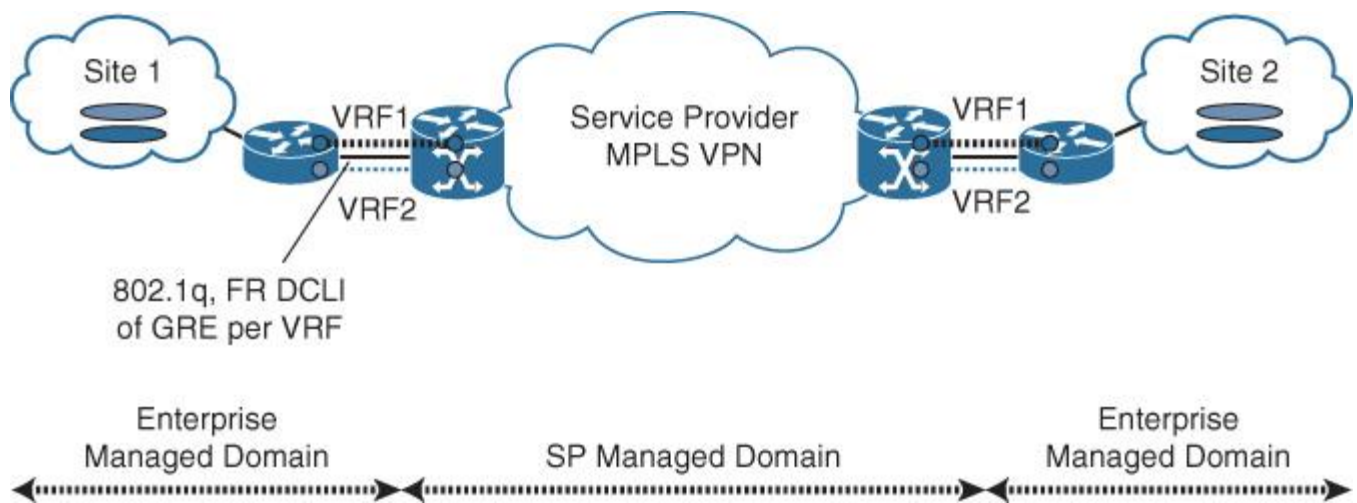


Figure 4-18 Multi-VRF CE

■ **Enable MPLS (Label Distribution Protocol [LDP]) with provider PE:** This approach is based on the Carrier Supporting Carrier design model (CSC, RFC 3107), where the CE node can send packets along with MPLS label to the provider PE, which ultimately can facilitate for enterprises the formation of their own multiprotocol BGP (MP-BGP) peering across the SP MPLS L3VPN backbone, as illustrated in [Figure 4-19](#).

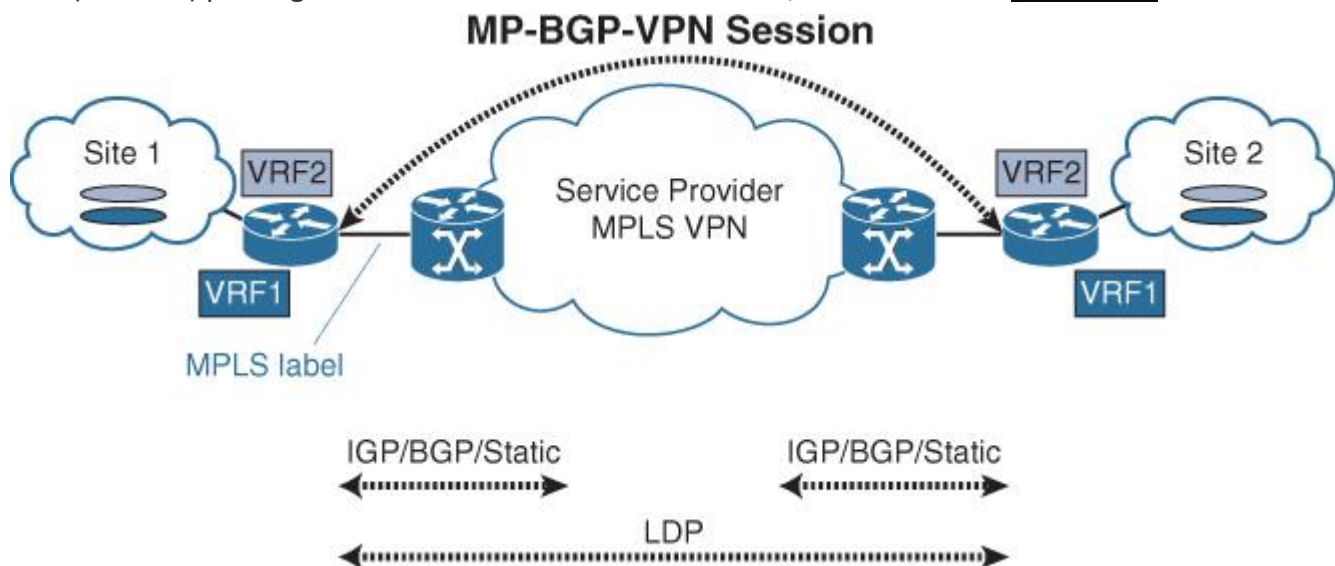


Figure 4-19 CSC model

[Table 4-11](#) compares these two design approaches from different design angles.

	CSC Model	Back-to-Back VRFs Model
Scalability	High	Low
Coordination with the SP	<i>Moderate</i>	<i>High</i>
Design complexity	Moderate	The larger, the more complex
Dependencies on SP (for example, multicast support)	High	High
Extra cost	No	Yes (SP might charge per additional VRF.)
Adding new virtual network/VRF requires coordination with SP	No	Yes
Requires label exchange with provider PE	Yes	No
Requires PE-CE routing instance per VRF	No	Yes
Control plane complexity	Moderate	High
Operational complexity	Moderate	High
Security and edge policy control	Moderate	High
QoS granularity	Moderate	High

Table 4-11 Multi-VRF CE Versus CSC Model for Enterprise WAN Virtualization

Over-the-Top WAN Virtualization Design Options (Service Provider Independent)

This section discusses the different design options that use various overlay approaches, which can facilitate the extension of an enterprise network virtualization over an unmanaged L3 SP WAN. Unlike the approaches discussed in the previous section, the design options discussed in this section are end-to-end controlled and deployed by the customer or enterprise side without any coordination/dependencies with the WAN SP (simply because all the methods are based on the concept of using different tunneling [overlay] mechanisms that typically encapsulate and hide all the traffic and virtualization setup from the underlying SP transport network):

■ **Point-to-point GRE tunnel per VRF:** This design option offers a simple private virtual network extension over GRE tunnels, where each GRE tunnel is assigned to a specific VRF per virtual network, without any need to coordinate with the WAN provider. However, this option can introduce operations and setup complexities in large deployments because of the large number of manual configurations of the point-point tunnels, each with its own control plane. In addition, this design option has the least scalability, because the number of tunnels can increase significantly when the number of sites and VRFs increases. For example, 60 sites with 3 VRFs each will require $(N - 1)$ tunnel per VRF, $(59 * 3) = 177$ tunnels to create. Nevertheless, this design option can be a good choice for traffic isolation between a very small number of sites (ideally two or three sites only) with a very limited number of VRFs (ideally two or three) VRFs, as illustrated in [Figure 4-20](#).

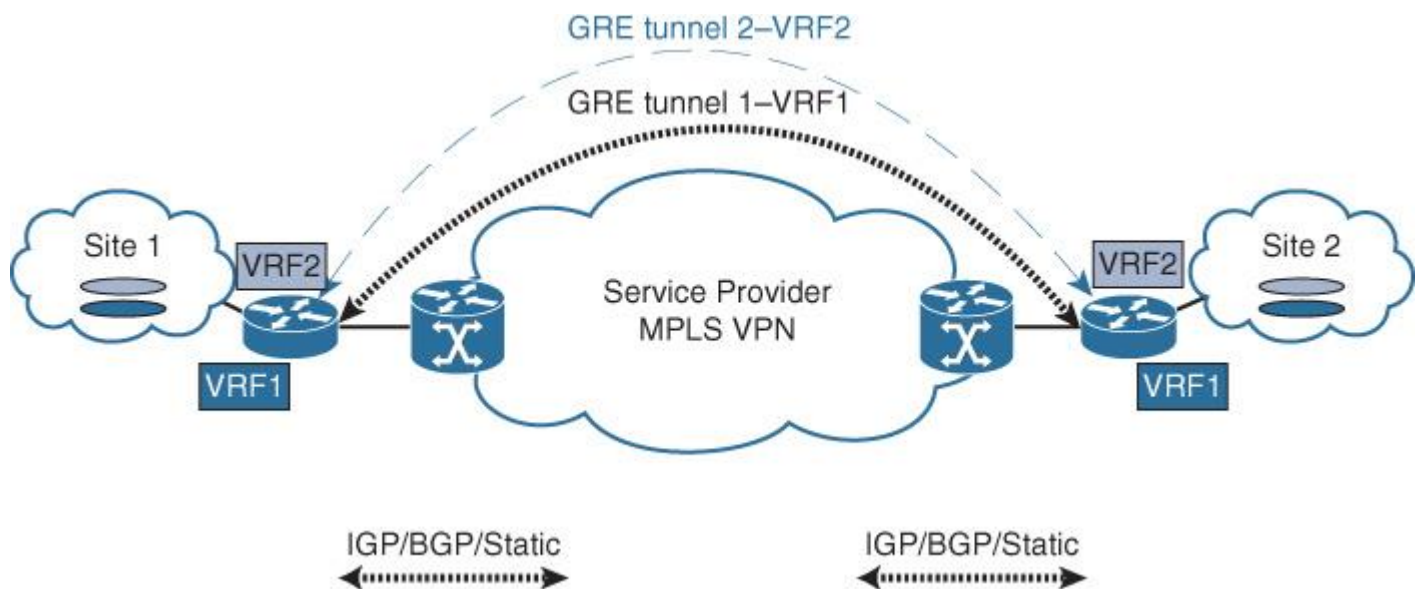


Figure 4-20 Point-to-Point GRE Tunnel per VRF

Note

For this design option and the subsequent ones, it is hard to generalize and provide a specific recommended number of remote sites or VRFs, because the decision has to be made based on these two variables when measuring the scalability of the design option. For example, evaluating this design option for a network that requires path isolation between 3 sites, where each site has 10 different virtual networks to transport, is different from when there are 3 sites with 2 virtual networks in each. In both cases, the number of sites is small; however, the number of VRFs (virtual networks) here becomes the tiebreaker:

■ **Dynamic multipoint GRE (DMVPN) per VRF:** This design option is typically based on using multipoint GRE tunnels (DMVPN) per virtual network, as illustrated in [Figure 4-21](#), which helps to overcome some of the scalability issues of the previous option to some extent. This design option also supports direct spoke-to-spoke traffic forwarding (bypassing the hub) per VRF. Furthermore, it supports deployments of a larger scale than those of the point-to-point GRE tunnels. However, it still has scalability and operational limitations and complexities when the network grows, because there will be a DMVPN cloud per VRF. This means that the greater the number of VRFs required, the greater the number of DMVPN clouds that need to be created and operated with a separate control plane for each. This design option ideally supports the following design combinations:

- Large number of remote sites with very small number of VRFs (ideally two)
- Small number of remote sites with small number of VRFs (ideally not more than three)

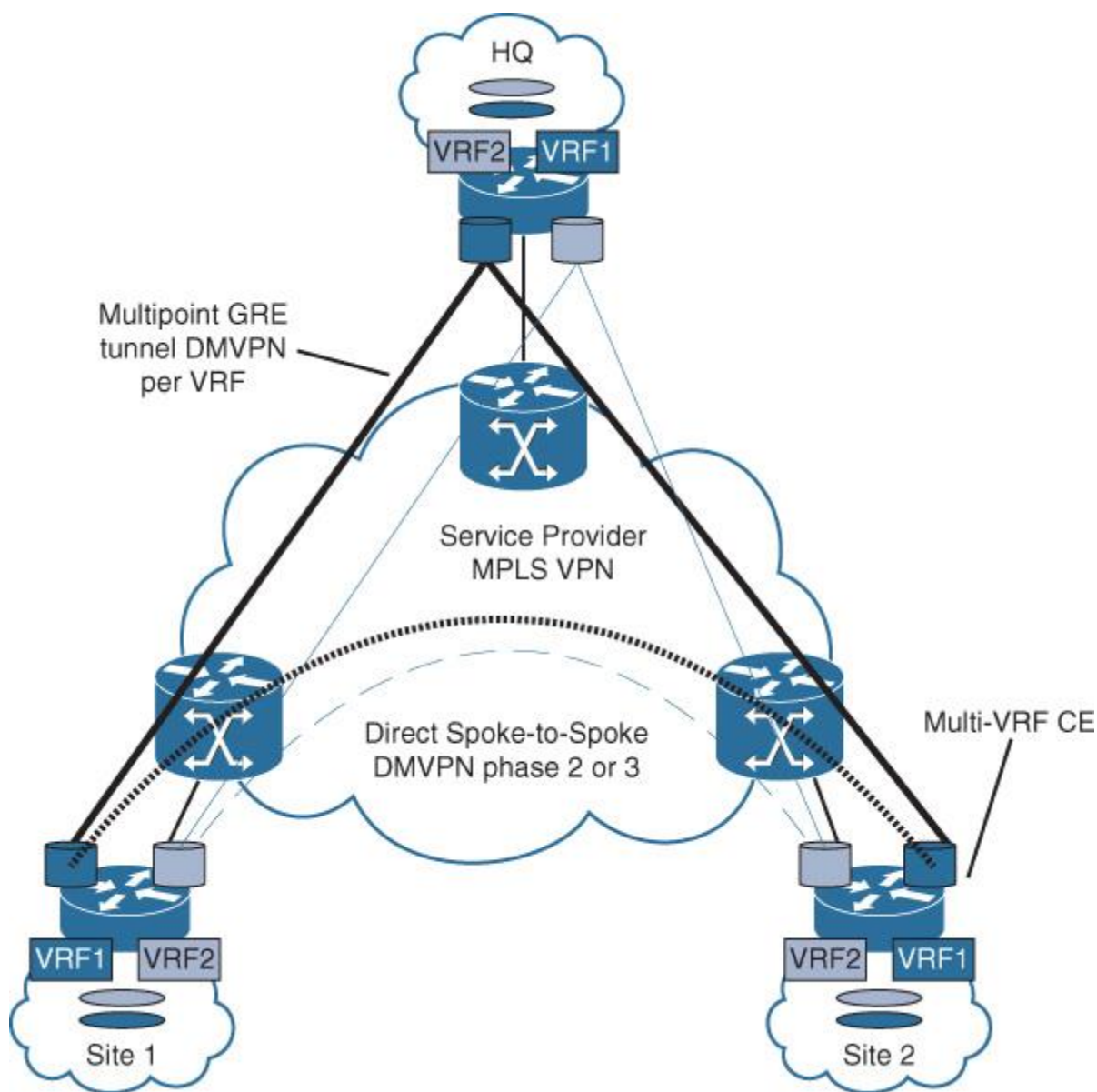
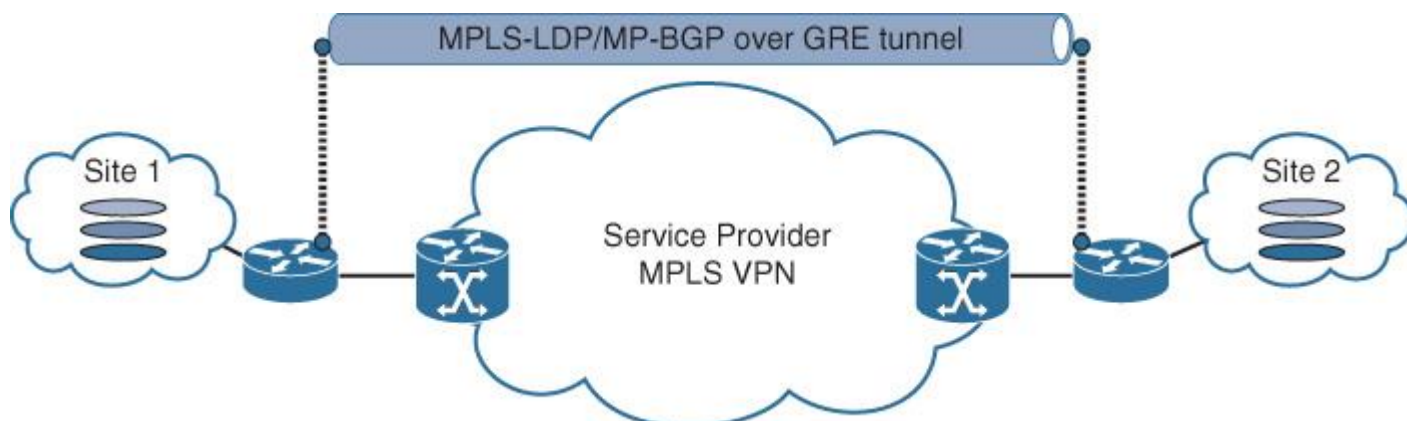


Figure 4-21 DMVPN per VRF

■ **MPLS over point-to-point GRE tunnel:** This design option is based on the concept of encapsulating MPLS labels in a GRE tunnel, as described in RFC 4023, which helps to overcome some of the limitations of the point-to-point GRE tunnel per VRF design option, by using MPLS with an MP-BGP VPNv4/6 session over one GRE tunnel (RFC 2547 MP-BGP control plane style), as depicted in [Figure 4-22](#). Consequently, there will be only one GRE tunnel required to carry LDP, IGP, and MP-BGP (VPNv4/6). Typically, there is no need to create a separate GRE tunnel per VRF with this design option. However, the number of remote sites is still a limiting factor in the scalability of this design option in the case where many remote sites need to be connected, either in a fully meshed manner or using hub-and-spoke overlay topology.



Furthermore, this design option can help simplify the interconnection of disjoint MPLS-enabled infrastructures over a native IP backbone. As illustrated in [Figure 4-23](#), MPLS over GRE is used to extend the reachability between two MPLS-enabled islands over a non-MPLS backbone (native IP).



3. Advanced DMVPN & Routing Fun in the Lab: Part 2 – The Forwarding Plane, <http://www.networkingwithfish.com>

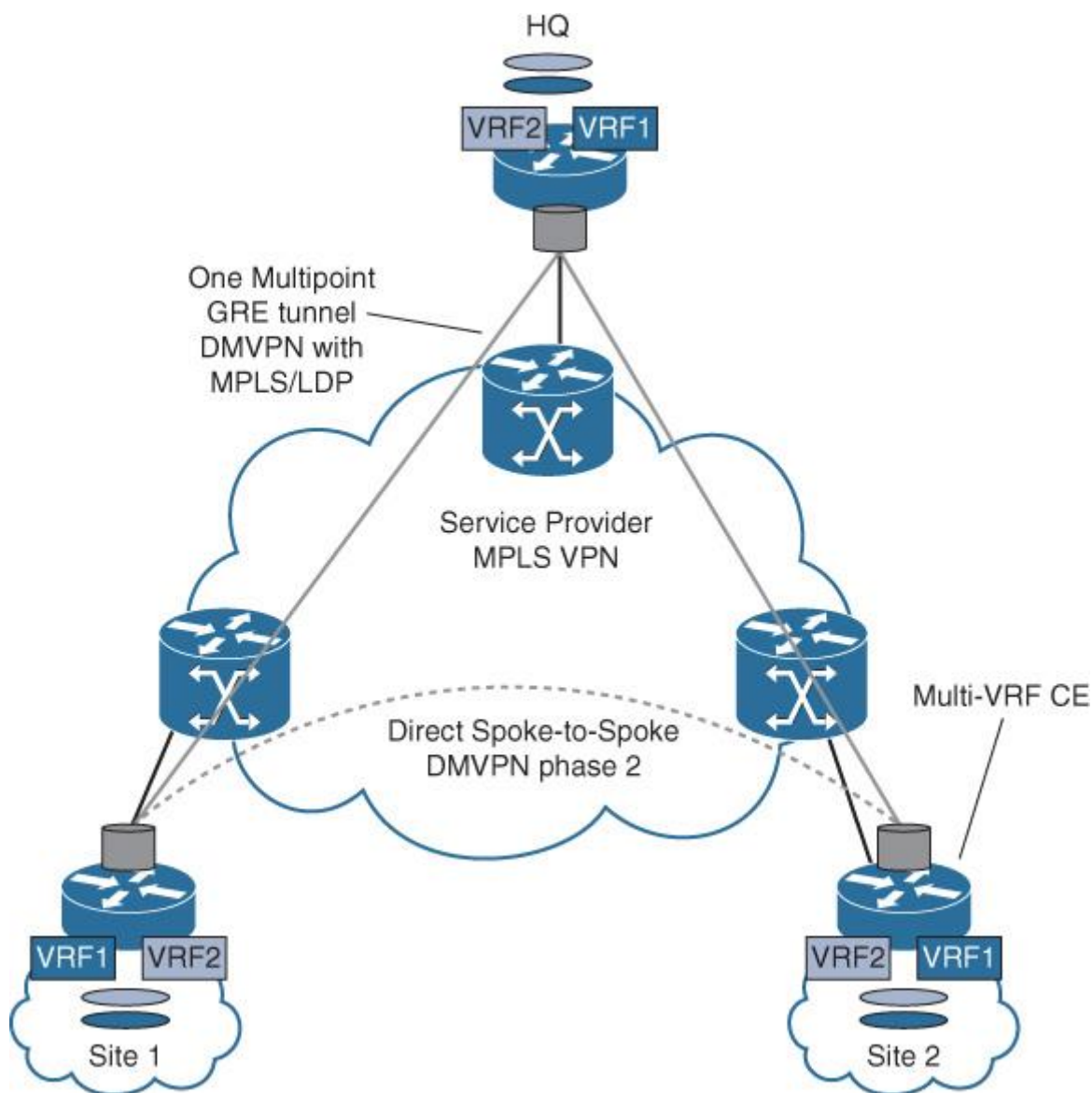


Figure 4-24 MPLS over DMVPN

■ **MPLS over multipoint GRE (using BGP for endpoint discovery):** MPLS over mGRE simplifies the design and implementation of overlaid (self-deployed) MPLS VPN using the standard RFC 2547 MP-BGP control plane, which offers dynamic tunnel endpoint discovery using BGP as the control plane. This solution requires only one IP address (typically a loopback address) of each of the enterprise CE routers to be advertised to the interconnecting SP cloud network, as depicted in [Figure 4-25](#). In addition, there is no requirement to manually configure any GRE tunnel or enable LDP/RSVP (Resource Reservation Protocol) on any interface. Instead, mGRE encapsulation is automatically generated with the dynamic endpoint discovery capability. The VPNv4 label and VPN payload are carried over the mGRE tunnel encapsulation. This solution offers a simplified and scalable any-to-any unicast (IPv4, IPv6 6VPE based) and multicast (MDT based) MPLS VPN communication model.

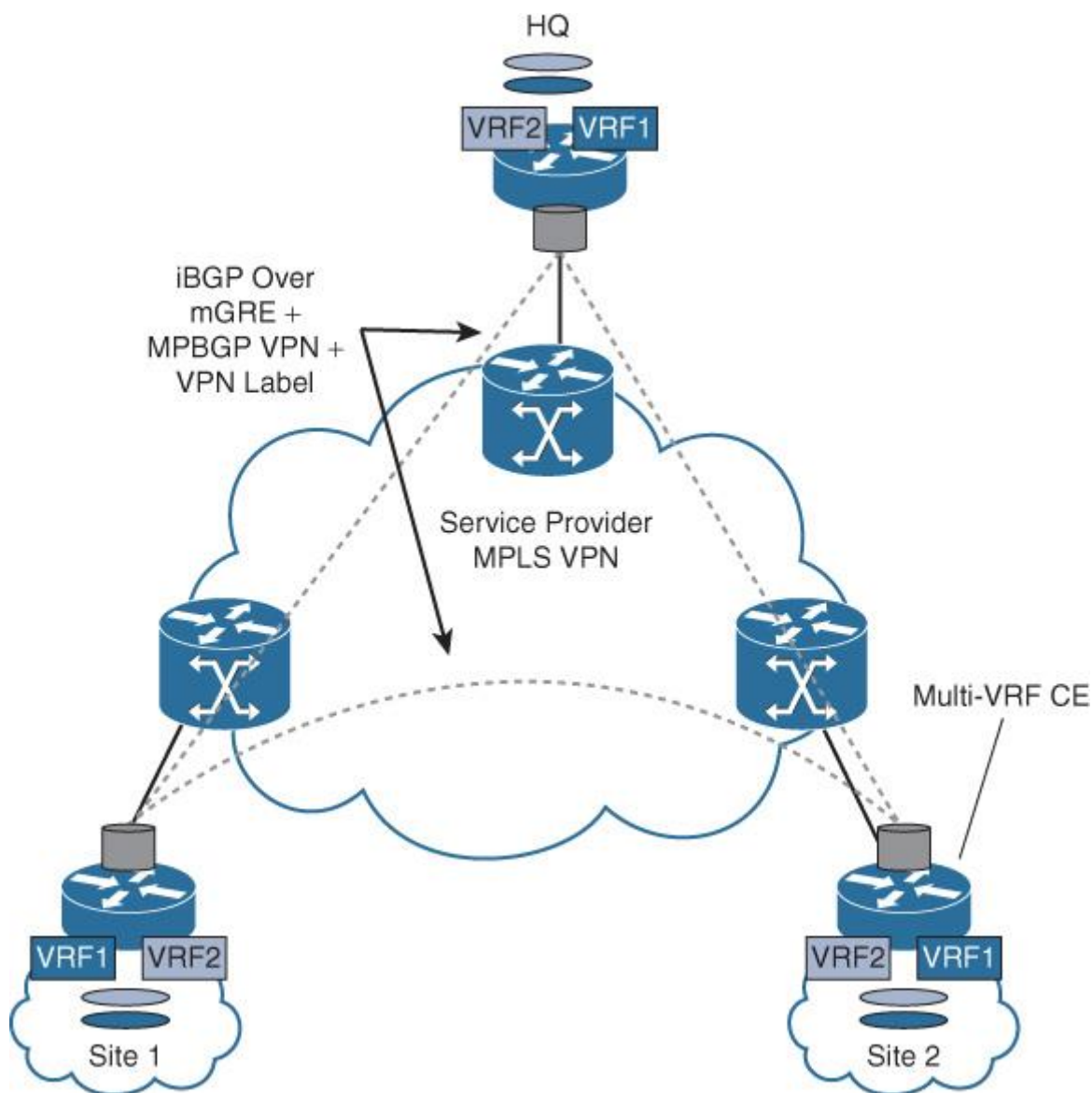


Figure 4-25 MPLS over mGRE BGP Autodiscovery Based

■ **EIGRP Over the Top (OTP):** As the name implies, EIGRP OTP offers enterprise customers the opportunity to form EIGRP adjacencies across unmanaged WAN transport (typically over an L3VPN MPLS provider cloud) using unicast packets for peering and exchanging route prefixes without being injected into the provider's MP-BGP VPNv4/v6 routing table. With this approach, EIGRP OTP offers simplified dynamic multipoint encapsulation using Locator/ID Separation Protocol (LISP) to encapsulate its data traffic. EIGRP OTP relies on EIGRP routing tables rather than on the LISP mapping system to populate IP routing information. Furthermore, multiple instances of EIGRP can be deployed, along with other network virtualization techniques, to offer multiple routing instances.

This design approach offers significant design flexibility for enterprise WAN connectivity because WAN networks will be seen as a virtual extension of the network and enterprise customers can simply and transparently extend their infrastructure reachability over the provider's network using one unified control plane protocol, as shown in [Figure 4-26](#).

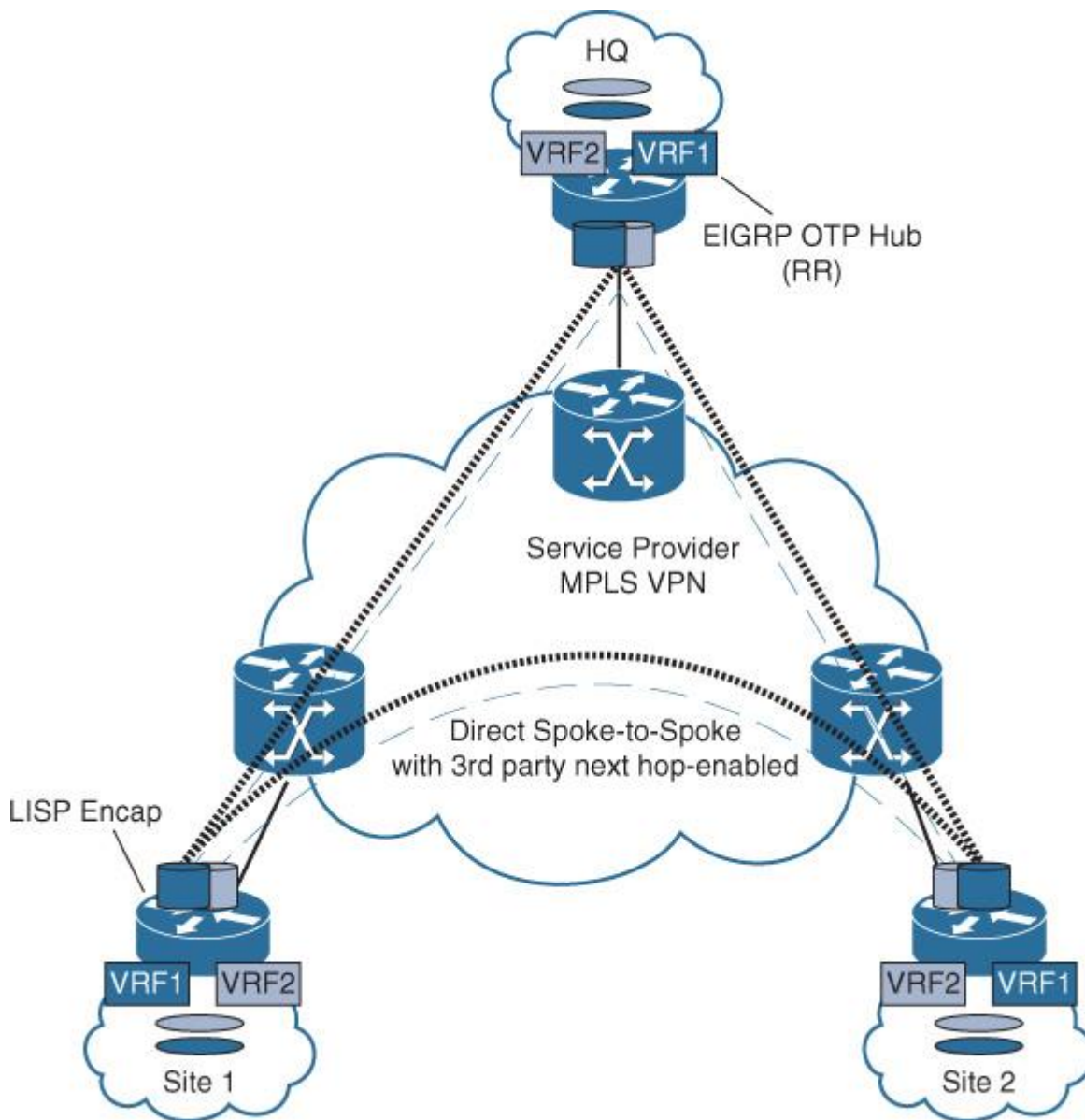


Figure 4-26 EIGRP OTP

Comparison of Enterprise WAN Transport Virtualization Techniques

Table 4-12 provides a summarized comparison, from different design aspects, between the different WAN virtualization techniques discussed in this chapter.

	SP Dependent	Control Plane	Number of VNs Scalability	Number of Remote Sites Scalability	Direct CE-to-CE Forwarding	Multi
VRF Lite	Yes	IGP/BGP	Limited	Very limited	Yes	SP de
CSC model	Yes	IGP + BGP/ MP-BGP	Scalable	Scalable	Yes	SP de
P2P GRE per Vrf	No	IGP/BGP	Limited	Very limited	Yes (full-mesh P2P tunnels)	Yes
P2P GRE + MPLS	No	IGP + MPMBGP	Scalable	Very limited	Yes (full-mesh P2P tunnels)	Yes
DMVPN per VRF	No	IGP/BGP	Limited	Scalable	Yes	Yes (l spoke path c
DMVPN + MPLS	No	IGP/BGP + MP-BGP	Scalable	Scalable	Yes (DMVPN phase 2)	Yes (l spoke path c
BGP mGRE + MPLS	No	MP-BGP	Scalable	Scalable	Yes	Yes
EIGRP OTP	No	EIGRP	Limited	Scalable	Yes (with third- party next hop enabled)	No

Table 4-12 WAN Transport Virtualization Techniques Comparison

Note

Operational complexity always increases when the network size increases and the WAN virtualization techniques used have limited scalability support and vice versa.

WAN Virtualization Design Options Decision Tree

[Figure 4-27](#) is a summarized decision tree of the different design options for enterprise WAN virtualization.

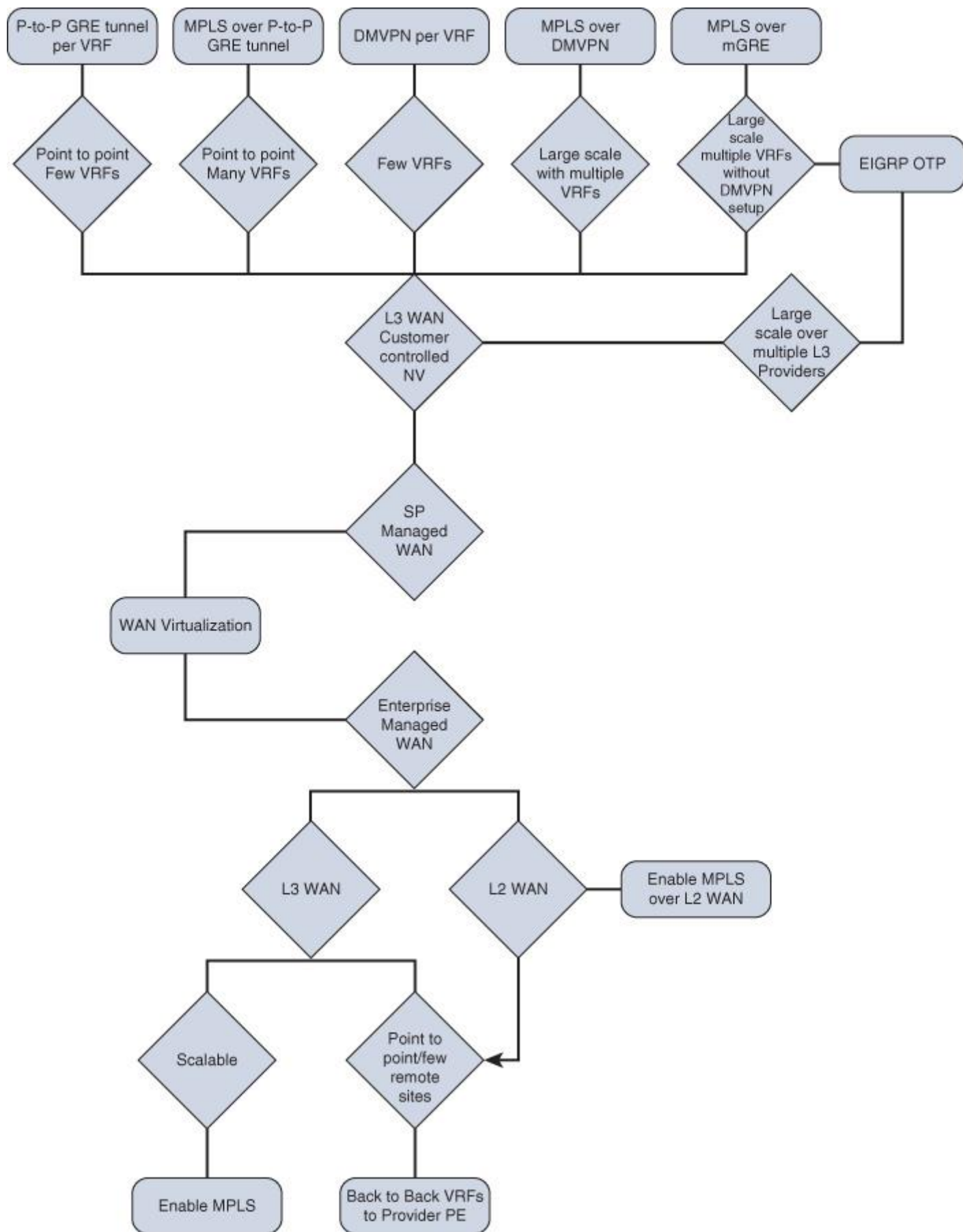


Figure 4-27 Enterprise WAN Virtualization Decision Tree

Enterprise WAN Migration to MPLS VPN Considerations

Network migration is one of the most challenging and critical projects of network design. Network designers sometimes spend a large amount of time focusing on the “end state” of their network designs. However, a

good and successful design must consider a migration strategy to move the network from its current state to the new state. Similarly, a good migration plan should address how network control plane protocols and applications will interact when the network is partially migrated.

It is impossible to provide general definite guidance for network migration because many variables drive and influence the migration plan and its strategy, such as business and application requirements, network size, and the technologies used. However, the following are some approaches and rules of thumb based on proven WAN migration experiences that should be considered when generating strategic or tactical migration plans:

- The phased migration approach is always recommended for large-scale networks with a large number of remote sites where multiple links or external exit points might exist.
- Logical and physical site architecture must be analyzed and taken into consideration during the planning phase, such as backdoor links, OSPF areas if the PE-CE protocol is OSPF, and BGP autonomous system (AS) numbering if BGP is used across the remote sites.
- In L3VPN, review the selected PE-CE routing protocol and consider how it can integrate with the existing routing setup and design without introducing any service interruption after moving to the new MPLS cloud.
- It is important to identify where the default route is generated and whether the remote sites will use a locally generated default route or over the MPLS cloud.
- Review the routing and identify the places that have summarization because, in some situations, this may lead to suboptimal routing or route summarization black holes.
- In L2VPN, there might be different topologies that can be provisioned by the provider (hub and spoke, full mesh, or partial mesh). A careful review of the topology and the effect of the path that traffic will take after the migration is very important, especially if you are moving from one topology to another. For example, you may be migrating the WAN from a hub-and-spoke topology over Frame Relay to a full-mesh L2 WAN (for example, VPLS).

Migrating from Legacy WAN to MPLS L3VPN WAN Scenario

This section provides an example of a phased migration approach based on a migration scenario of an international media company that is currently using two separate Layer 2 (Frame Relay) clouds over hub-and-spoke topology across two different geographic regions (the United States and Europe). In addition, the hub sites of both the U.S. and Europe regions are interconnected over a Layer 2 virtual leased line (VLL) provisioned by an international Carrier Ethernet, as illustrated in [Figure 4-28](#). After this media company finished a recent migration project of their video and broadcasting systems from a legacy time-division multiplexing (TDM)-based solution to an IP-based solution, they started facing degraded video quality because of the increased number of packets lost and delayed across the WAN. The obvious cause of this issue is the large number of video streams across the low-capacity Frame Relay WAN links and because of the long trip each video stream has to take to reach its intended destination. For instance, traffic between two sites located within the same region has to traverse the hub site first. In addition, traffic between sites across different regions has to traverse the local and remote hub sites over the VLL link, as shown in [Figure 4-28](#). Therefore, to overcome these limitations, a decision has been made to migrate the current hub-and-spoke Frame Relay WAN to an any-to-any topology over a single global MPLS L3VPN provider, to meet current traffic flow requirements with regard to the available bandwidth and delay and to achieve more consistent end-to-end QoS.